

Upstream open reading frames buffer translational variability during *Drosophila* evolution and development

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Yuanqiang Sun, Yuange Duan, Peixiang Gao, Chenlu Liu, Kaichun Jin, Shengqian Dou, Wenxiong Tang, Hong Zhang, Jian Lu 

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China • Eye Institute of Shandong First Medical University, State Key Laboratory Cultivation Base, Shandong Provincial Key Laboratory of Ophthalmology, Qingdao, China • College of Ecology, Lanzhou University, Lanzhou, China • Beijing Advanced Center of RNA Biology (BEACON), Peking University, Beijing, China • Southwest United Graduate School, Kunming, China

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This study reveals the **important** role of upstream open reading frames (uORFs) in limiting the translational variability of downstream coding sequences. Through a combination of computational simulations, comparative analyses of translation efficiency across different developmental stages in two closely related *Drosophila* species, and manipulative, experimental validation of translation buffering by an uORF for a gene, the authors provide **convincing** evidence supporting their conclusions. This work will be of broad interest to molecular biologists and geneticists.

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Abstract

Protein abundance tends to be more evolutionarily conserved than mRNA levels both within and between species, yet the mechanisms underlying this phenomenon remain largely unknown. Upstream open reading frames (uORFs) are widespread *cis*-regulatory elements in eukaryotic genomes that regulate translation, but it remains unclear whether and how uORFs contribute to stabilizing protein levels. In this study, we performed ribosome translation simulations on mRNA to quantitatively assess the extent to which uORF translation influences the translational variability of downstream coding sequences (CDSs) across varying contexts. Our simulations revealed that uORF translation dampens CDS translational variability, with buffering capacity increasing in proportion to uORF efficiency, length, and number. We then compared the translomes at different developmental stages of two *Drosophila* species, demonstrating that uORFs buffer mRNA translation fluctuations during both evolution and development. Experimentally, deleting a uORF in the *bicoid* (*bcd*) gene—a prominent example of translational buffering—resulted in extensive changes in gene expression and phenotypes in *Drosophila melanogaster*. Additionally, we observed uORF-mediated buffering between primates and within human populations. Together, our results

reveal a novel regulatory mechanism by which uORFs stabilize gene translation during development and across evolutionary time.

Introduction

Organisms have evolved various strategies for the spatiotemporal regulation of gene expression ^{1–3}. This is important because aberrant gene expression can result in phenotypic defects or diseases, while the variation and evolution of gene expression patterns frequently promote phenotypic diversification and adaptation ⁴. Although variations in mRNA abundance are widely observed within or between species, protein abundance tends to show stronger evolutionary constraint ^{5,6}, as observed in yeasts ^{7–9}, primates ^{10,11}, and other organisms ^{12–14}. Nevertheless, the molecular mechanisms by which the conservation of protein abundance across species is achieved are largely unknown ^{5,6}.

Eukaryotic mRNA translation is a crucial step in gene expression and is highly regulated by multilayered mechanisms ^{15–17}. Upstream open reading frames (uORFs), which are short open reading frames in the 5-terminal untranslated regions (5' UTRs) of eukaryotic mRNAs, play crucial roles in regulating mRNA translation. Approximately 50% of eukaryotic genes contain uORFs ¹⁸, and their evolution has been tightly shaped by natural selection ^{19–23}. The functions of uORFs have been explored in various contexts, including development ^{23–30}, disease ^{31–35}, and stress responses ^{36–43}. The prevailing consensus is that uORFs typically repress downstream coding sequence (CDS) translation by sequestering ribosomes, a process influenced by factors such as uORF length, position, and sequence context ^{34,43–48}. However, under stress conditions, certain uORFs can facilitate CDS translation by promoting ribosome reinitiation, illustrating their context-dependent functions ^{42,43,48–52}.

Gene expression noise, which arises from the inherent stochasticity of biological processes such as transcription and translation, is generally detrimental to organismal fitness ⁵³ and is primarily determined at the translational level ⁵⁴. Recent studies suggest uORFs might play essential roles in buffering translational noise and stabilizing protein expression. For example, Wu et al. (2022) ⁵⁵ demonstrated that uORFs reduce protein production rates to stabilize TOC1 protein levels, ensuring precise circadian clock function in plants ⁵⁵. Similarly, Bottorff et al. (2022) ⁵⁶ used a human cell reporter system to show that a ribosome stall in cytomegaloviral *UL4* uORFs buffers against CDS translation reductions ⁵⁶. Under stress, translation initiation is typically downregulated, yet most human mRNAs resistant to this inhibition contain translated uORFs, with a single uORF often being sufficient for resistance ⁴². The computational model of Initiation Complexes Interference with Elongating Ribosomes (ICIER) suggests that derepression of downstream translation is a general mechanism of uORF-mediated stress resistance ⁴³. Despite these findings, the current understanding of uORFs in stabilizing translation is limited to single-gene cases or stressed conditions. It remains unclear whether and how uORFs affect gene translation variability on a genome-wide scale during evolution and development, and whether the identified mechanisms are universal or vary substantially among different taxa. To address these questions, a combination of modeling, genome-wide analyses, and comparative studies across species is required.

In this study, we first adapted the ICIER framework ⁴³ to simulate the translating ribosome on a mRNA to quantitatively measure the extent to which uORF translation reduces the translational variability of the downstream CDS under different translation contexts. We then compared the translomes of two closely related *Drosophila* species, *D. melanogaster* and *D. simulans*, and further supported the notion that uORFs could buffer the fluctuations of CDS translation during the development and evolution of *Drosophila*. The patterns also reappeared among primates and human populations. We next knocked out the *bicoid* (*bcd*) uORF, a case showing the significant

buffering effect in our data, and observed broad changes in the embryonic transcriptome and phenotypic defects in *D. melanogaster*. Together, our results demonstrate a novel role for uORFs by maintaining translation stabilization during *Drosophila* evolution and development.

Results

An extended ICIER model for quantifying uORF buffering in CDS translation

To quantitatively assess how uORF translation modulates the variability of downstream CDS translation, we adapted the ICIER model⁴³, originally grounded in the totally asymmetric simple exclusion process (TASEP). TASEP has been extensively utilized to model the stochastic nature of ribosome movement along mRNA, capturing the effects of ribosome traffic jams, where ribosomes may slow down or stall when a site ahead is occupied^{57–60}. The ICIER model extends this by simulating the interplay between scanning (40S) and elongating (80S) ribosomes, particularly focusing on how uORFs impact the overall translation rate of the main CDS⁴³.

We extended the original ICIER model⁴³ with several major modifications (**Figure 1A**). First, while the original ICIER model only considered the scenario where the elongating ribosome (80S) causes downstream scanning ribosomes (40S) to dissociate from the mRNA when they move along the mRNA and collide, recent findings have shown that upstream dissociation can also play a critical role in uORF-mediated regulation⁵⁶. To incorporate this, we accounted for more complex ribosome interactions, including three possible scenarios where the 80S collides with 40S: i) 80S only causes the downstream 40S to dissociate from the mRNA with a probability of K_{down} (“downstream dissociation”, K_{down} ranging from 0 to 1), following the original ICIER model; ii) the 80S only causes the upstream 40S to dissociate from the mRNA with a probability of K_{up} (“upstream dissociation”, $K_{up} > 0$ and $K_{down} = 0$); and iii) a combination of the downstream and upstream dissociation models (“double dissociation”, $K_{up} > 0$ and $K_{down} > 0$).

Second, the original ICIER model only considered the ribosome collision and dissociation in the uORF and counted the 40S scanning ribosome escaping from the uORF as a proxy of the CDS translation rate. In our extended model, we also considered the ribosome collision and dissociation events in the CDS downstream of the uORF. We recorded the number of 80S ribosomes that completed translation at the stop codon of a CDS (N_{EC}) or uORF (N_{Eu}) during a given time interval, using these counts as proxies to quantify the translation rate of CDS or uORF. These indices allowed us to directly and quantitatively measure the impact of uORFs on CDS translation.

Third, while the original ICIER model accounted for only a single uORF within an mRNA, we extended it to consider two uORFs coexisting on an mRNA molecule, allowing us to explore the possible combinatorial effects of uORF-mediated translation regulation in a more complex yet more common scenario, as previous studies have shown that uORFs tend to be clustered within genes¹⁸.

These extensions allow for a more comprehensive exploration of uORF-mediated translational buffering, offering deeper insights into how these regulatory elements might stabilize protein synthesis across varying translation contexts.

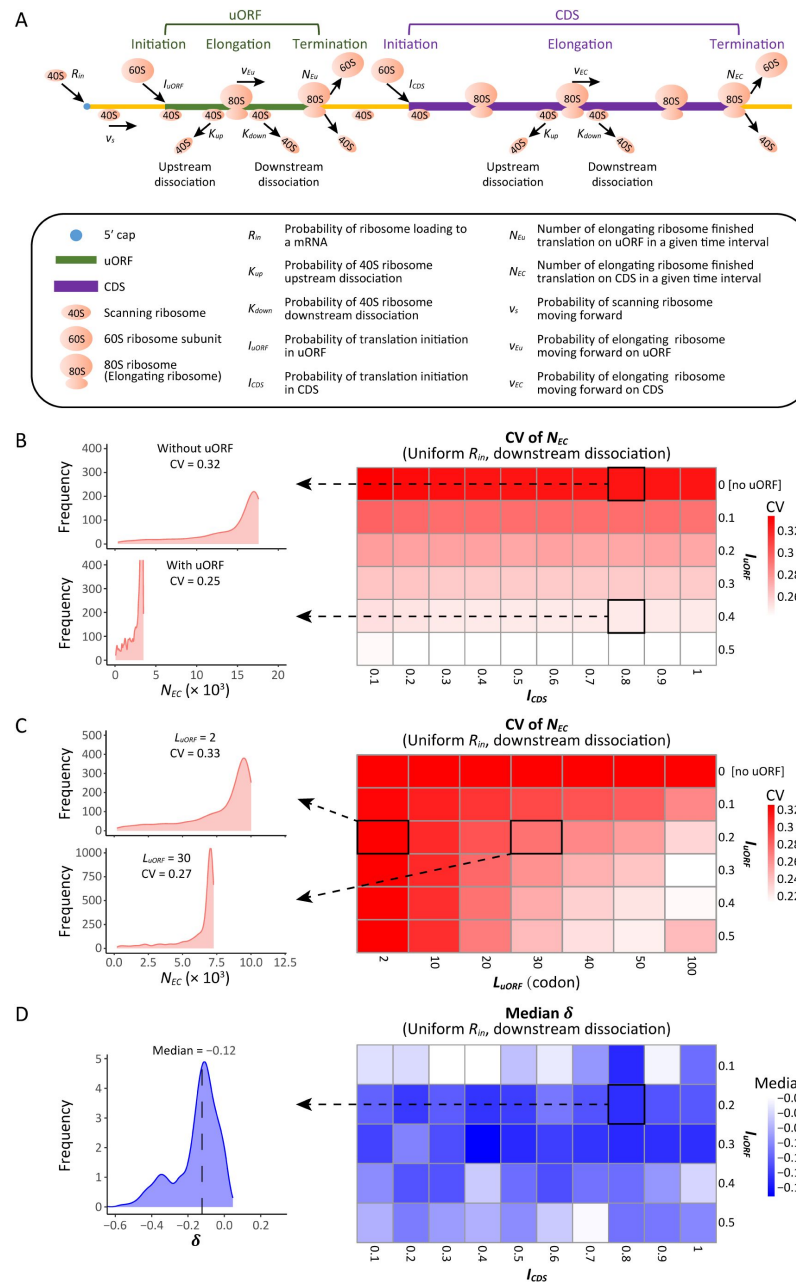


Figure 1.

Modeling simulation of uORF-mediated translation buffering.

(A) Model schema of the modified ICIER model (on the top); the parameters are listed in the box below the schema. (B) Heatmap showing the CVs of CDS translation rate (N_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. The left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $I_{CDS} = 0.8$ & $I_{uORF} = 0$ (top panel, without uORF) and $I_{CDS} = 0.8$ & $I_{uORF} = 0.4$ (bottom panel, with uORF). (C) Heatmap showing CVs of CDS translation rate (N_{EC}) under different L_{uORF} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. The left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $L_{uORF} = 2$ & $I_{uORF} = 0.2$ (top panel) and $L_{uORF} = 30$ & $I_{uORF} = 0.2$ (bottom panel). (D) Heatmap showing median δ [$\log_2(\Delta N_{EC}/\Delta N_{EU1})$] under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. The left panel elicited by the dotted line from a specific square of right heatmap was an example showing the distribution of δ under $I_{CDS} = 0.8$ & $I_{uORF} = 0.2$. The vertical dashed line indicated the median value of δ .

uORF-mediated buffering of CDS translation across different parameter settings

To systematically investigate the extent to which uORF translation modulates the variability of downstream CDS translation, we conducted simulations across various parameter settings using three dissociation models: upstream, downstream, and double dissociation, each reflecting different possible interactions between scanning and elongating ribosomes on the mRNA. We considered a range of parameters crucial to the translation process (Table S1), including the length of the 5' leader before the uORF (fixed at 150 nucleotides), the length of the uORF itself (ranging from 2 to 100 codons), the distance between the uORF stop codon and the CDS start codon (150 nucleotides), the length of the CDS (500 codons), and the length of the 3' UTR (150 nucleotides). Additionally, we modeled the probabilities associated with ribosome movement and initiation, such as the probability of a 40S ribosome moving to the next nucleotide ($v_s = 0.3$), the probability of an 80S ribosome moving to the next position within the uORF ($v_{Eu} = 0.3$) and within the CDS ($v_{EC} = 0.5$), and the probability of loading a new 40S ribosome at the 5' end of the mRNA (R_{in}). We also explored different probabilities of translation initiation at both the uORF start codon (I_{uORF}) and the CDS start codon (I_{CCS}). Key parameters were adapted from the original ICIER model⁴³, ensuring a robust basis for comparison while allowing exploration of additional variables that influence uORF-mediated translational buffering.

In our simulations, R_{in} values were varied to simulate fluctuations in translational resources, such as ribosome availability, that could arise from genetic differences or environmental changes during evolution or development. By generating 1,000 R_{in} values following either uniform or exponential distributions (ranging from 0 to 0.1, **Figure S1A**), we aimed to capture the natural variability in ribosome loading rates that might occur across different species, individuals, or developmental stages.

These values were then fed into the simulation models to evaluate their impact on both the level and variability of translation rate for CDSs, with the number of ribosomes completing translation on CDSs (N_{EC}) in a given time interval used as a proxy for translational rate (**Figure S1B**).

Across all model settings, uORF translation ($I_{uORF} > 0$) consistently reduced CDS translation rate (N_{EC}) by about 30% to 80% as I_{uORF} increased from 0.1 to 0.5, compared to scenarios where the uORF was absent or untranslated ($I_{uORF} = 0$) (**Figure S2** and **S3**). This confirms the inhibitory effect of uORFs on downstream CDS translation. Notably, the coefficient of variation (CV), a measurement of variability for translation rate (N_{EC}), was lower when the uORF was translated (**Figure 1B**). These CV values further decreased by approximately 10% to 25% as I_{uORF} increased from 0.1 to 0.5 (**Figure 1B**), and this buffering effect persisted across different parameter settings (**Figure S4** and **S5**). Moreover, the CV of translation rate (N_{EC}) further decreased by about 6% to 30% as uORF length (L_{uORF}) increased from 2 to 100 codons (**Figure 1C** and **S6-7**). These simulation results indicate that uORF translation can reduce variability in downstream CDS translation, with the buffering capacity positively correlated with both uORF translation initiation efficiency and length under the applied simulation conditions.

To more quantitatively investigate the relationship between changes in translation rate in uORF (N_{EU}) and CDS (N_{EC}), for each parameter setting, we used the median value (R_{inm}) of the 1,000 R_{in} inputs as the baseline, calculating the corresponding N_{ECm} and N_{EUm} values. We then calculated the changes of N_{EC} (ΔN_{EC}) and N_{EU} (ΔN_{EU}) relative to N_{ECm} and N_{EUm} for each R_{in} value, respectively. Across different parameter settings, ΔN_{EU} was consistently and significantly positively correlated with ΔN_{EC} ($P < 0.001$, Spearman's correlation) (**Figure S8-9**), indicating that fluctuations of the ribosome loading rate influence the translation of both uORFs and CDSs in the same direction. Nevertheless, our simulations showed that variations in R_{in} led to a larger change in N_{EU} than in N_{EC} , as the median value of δ [defined by $\log_2(\Delta N_{EC}/\Delta N_{EU})$] was consistently

less than 0 across various I_{uORF} and I_{CCS} combinations (Figures 1D and S10-11). This finding suggests that translational fluctuation in uORFs is greater than that in downstream CDSs, indicating that uORFs can buffer against upstream fluctuations.

Simulations of mRNAs with two uORFs revealed patterns consistent with a buffering role for uORFs (Figure S12). To compare the buffering effects of a single uORF versus two uORFs, we calculated the ratio of the CV of N_{EC} with two uORFs to that with a single uORF. A ratio less than 1 suggests that two uORFs provide greater buffering than a single uORF. For comparability, we examined the CV of N_{EC} where the I_{uORF} in the single-uORF model equals I_{uORF1} (I_{uORF} of the first uORF) in the two-uORF model, both ranging from 0 to 0.5. The CV ratio consistently remained below 1 across a range of I_{uORF2} (I_{uORF} of the second uORF) (Figure S13), indicating that two uORFs offer stronger buffering than a single uORF.

Collectively, these simulations collectively suggest: (1) uORF-mediated translational control buffers against CDS translation variability, (2) uORFs exhibit greater translational fluctuations than downstream CDSs, and (3) the buffering capacity of uORFs positively correlates with their translation initiation efficiency and length. Subsequently, we sought to validate these simulation results in a biological context by confirming the uORF-mediated buffering effect during organismal evolution and development, as both processes face environmental and/or genetic changes that frequently disturb mRNA translation.

Generating matched translome data from two *Drosophila* species for comparative analysis

To validate the uORF-mediated translational buffering during *Drosophila* evolution, we performed a comparative analysis of the translomes of two closely related *Drosophila* species, *D. melanogaster* and *D. simulans*, which diverged approximately 5.4 million years ago⁶¹. We generated high-throughput sequencing data for *D. simulans*, including transcriptome (mRNA-Seq) and translome (Ribo-Seq) profiles from various developmental stages and tissues, including embryos at 0–2 h, 2–6 h, 6–12 h, and 12–24 h, third-instar larvae, P7–8 pupae, female and male bodies, and female and male heads. In total, we obtained approximately 786 million high-quality reads for *D. simulans* (Table S2). These datasets were designed to be directly comparable to the previously published *D. melanogaster* data²³, with identical embryonic stages and tissue types. For each sample, we followed established procedures^{62–66} to calculate the translational efficiency (TE) for each feature (CDS or uORF). TE serves as a proxy for the translation rate at which ribosomes translate mRNA into proteins, typically quantified by comparing the density of ribosome-protected mRNA fragment (RPF) to the mRNA abundance for that feature (see **Materials and Methods**). This comprehensive comparative translome analysis, utilizing matched developmental stages and tissues between the two species, allowed us to assess the uORF-mediated translational buffering effects during evolution.

Translational conservation and dominance of uORFs between *Drosophila* species

Given that the translation of uORFs is a crucial determinant of their functional impact, we first characterized the translational profiles of uORFs in *D. melanogaster* and *D. simulans* to explore the evolutionary roles of uORFs in translation regulation. We focused on canonical uORFs that initiate with an ATG start codon in the 5' UTR and terminate with a stop codon (TAA, TAG, or TGA). Because the ATG start codon is the defining feature of a canonical uORF and tends to be more conserved than its downstream sequence⁶⁷, we defined uORF conservation based on the presence of the ATG start codon in the 5' UTR of *D. melanogaster* and its orthologous positions in *D. simulans*, regardless of differences in the stop codon. Using this criterion, we identified 18,412 canonical uORFs with conserved start codons between the two species. Additionally, we identified 2,789 canonical uORFs specific to *D. melanogaster* and 2,440 canonical uORFs specific to *D.*

simulans. The TE values of the conserved uORFs were highly correlated between the two species across all developmental stages and tissues examined, with Spearman correlation coefficients (ρ) ranging from 0.478 to 0.573 (**Figure 2A**). In contrast, TE of CDSs exhibited a significantly higher correlation between the two species in the corresponding samples compared to that of uORFs, with Spearman's ρ ranging from 0.588 to 0.806 ($P = 0.002$, Wilcoxon signed-rank test; **Figure 2A**). This observation is consistent with our simulation results, which indicate that uORFs experience greater translational fluctuations than their downstream CDSs. Notably, conserved uORFs exhibited significantly higher TEs compared to species-specific uORFs in both species. The median TE of conserved uORFs was 1.62 times that of non-conserved uORFs in *D. simulans*, while the corresponding ratio in *D. melanogaster* was 1.52 (**Figure 2B**).

In *D. melanogaster*, 7,259 (52.2%) genes had no uORFs, 2,687 (19.3%) had a single uORF, and 3,961 (28.5%) contained multiple uORFs. Among genes with multiple uORFs, we defined the uORF with the highest TE as the dominant uORF for that gene, as TE is one of the most relevant metrics for assessing uORF function^{45,67}. The median TE of the dominant uORF was 4.84 times that of the second-highest uORF within the same gene in *D. melanogaster*, and the corresponding ratio was 5.21 times in *D. simulans* (**Figure S14**). To assess the consistency of this dominance across different tissues and developmental stages, we identified 3,072 multiple-uORF genes in *D. melanogaster* with at least one translated uORF (TE > 0.1 in at least five stages/tissues). Of these, 569 genes consistently used the same dominant uORF across the measured samples, significantly higher than the number expected under randomness (5 genes, 95% confidence interval: 1–10) based on shuffling the TEs of uORFs 1,000 times (**Figure S15**). This trend was also observed in *D. simulans* and persisted under different thresholds for defining “translated uORFs” (**Figure S15**). These results suggest that genes with multiple uORFs tend to retain the same dominant uORF across developmental stages, indicating that the dominant uORF may serve as the key translational regulator of the downstream CDS. Moreover, we found that the dominant uORFs showed a higher proportion of conserved uATGs than the other translated uORFs (median proportion 82.5% versus 78.8%, $P < 0.001$) (**Figure 2C**). Additionally, the absolute values of TE fold-change ($|\log_2 \text{TE-FC}|$) between the two species were, on average, 23.2% smaller for dominant uORFs than for other conserved uORFs (**Figure 2D**), suggesting that dominant uORFs are more likely under stronger stabilizing selection. These findings suggest that, in genes with multiple uORFs, the dominantly translated uORF may play a more important role in regulating CDS translation than the other uORFs.

uORFs buffer interspecific translational divergence of CDSs

To investigate the relationship between translational changes in uORFs and their downstream CDSs across species, we analyzed the translation efficiency (TE) of uORFs and corresponding downstream CDSs in *D. melanogaster* and *D. simulans*. Consistent with our simulations that the translation of a uORF is tightly linked to that of downstream CDS, uORFs exhibited a modest, yet statistically significant, positive correlation with the TE of their downstream CDSs across all samples analyzed ($P < 0.001$, Spearman's correlation) (**Figure 3A**). We then compared the interspecific TE change of a uORF ($\beta_u = \text{TE}_{\text{uORF}, \text{sim}} / \text{TE}_{\text{uORF}, \text{mel}}$) with that of its corresponding CDS ($\beta_c = \text{TE}_{\text{CDS}, \text{sim}} / \text{TE}_{\text{CDS}, \text{mel}}$) between *D. melanogaster* and *D. simulans*. We found β_u is significantly positively correlated with β_c across all samples (P values < 0.001, Spearman's correlation) (**Figure S16**). These results align well with our simulations, which showed that fluctuations in translational factors (such as ribosomes) influence both uORF and CDS translation in the same direction (**Figure S8–S9**).

While the direction of TE changes for uORFs and CDSs tends to be consistent, our simulations suggest that the magnitude of TE changes in CDSs is generally smaller than that in uORFs, due to the buffering effect of uORF (**Figure S10–S11**). To validate this, we first identified uORFs and CDSs with significant interspecific TE differences by assessing whether β_u or β_c significantly deviated from 1, using an established statistical framework²³. This analysis uncovered 1,151 to 4,189 CDSs with significant interspecific TE changes (FDR < 0.05) (**Table 1**), with genes involved

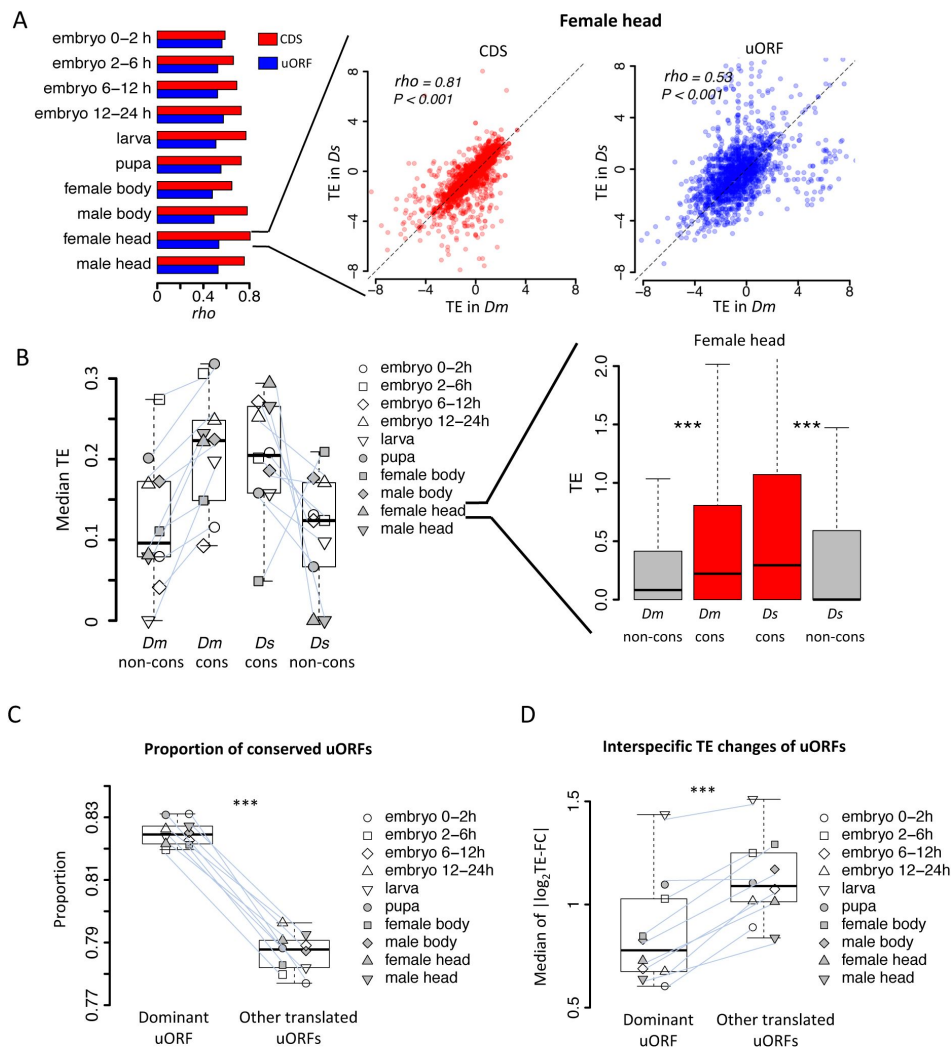


Figure 2.

Conservation and translation of uORFs between *D. melanogaster* and *D. simulans*.

(A) Spearman's correlation coefficients (ρ , represented by the bars) of TEs between *Dm* (*D. melanogaster*) and *Ds* (*D. simulans*) for CDS (red) and uORFs (blue). All P -values for Spearman's correlation are less than 0.001. The P value for the comparison between ρ values of CDSs TE and uORFs TE is 0.002 (Wilcoxon signed-rank test). Data for the female head sample is shown as an example in the right panel. The x- and y-axes represent the TEs in *Dm* and *Ds*. (B) The median of TE of conserved and species-specific uORFs in each sample. Each dot represents the median TE of a sample for a specific uORF class. Data from the female head sample is shown as an example in the right panel. P values were obtained from Wilcoxon rank-sum tests. $***, P < 0.001$. (C) Fraction of conserved uORFs among dominant uORFs and other translated uORFs in each sample. The paired samples in *Dm* and *Ds* were linked together. The P value was obtained by the paired Wilcoxon signed-rank test. $***, P < 0.001$. (D) Absolute values of the interspecific TE fold changes ($\log_2 \text{TE-FC}$) of dominant uORFs and the other translated uORFs in each sample. The paired samples in *Dm* and *Ds* were linked together. The median value of each sample is shown. The P value was obtained via the paired Wilcoxon signed-rank test. $***, P < 0.001$. Data from the female head sample were used as an example in the right panel.

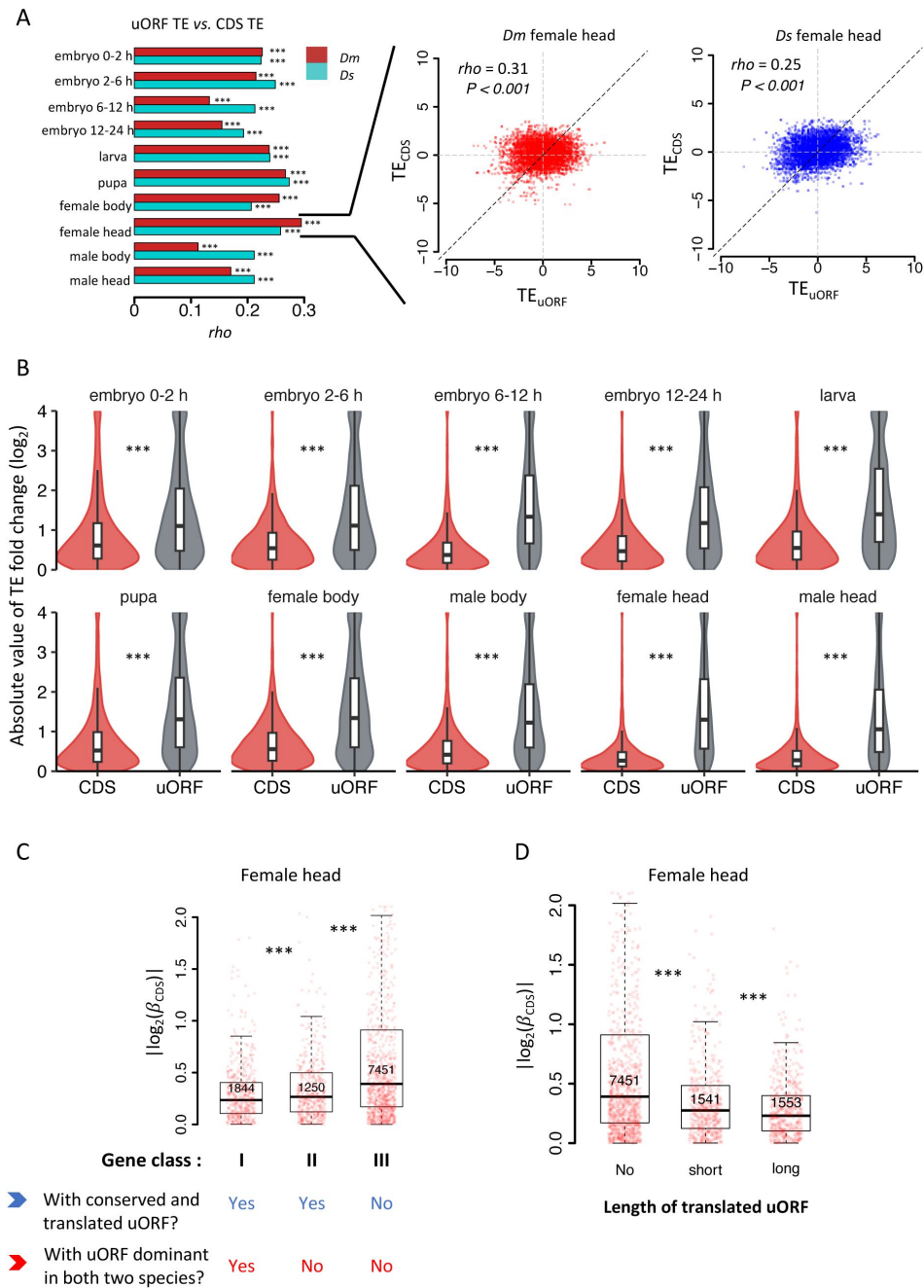


Figure 3.

uORFs reduce CDS translational divergence between *D. melanogaster* and *D. simulans*.

(A) The correlation of uORF TEs and the corresponding CDS TEs in 10 samples of *Dm* (*D. melanogaster*) and *Ds* (*D. simulans*). The bars represent Spearman's correlation coefficient (ρ). In all samples, we obtained both P values < 0.001 . Data for the female head sample of *Dm* and *Ds* are shown as examples in the right panel. (B) The absolute values of interspecific TE changes for CDS and uORF in each sample between two species. For visualization purposes, all values greater than 4 were assigned a value of 4. ***, $P < 0.001$, Wilcoxon rank-sum test. (C) Genes expressed in female heads (mRNA RPKM > 0.1 in both species) were classified into three classes according to whether a gene had a conserved and dominantly translated uORF or not. Boxplots showing interspecific CDS TE variability $|\log_2(\beta_C)|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$. (D) Genes expressed in female heads were classified into three classes according to the length of translated uORFs. Boxplots showing interspecific CDS TE variability $|\log_2(\beta_C)|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$.

in development, morphogenesis, and differentiation being significantly enriched during embryonic stages (Figure S17). Conversely, genes related to metabolism, response to stimuli, and signaling were enriched in larval, pupal, and adult stages (Figure S17). Additionally, we identified 144 to 1,193 uORFs with significant TE differences between species, accounting for approximately 1–15% of expressed uORFs (Table 1). Note that due to their shorter length and generally lower TE, uORFs had considerably lower read counts than CDSs, limiting the statistical power to detect significant interspecific TE differences for uORFs. This trend consistently holds whether analyzing all expressed uORFs (Figure S18A) or only highly expressed genes (Figure S18B). Thus, the fewer uORFs showing significant TE divergence likely reflects lower read counts and statistical sensitivity rather than reduced translational variability relative to CDSs. In fact, the absolute values of $\log_2(\text{fold change})$ of TE for uORFs between *D. melanogaster* and *D. simulans* were significantly greater than those observed for corresponding CDSs across all samples ($P < 0.001$, Wilcoxon signed-rank test; Figure 3B), suggesting that the magnitude of TE changes in CDSs is generally smaller than that in uORFs, due to the buffering effect of uORF.

We further quantitatively compared the magnitude of interspecific TE changes between uORFs and their corresponding CDSs using a previous method²³. We defined $\gamma = \beta_c / \beta_u$ for a uORF-CDS pair within the same mRNA and tested whether γ was significantly different from 1 to identify pairs with differential TE changes between uORFs and CDSs (Figure S19). When $\gamma < 1$ and $\beta_u > 1$, or $\gamma > 1$ and $\beta_u < 1$, it indicates that the TE change for the CDS is smaller than that for the uORF (Figure S19). Among CDS-uORF pairs where $\beta_u > 1$, nearly all (8–487) showed a significant $\gamma < 1$ in each sample, except for one pair from the pupal stage where $\gamma > 1$ (Table 1). This suggests that the magnitude of TE changes in CDSs was generally smaller than in uORFs when uORF TE increased, and vice versa when uORF TE decreased ($\beta_u < 1$) (Table 1). These comparative translome analyses indicate that uORFs buffered downstream CDS translation changes during the evolutionary divergence of *D. melanogaster* and *D. simulans*.

uORF buffering is influenced by its conservation, dominance, and length

To investigate how the conservation level and translation patterns of uORFs influence their buffering capacity on CDS translation, we categorized genes expressed in each pair of samples into three classes: Class I, genes with conserved uORFs that are dominantly translated (i.e., exhibiting the highest TE among all uORFs within the same gene) in both *Drosophila* species; Class II, genes with conserved uORFs that are translated in both species but not dominantly translated in at least one; and Class III, the remaining expressed genes. We then compared the absolute values of interspecific TE changes of the CDS ($|\beta_c|$) across these three categories. Significant differences in $|\beta_c|$ were observed, with a consistent hierarchy of Class I < II < III across all pairs of samples (Figure 3C and S20). On average, Class I genes exhibited an average of 8.18% and 23.8% lower $|\beta_c|$ values compared to Class II and Class III, respectively. This indicates that conserved and dominantly translated uORFs exert a stronger buffering effect on CDS translation.

To further validate the simulation results suggesting that longer uORFs have a stronger buffering effect (Figure 1C and S6–7), we divided genes expressed in each pair of samples into three groups: those without translated uORFs (No), those with short uORFs (short, total length below the median), and those with long uORFs (long, total length above the median). Consistently, longer uORFs were associated with stronger buffering effects on CDS translation across all pairs of samples (Figure 3D and S21). Specifically, genes with longer uORFs showed 12.7% and 26.5% lower $|\beta_c|$ values compared to genes with short uORFs or no uORFs, respectively.

Overall, these findings underscore that the buffering capability of a uORF is positively correlated with its conservation level, translation dominance, and length.

Sample types	# of expressed uORFs *	$\beta_u \neq 1$ (%) **	# of expressed CDSs *	$\beta_c \neq 1$ (%) **	uORF-CDS pairs with $\beta_u > 1$			uORF-CDS pairs with $\beta_u < 1$		
					Total	$\gamma > 1$	$\gamma < 1$	Total	$\gamma > 1$	$\gamma < 1$
0–2 h embryo	7,704	1,193 (15.49)	7,934	4,189 (52.80)	770	0	366	423	69	0
2–6 h embryo	7,822	567 (7.25)	9,795	3,063 (31.27)	249	0	63	318	135	0
6–12 h embryo	10,400	1,040 (10.00)	10,643	2,924 (27.47)	973	0	641	67	22	1
12–24 h embryo	11,365	535 (4.71)	11,158	3,537 (31.70)	454	0	234	81	8	2
Larva	10,008	464 (4.64)	11,831	3,554 (30.04)	110	0	26	354	177	0
Pupa	12,309	635 (5.16)	12,209	4,087 (33.48)	136	1	15	499	210	0
Male body	10,894	197 (1.81)	12,284	2,432 (19.80)	122	0	29	75	4	0
Male head	10,904	144 (1.32)	10,447	1,151 (11.02)	119	0	31	25	5	0
Female body	9,809	340 (3.47)	11,002	3,605 (32.77)	279	0	17	61	3	0
Female head	10,935	332 (3.04)	10,545	1,270 (12.04)	324	0	152	8	3	0

* Only uORFs and CDSs with an mRNA RPKM > 0.1 in both *D. melanogaster* and *D. simulans* were considered in each sample pair in the analysis.

** $\beta_u = TE_{uORF,sim}/TE_{uORF,mel}$ is the fold change of TE_{uORF} in *D. simulans* relative to *D. melanogaster* for each sample. $\beta_c = TE_{CDS,sim}/TE_{CDS,mel}$ is the fold change of TE_{CDS} in *D. simulans* relative to *D. melanogaster* for each sample. $\gamma = \beta_c/\beta_u$. For each CDS–uORF pair, either $\beta_u > 1$ and $\gamma < 1$ or $\beta_u < 1$ and $\gamma > 1$ means that the magnitude of TE change is lower for a CDS than for a uORF. The statistical significance of β_c , β_u , and γ were determined according to an FDR < 0.05.

Table 1.

Numbers of genes showing different magnitudes of TE changes between uORFs and CDS at the interspecific level.

uORFs buffer translational fluctuations during *Drosophila* development

Gene expression undergoes dynamic changes during *Drosophila* development, with significant alterations in the translation program to meet developmental demands, including shifts in ribosome loading rates^{24,68,69}. Therefore, we extended our analysis to investigate the role of uORFs in buffering these translational fluctuations, hypothesizing that uORFs could mitigate them during development. According to our hypothesis, if a gene has a translated uORF in *D. melanogaster* but not in its orthologous gene in *D. simulans*, then the translation of this gene is likely more stable across developmental stages in *D. melanogaster* than its ortholog in *D. simulans*, and vice versa.

To test this hypothesis, we compared the CV of CDS TE across 10 developmental stages in *D. melanogaster* and in *D. simulans*, respectively. Genes with translated uORFs (TE > 0.1 in at least one sample) in *D. melanogaster* exhibited significantly 22.5% smaller CVs in *D. melanogaster* than their orthologs lacking these uORFs in *D. simulans*, indicating more stable translation (Figure 4A). Consistently, for genes with translated uORFs in *D. simulans* but not in *D. melanogaster*, the CVs were 13.3% lower in *D. simulans* compared to their orthologs lacking these uORFs in *D. melanogaster* (Figure 4B). Consistent results were observed when a uORF is required to be translated (TE > 0.1) in all 10 samples (Figure S22A), in 4 embryonic stages (Figure S22B), or in 6 stages including embryos, larva, and pupa (Figure S22C). Moreover, within each species, genes with translated uORFs also showed less variability in CDS TE across developmental stages compared to those without translated uORFs, with a 31.8% reduction in *D. melanogaster* and a 28.9% reduction in *D. simulans* (Figure 4C). This effect was consistent across different thresholds for defining “translated uORFs” (Figure S23).

These results suggest that uORFs function as translational buffers, reducing gene translation fluctuations during *Drosophila* development.

Knocking out the uORF of *bcd* increased *bcd* CDS translation in *D. melanogaster*

After verifying the uORFs-mediated translational buffering during *Drosophila* evolution and development, we next aimed to directly explore the biological function of these buffering-capable uORFs in vivo. We first applied stringent criteria to identify uORFs with significant buffering effects on CDS translation between *D. melanogaster* and *D. simulans*. Specifically, we looked for 1) uORFs with significant TE changes ($|\log_2(\beta_u)| > 1.5$, adjusted $P < 0.05$), 2) negligible changes in its corresponding CDS translation ($|\log_2(\beta_c)| < 0.05$, adjusted $P > 0.05$), and 3) a significant difference between the magnitude of these changes ($|\log_2(\gamma)| > 1.5$, adjusted $P < 0.05$). We identified 131 uORF-CDS pairs in 103 genes that meet these criteria in at least one stage/tissue (Table S3), with a majority of the genes (67%, 69 out of 103) from embryonic stages (Figure S24A), suggesting a crucial role for uORFs in maintaining translational stability during early development. Among these genes, one notable case is the *bicoid* (*bcd*) gene, a master regulator of anterior-posterior axis patterning during early embryogenesis^{70–72}. The *bicoid* gene contains a 4-codon uORF (excluding the stop codon) in its 5' UTR, and branch length score (BLS) analysis¹⁸ showed that the start codon (uATG) of the *bcd* uORF is highly conserved across the *Drosophila* phylogeny, with a BLS of 0.90 (on a scale from 0 to 1, where higher values indicate greater conservation) (Figure 5A and S24B). Ribo-Seq data revealed that in 0–2 h embryos, the TE of the *bcd* uORF varied more than threefold, while the TE of the CDS was virtually the same between *D. melanogaster* and *D. simulans* (Figure 5B and Table S3). This suggests that the *bcd* uORF significantly buffers translation during early development.

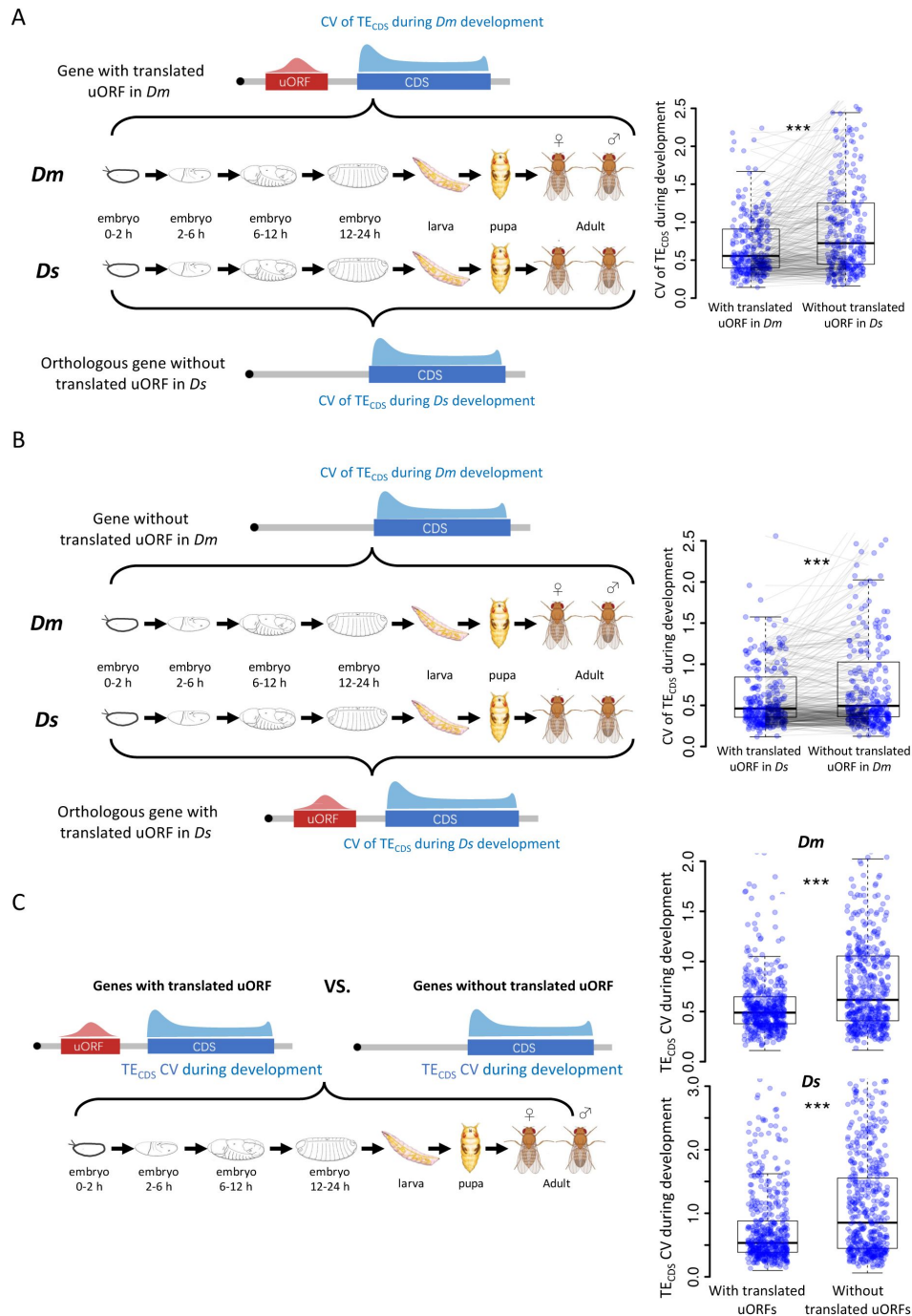


Figure 4.

uORFs could reduce CDS translational fluctuation during *Drosophila* development.

(A) The CV of TE_{CDS} across 10 *Dm* (*D. melanogaster*) samples and 10 *Ds* (*D. simulans*) samples. The selected gene with uORFs translated ($TE > 0.1$) in at least one *Dm* sample but its homologous gene without translated uORF in *Ds* samples. Each pair of dots linked by a gray line represents a pair of homologous genes in *Dm* and *Ds*. ***, $P < 0.001$, Wilcoxon signed-rank test. (B) The CV of TE_{CDS} across 10 *Dm* samples and 10 *Ds* samples. The selected gene with uORFs translated ($TE > 0.1$) in at least one *Ds* sample but its homologous gene without translated uORF in *Dm* samples. Each pair of dots linked by a gray line represents a pair of homologous genes in *Dm* and *Ds*. ***, $P < 0.001$, Wilcoxon signed-rank test. (C) Within each *Drosophila* species, the CV of TE_{CDS} of genes with translated uORFs compared to genes without the translated uORFs. The P values are obtained by the Wilcoxon rank-sum test. ***, $P < 0.001$.

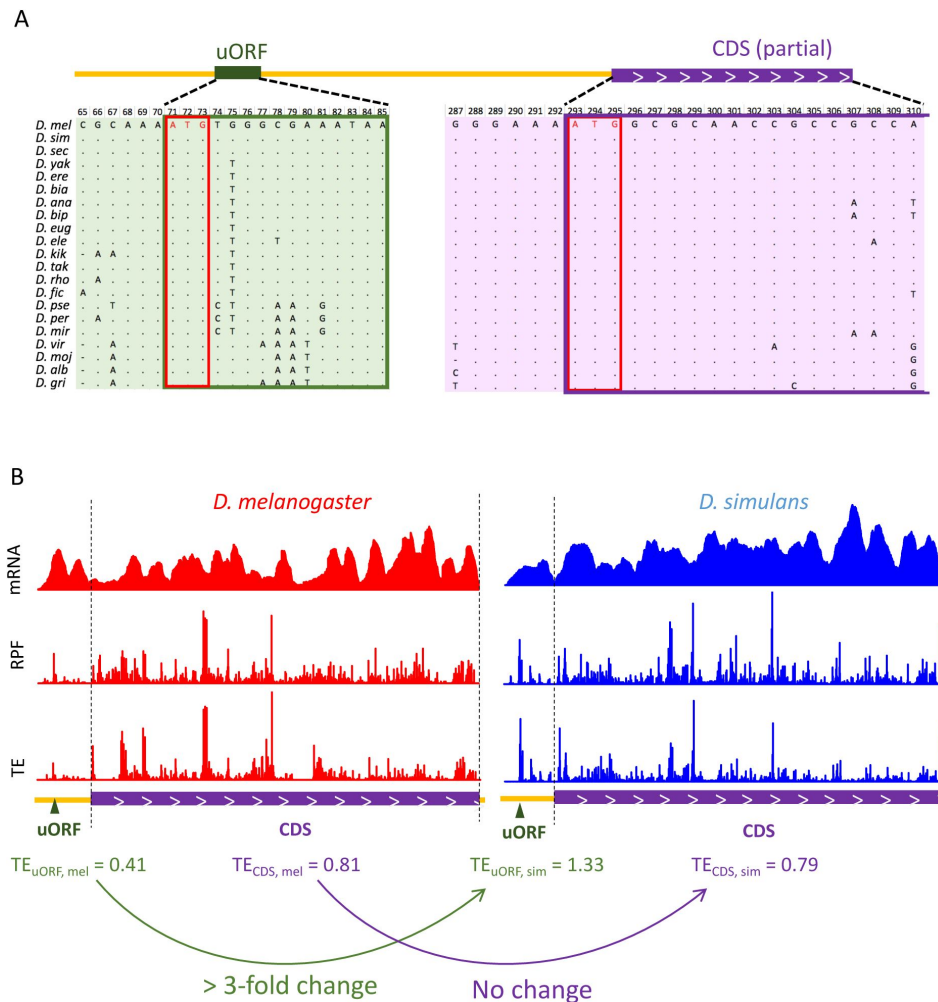


Figure 5.

The strong buffering effect of the *bcd* uORF on CDS translation between the two *Drosophila* species.

(A) Multiple sequence alignment of the *bcd* uORF and partial CDS in *D. melanogaster* and 20 other *Drosophila* species. The uORF and CDS are boxed in green and purple, respectively. The start codons of the uORF and CDS are boxed in red. (B) The coverage of mRNA-Seq (top), Ribo-Seq (middle), and TEs (bottom) of the *bcd* uORF and CDS in 0-2 h embryos of *D. melanogaster* (red) and *D. simulans* (blue). The uORF and CDS are denoted at the lower panel with dark green triangles and purple boxes, respectively. The 2 dashed lines mark the CDS region. The uORF TE, CDS TE and their interspecific changes were labeled at the bottom.

To investigate the regulatory role of the *bcd* uORF, we used CRISPR-Cas9 to knock out its start codon in *D. melanogaster*, generating two mutant homozygotes (uKO1/uKO1 and uKO2/uKO2) with a genetic background matched to that of the wild-type (WT) (Figures 6A and S25). To determine whether these mutations enhanced *bcd* CDS translation, we performed ribosome fractionation followed by qPCR^{73–75}, comparing *bcd* mRNA levels in polysome and monosome fractions (P-to-M ratio) from 0–2 h embryos of uKO1/uKO1, uKO2/uKO2, and WT flies (Figure 6B). A larger P-to-M ratio means more mRNAs are enriched in the polysome fractions and bound by more ribosomes, thus indicative of higher translation efficiency. P-to-M ratios were higher in the mutants compared to WT at 29°C (Figure 6C), with a similar but less pronounced trend at 25°C, where the difference between uKO2/uKO2 and WT was not statistically significant (Figure 6C). These findings, along with the known impact of temperature on gene expression and phenotypic plasticity^{76,77}, suggest that the regulatory function of the uORF and overall translation efficiency are temperature-sensitive, highlighting a complex interplay between environmental conditions and gene regulation. To further confirm the *bcd* uORF's regulatory function, we conducted dual-luciferase reporter assays. The 5' UTR from *bcd* mutant (uKO1 or uKO2) or WT was cloned into a *Renilla* luciferase reporter construct (Figure 6D). Luciferase activity was significantly higher in uKO1 and uKO2 compared to the WT 5' UTR, confirming the repressive role of the *bcd* uORF.

***bcd* uORF mutants show wide transcriptomic alteration during *Drosophila* embryogenesis**

Since Bcd regulates the expression of many zygotic genes^{70–72}, we anticipated that the increased translation of *bcd* resulting from disrupting the uORF would influence *Drosophila* transcriptomes and phenotypes. To verify this notion, we performed RNA sequencing on embryos from WT and uKO2/uKO2 flies at four developmental stages (0–2 h, 2–6 h, 6–12 h, and 12–24 h) under both 25°C and 29°C conditions, using two biological replicates (Table S4 and Figure S26). Differential expression analysis revealed widespread alterations in gene expression between WT and mutant embryos, with the number of differentially expressed genes (DEGs) increasing over developmental time and at higher temperature (Figure 6E). At 25°C, we identified 674, 817, 1047, and 2,041 DEGs in 0–2 h, 2–6 h, 6–12 h, and 12–24 h embryos, respectively; while at 29°C, we detected 3,884, 2,358, 2,901, and 4,164 DEGs in the corresponding stages (Figure 6E). The majority of DEGs were stage-specific, with only a small fraction consistently differentially expressed across all four stages. Functional enrichment analysis of the DEGs revealed distinct biological pathways affected at each stage, including cell morphogenesis and pattern specification in 0–2 h embryos, metabolic processes and tissue development in 2–6 h and 6–12 h embryos, and mitochondrial respiration in 12–24 h embryos (Figure 6F). Notably, direct targets of Bcd⁷⁸ were significantly enriched among the DEGs in three out of the four stages, with the exception of 2–6 h embryos (Figure S27). RT-qPCR validation of 20 target genes of Bcd confirmed the reliability of the RNA-seq differential expression analysis (Figure S28). Together, these findings demonstrate that disruption of the *bcd* uORF leads to widespread transcriptional changes during *Drosophila* development, affecting processes ranging from embryogenesis to postembryonic metabolism.

***bcd* uORF mutants display decreased hatching rates and starvation resistance**

Given the widespread transcriptome alterations, we anticipated phenotypic abnormalities in the *bcd* uORF mutants. As expected, both uKO1/uKO1 and uKO2/uKO2 mutants exhibited significantly lower hatching rates compared to WT [$P = 1.4 \times 10^{-6}$ and 7.9×10^{-6} , respectively, Wilcoxon rank-sum test (WRST); Figure 7A]. At 25°C, uKO1/uKO1 mutants produced fewer offspring than WT flies ($P < 0.001$, WRST, Figure 7B). Given that *bcd* is a maternal gene, we expected reciprocal crosses between uKO1/uKO1 mutants and WT flies to produce different outcomes. Indeed, crossing

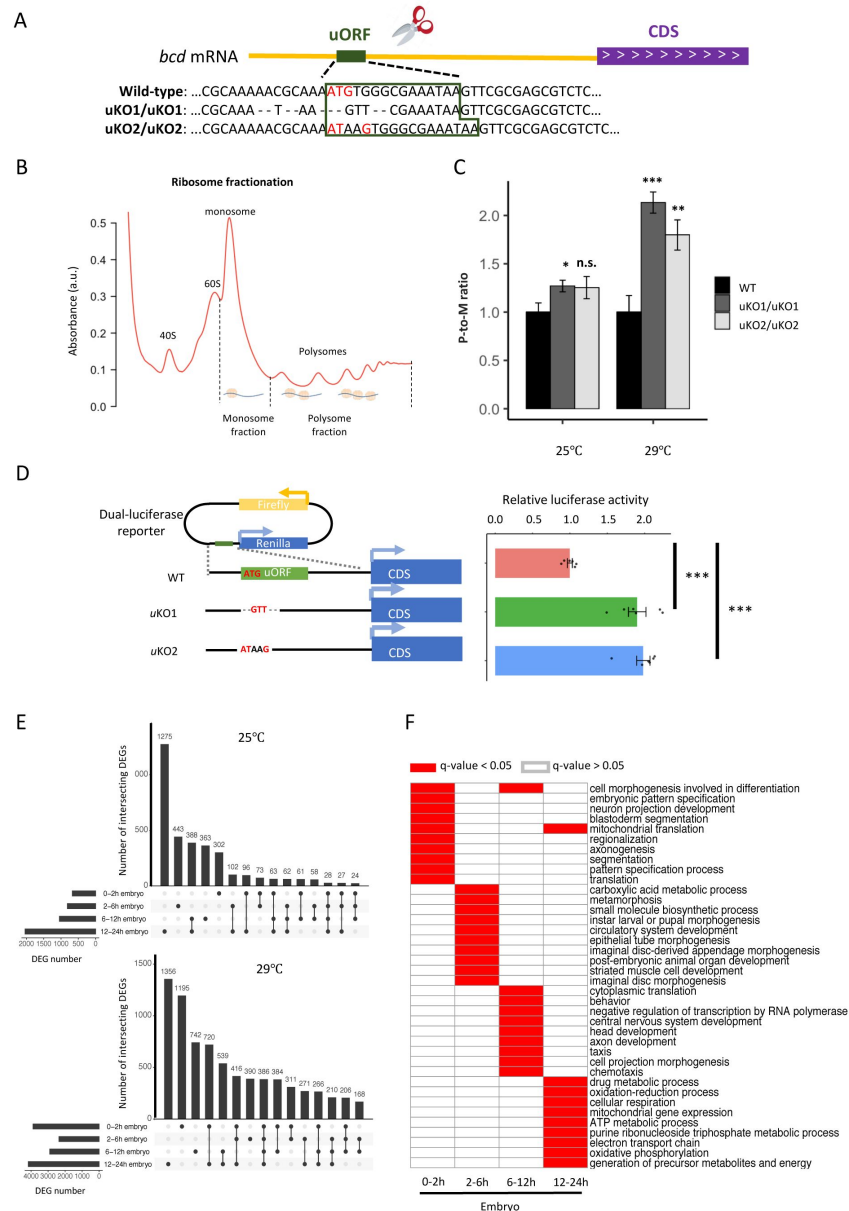


Figure 6.

Knocking out the *bcd* uORF increases CDS translation and perturbs the transcriptome during *D. melanogaster* embryogenesis.

(A) Genotypes of WT and two uORF knock-out strains (uKO1 and uKO2) generated by CRISPR-Cas9 technology. The uORF is boxed in dark green, and the red ATG represents the start codon of the uORF in the *D. melanogaster* genome. (B) Two ribosome fractions (monosome and polysome) of 0-2 h embryos were separated in a sucrose density gradient. Relative RNA abundance in the monosome and polysome fractions was quantified by real-time quantitative PCR. (C) P-to-M ratio of *bcd* mRNA (*bcd* mRNA abundance in polysome fraction/*bcd* mRNA abundance in monosome fraction) at 25°C (left) and 29°C (right). The P-to-M ratios of mutants were normalized to WT controls at 25°C and at 29°C, respectively. Error bars represent the S.E. of six biological replicates. Asterisks indicate statistical significance (*, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; n.s., $P > 0.05$). (D) Dual-luciferase assay for *bcd* WT uORF and mutated uORF. The reporter structures of the WT and uORF mutants are illustrated on the left. The uORF mutant sequence was the same as that in the fly mutant created with CRISPR-Cas9 technology. The relative activity of *Renilla* luciferase was normalized to that of firefly luciferase. Error bars represent the S.E. of six biological replicates. Asterisks indicate statistical significance (***, $P < 0.001$). (E) The number of DEGs in each stage and their intersection with each other at 25°C (top) and 29°C (bottom). (F) Gene ontology analysis of DEGs at 29°C in each stage. The biological process (BP) terms with q-values < 0.05 in each stage are indicated in red and others are indicated in white.

uKO1/uKO1 males with WT females resulted in offspring numbers similar to those from WT crosses, while crossing uKO1/uKO1 females with WT males yielded offspring numbers comparable to crosses between uKO1/uKO1 mutants (**Figure 7B**). This verified that the reduction in offspring is due to maternal defects in the uKO1/uKO1 mutants. Similar patterns were observed for uKO2/uKO2 mutants at 25°C (**Figure 7B**). Notably, the fecundity reduction in uORF-KO mutants was more pronounced at 29°C (**Figure 7C**). Furthermore, crossing uKO1/uKO1 and uKO2/uKO2 mutants also produced significantly fewer progeny than WT flies (**Figure 7C**), ruling out genetic background or off-target effects. Collectively, these data demonstrate that disrupting the *bcd* uORF significantly impaired hatchability and fertility.

We also found that both uKO1/uKO1 and uKO2/uKO2 female mutants perished significantly faster than WT flies under starvation conditions (**Figure 7D**). Males showed similar tendencies, although the difference was not statistically significant for uKO1/uKO1 mutants (**Figure 7D**). These data suggest that the knockout of *bcd* uORF diminished starvation resistance in adults, likely due to embryogenesis abnormalities induced by the *bcd* uORF deletion, even in those that successfully developed to adulthood.

Conservation of uORF-mediated translational buffering in primates

To explore the generality of uORF-mediated translational buffering across evolutionary clades, we analyzed previously published transcriptome and translome data from three tissues (brain, liver, and testis) in humans and macaques⁷⁹. We identified 33,680 canonical uORFs in humans and 29,516 in macaques, with 24,385 conserved between the two species. Despite the larger number of uORFs in primates compared to *Drosophila* due to differences in genome size and gene number, the median TE of conserved uORFs was 1.79 times that of non-conserved uORFs in humans, and the corresponding ratio was 3.43 in macaques (**Figures 8A** and S29). TEs of uORFs were positively correlated between humans and macaques across all tissues ($P < 0.001$, **Figure 8B**). Additionally, significant positive correlations were observed between the TEs of uORFs and their corresponding coding sequences (CDSs) in all tissues (**Figure S30**). Although interspecific TE divergence of uORFs (β_u) and CDSs (β_c) were positively correlated ($P < 0.001$, **Figure 8C**), uORFs generally exhibited larger divergence (Table S5). Notably, longer uORFs showed stronger buffering effects on CDS translation, reducing interspecific TE divergence by 12.9% and 38.0% compared to genes with short or no uORFs (**Figures 8D** and S31).

As in *Drosophila*, we categorized expressed human genes into three classes based on uORF conservation and translation: Class I, genes with conserved and dominantly translated uORFs in both humans and macaques; Class II, genes with conserved uORFs translated in both species but not dominantly in at least one; and Class III, the remaining expressed genes (**Figures 8E** and S32). Consistent with findings in *Drosophila*, significant differences in $|\beta_c|$ between humans and macaques were observed in the order of Class I < II < III, with Class I genes showing 9.8% and 17.1% lower $|\beta_c|$ values compared to Class II and Class III genes, respectively (**Figures 8E** and S32). These similarities between primates and *Drosophila*—two clades that diverged over 700 million years ago⁸⁰—suggest that uORF-mediated translational buffering is a widespread mechanism for stabilizing gene translation across evolutionary clades.

We also analyzed matched mRNA-Seq and Ribo-Seq data from 69 human lymphoblastoid cell lines^{81,82} to test whether uORFs buffer against translational variability across different individuals within humans. Genes with translated uORFs exhibited, on average, 8.65% lower CVs in CDS TE across individuals than genes without translated uORFs (**Figure 8F**), with longer uORFs showing stronger buffering effects (**Figure S33**). Collectively, these findings suggest that uORFs play a crucial role in reducing translational variability both across evolutionary clades and within species.

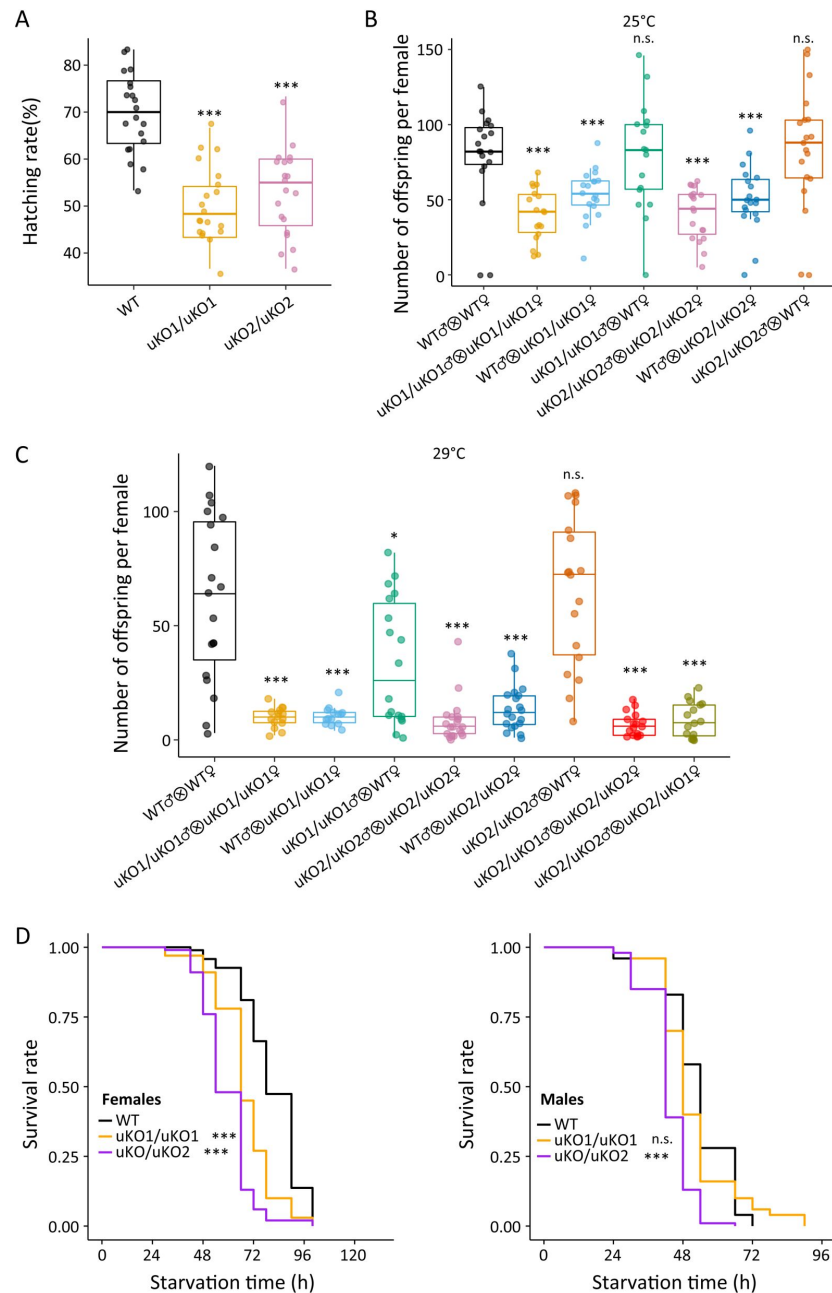


Figure 7.

Knockout of the *bcd* uORF reduces offspring number and starvation resistance.

(A) Comparison of the hatching rates (%) of mutant and WT offspring ($n = 20$, Wilcoxon rank-sum test; ***, $P < 0.001$). (B) The offspring number per maternal parent in different crosses over 10 days at 25°C. Asterisks indicate significant differences between various crosses and crosses of WT females with WT males ($n = 20$, Wilcoxon rank-sum test; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; n.s., $P > 0.05$). The different crosses were denoted as the x-axis labels. (C) The offspring number per maternal parent in different crosses over 10 days at 29°C. (D) Survival curves of WT and mutant adult flies of females (left) and males (right) under starvation conditions. The black line represents the WT, the red line represents the uKO1/uKO1 mutant, and the blue line represents the uKO2/uKO2 mutant. Asterisks indicate significant differences compared to the WT. ($n = 200$, log-rank test; ***, $P < 0.001$; n.s., $P > 0.05$).

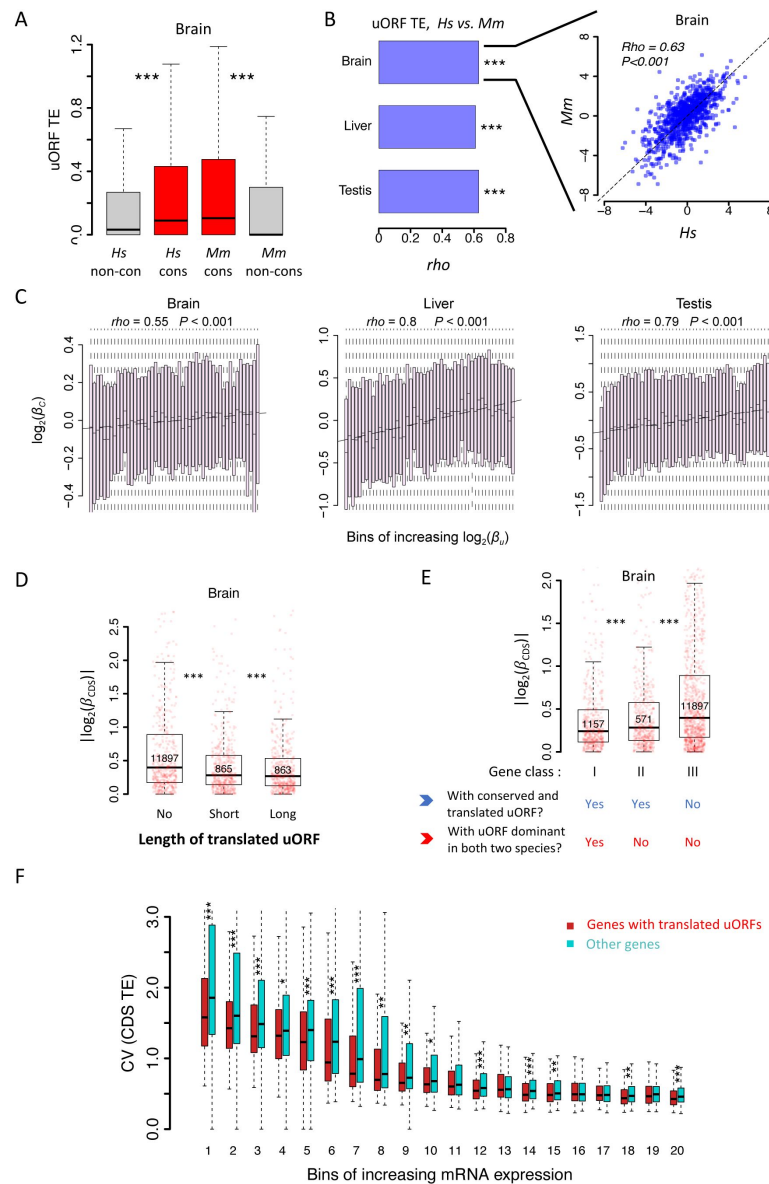


Figure 8.

uORFs function as translational buffers in primates.

(A) Boxplots showing the TEs of conserved and species-specific uORFs between *Hs* (*H. sapiens*) and *Mm* (*M. mulatta*). Data for the brain is shown as an example. Wilcoxon rank-sum tests. ***, $P < 0.001$. (B) Spearman's correlation coefficient (ρ) of uORFs' TE between humans and macaques. The ρ values in the brain, liver, and testis were shown as bar plots. ***, $P < 0.001$. Data for the brain is shown as an example in the right panel. (C) Correlation between interspecific uORF TE changes ($\log_2\beta_u$) and corresponding CDS TE changes ($\log_2\beta_c$) in three tissues. The x-axis was divided into 50 equal bins with increasing β_u . (D) Genes expressed in brains were classified into three classes according to the total length of translated uORFs. Boxplots showing interspecific CDS TE variability $|\log_2\beta_c|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$. (E) Genes expressed in brains (mRNA RPKM > 0.1 in both species) were classified into three classes according to whether a gene had a conserved and dominantly translated uORF (TE > 0.1) in both species or not. Boxplots showing interspecific CDS TE variability $|\log_2\beta_c|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$. (F) Boxplot showing the coefficients of variation (CVs) of CDS TE among the 69 lymphoblastoid cell lines (LCLs). Expressed genes (mean mRNA RPKM > 0.1) were divided into 20 bins with increased mRNA expression levels. In each bin, the genes were divided into two fractions according to whether the gene had a translated uORF or not. Wilcoxon rank-sum tests. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

Discussion

Translational control is vital for maintaining protein homeostasis and cellular activities ^{83,84}. However, the mechanisms underlying the conservation of protein abundance across species remain largely unknown ^{5,6}. In this study, we extended the ICIER model and conducted simulations to explore uORFs' regulatory roles in buffering translation during evolution. Our simulations demonstrated that uORF translation reduces variability in downstream CDS translation, with buffering capacity positively correlated with uORF translation efficiency, length, and number. Comparative translome analyses across developmental stages of two *Drosophila* species provided evidence that uORFs mitigate interspecific differences in CDS translation. Similar patterns were observed between humans and macaques, and between human individuals, suggesting that uORF-mediated buffering is an evolutionarily conserved mechanism in animals. Additionally, in vivo experiments showed that knocking out the *bcd* uORF led to aberrant embryogenesis and altered starvation resistance, with more pronounced phenotypic effects at higher temperatures, supporting the role of uORFs in fine-tuning translation and phenotypic outcomes.

While the prevailing consensus on uORFs' functions is that they repress the translation of downstream CDS by sequestering ribosomes ^{34,43,48}, recent studies have suggested uORFs might play essential roles in stabilizing protein expression ^{43,55,56}. While these studies have primarily focused on the role of uORFs in single-gene cases or under stress conditions, our work expands this knowledge by demonstrating that uORFs act as a general mechanism for buffering translational variability on a genome-wide scale across multiple species and developmental stages. Our findings also suggest that uORF-mediated translational buffering is not merely a response to environmental stress but an evolutionarily conserved mechanism integral to maintaining protein homeostasis across species. Organisms can buffer phenotypic variation against environmental or genotypic perturbations through “canalization” ⁸⁵. The discovery that uORFs are crucial for *Drosophila* development links them to canalization, providing evidence that uORFs reduce interspecific CDS translation differences and enhance phenotypic stability. While previous studies primarily focused on the stabilization of transcriptional levels ^{86–90} or protein levels ^{55,88,91–93}, our study opens new avenues for exploring how organisms maintain phenotypic robustness at the translational level through molecular mechanisms like uORF-mediated buffering. This study lays the groundwork for future research into the evolutionary and functional roles of uORFs, particularly in the context of canalization and adaptive evolution.

Previous studies have shown that a significant fraction of fixed uORFs in the populations of *D. melanogaster* and humans were driven by positive Darwinian selection ^{63,67}, suggesting active maintenance through adaptive evolution rather than purely neutral or deleterious processes. While uORFs have traditionally been recognized for their capacity to attenuate translation of downstream CDSs, accumulating evidence now underscores their critical role in stabilizing gene expression under fluctuating cellular and environmental conditions ^{43,55,56}. Whether the favored evolutionary selection of uORFs acts primarily through their role in translational repression or translational buffering remains a compelling yet unresolved question, as these two functions are inherently linked. Indeed, highly conserved uORFs tend to be translated at higher levels, resulting not only in stronger inhibition of CDS translation ^{34,45,67}, but also in a more pronounced buffering effect, as demonstrated in this study. This buffering capacity of uORFs potentially provides selective advantages by reducing fluctuations in protein synthesis, thus minimizing gene-expression noise and enhancing cellular homeostasis. This suggests that selection may favor uORFs that contribute to translational robustness, a hypothesis supported by findings in yeast and mammals showing that uORFs are significantly enriched in stress-response genes and control the translation of certain master regulators of stress responses ^{41,42,94,95}. Our study suggests that translational buffering, rather than translational repression alone, can also

drive evolutionary selection favoring uORFs, although it remains challenging to empirically disentangle these functions. Future comparative genomic analyses, coupled with experimental approaches such as ribosome profiling and functional mutagenesis, will be crucial in elucidating the precise evolutionary forces driving uORF conservation and adaptation.

The original ICIER model was devised to elucidate how a single uORF can confer resistance to global translation inhibition on its host gene during stress conditions ⁴³. It proposed that an elongating ribosome (80S) causes downstream 40S subunits to dissociate from the mRNA after collision. Our revised model expands this scope to include bidirectional dissociation, where an 80S ribosome can release both upstream and downstream 40S subunits, reflecting recent findings that 80S primarily causes upstream 40S dissociation during 40S/80S collisions ⁵⁶. Mechanistically, increasing the loading rate of 40S ribosomes onto the mRNA (R_{in}) elevates the density of 40S and 80S ribosomes scanning the uORF, thereby raising the likelihood of 40S-80S collisions and subsequent 40S dissociation. This leads to a reduced flow of 40S ribosomes reaching the CDS relative to the initial increase in 40S at the 5' end of the mRNA and uORF. Conversely, a decrease in 40S ribosome loading rate reduces the density of scanning ribosomes, diminishing the probability of 40S-80S collisions and 40S dissociation, and partially restoring the flow of 40S ribosomes to the CDS. Overall, the 40S-80S collision mechanism and subsequent 40S dissociation at the uORF moderate downstream CDS translation less than changes in ribosome loading at the 5' end of the mRNA or uORF. The revised model underscores the analogy of uORFs as “molecular dams”, adeptly modulating the stream of ribosomes and cushioning downstream CDS translation against fluctuations (**Figure 9A**). Additionally, our study broadens its perspective to account for the potential interactive effects of multiple uORFs within a gene, recognizing the prevalent clustering of these regulatory elements and their combinatorial influence on translational control. In essence, our study presents a nuanced and comprehensive expansion of the ICIER model, paving the way for a deeper understanding of uORFs as pivotal elements in the maintenance of protein synthesis stability under variable translational conditions. However, we realize the existence of alternative mechanisms governing uORF function, such as those found in traditional repression models ⁵⁵ or ribosome queuing and reinitiation strategies ⁵⁶. As such, further computational studies are needed to dissect and understand the full spectrum of uORF-mediated translational regulation.

Ribosome slowdown or stalling on mRNA due to rare codons ^{56,96–98} or nascent blocking peptides ^{99–102} frequently triggers ribosome collisions genome-wide ^{103–105}. Such collisions, especially among elongating 80S ribosomes, often activate ribosome quality control (RQC) pathways that recognize collision interfaces on the 40S subunit, leading to ribosomal subunit dissociation and degradation ^{106–108}. In mammals, ZNF598 specifically identifies collided ribosomes to initiate ubiquitin-dependent protein and mRNA quality control pathways ^{109–113}. Analogously, yeast employs Hel2-mediated ubiquitination of uS10, initiating dissociation via the RQC-trigger complex (RQT) ¹¹⁴. Furthermore, the human RQT (hRQT) complex recognizes ubiquitinated ribosomes and induces subunit dissociation similarly to yeast RQT ¹¹⁵. However, transient ribosome collisions can evade RQC by promoting resumed elongation through mechanical force provided by trailing ribosomes, thereby mitigating stalling ¹¹⁶. Beyond 80S collisions, evidence increasingly highlights a distinct collision type involving scanning 40S subunits or pre-initiation (43S) complexes. Recently, an initiation RQC pathway (iRQC) targeting the small ribosomal subunit (40S) has been described, particularly involving collisions between scanning 43S complexes or between stalled 43S and elongating 80S ribosomes (**Figure 9B**) ^{117,118}. During iRQC, E3 ubiquitin ligase RNF10 ubiquitinates uS3 and uS5 proteins, resulting in 40S degradation ¹¹⁸. This mechanism aligns closely with our ICIER model, proposing collision-driven 43S dissociation in the 5' UTRs. Future studies exploring these mechanisms in greater detail will clarify how uORFs modulate translational regulation through buffering effects.

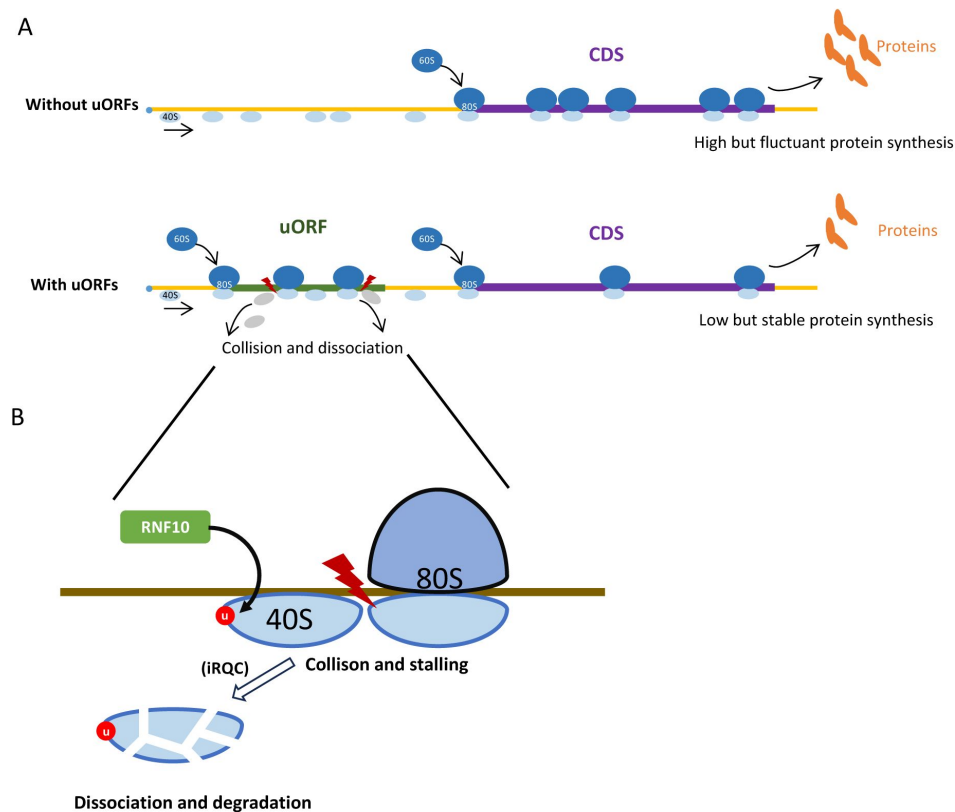


Figure 9.

Model illustrating the uORF-mediated translational buffering.

(A) uORFs can reduce and stabilize the downstream CDS translation through “ribosomes collision and dissociation” mechanism in an mRNA containing uORFs. In contrast, the CDS translation is higher and more fluctuant in an mRNA without uORFs. (B) The collided and stalled 40S subunit dissociates from mRNA, is ubiquitinated by the E3 ubiquitin ligase RNF10 and degraded by the initiation ribosome quality control (iRQC) pathway.

The *bcd* gene, crucial for *Drosophila* embryogenesis, has been well studied [71](#)[72](#)[119](#)[123](#), but its uORF remains underexplored. Our study showed that deleting the *bcd* uORF led to abnormal phenotypes and reduced fitness, underscoring the importance of uORF-mediated translation. These effects were more severe under heat stress (29°C) than at normal temperatures (25°C), suggesting uORFs play a vital role in buffering phenotypic variation. We demonstrated that the *bcd* uORF represses CDS translation using sucrose gradient fractionation followed by qPCR—an approach that directly measures translation efficiency while minimizing confounding from RNA/protein degradation. However, detecting Bcd protein levels with antibodies across developmental stages or conditions in the mutants and wild-type controls would provide an even stronger validation of our model and should be explored in future studies. Rescue experiments, typically used to confirm that phenotypic changes are due to specific genetic edits, faced challenges in our study and previous uORF-KO experiments in animals [124](#)[125](#) and plants [126](#)[129](#). We altered the start codon of the *bcd* uORF to minimize effects on the 5' UTR, without impacting the coding sequence, making traditional rescue experiments impractical. To control for genetic background variations, we backcrossed two *bcd* uORF-KO mutant strains with *w*¹¹¹⁸ flies for nine generations, using *w*¹¹¹⁸ as a control. Both uORF-KO mutants (uKO1/uKO1 and uKO2/uKO2) consistently showed reduced progeny, ruling out background or off-target effects. Crosses between the two mutants produced fewer offspring than WT crosses, confirming that the reduced progeny was due to the uORF deletions, not genetic background. As *bcd* is maternally inherited, we expected defective phenotypes only in offspring from mutant females crossed with wild-type males. This was confirmed: uKO1/uKO1 males crossed with wild-type females had normal offspring, while uKO1/uKO1 females crossed with wild-type males had reduced offspring, similar to mutant crosses. Similar results were observed for uKO2/uKO2 mutants. These findings confirm that disrupting *bcd* uORFs significantly reduces hatchability and fertility, effects not attributable to genetic background. Our results highlight the critical role of *bcd* uORFs in development and fecundity, and provide a basis for future studies on uORF regulation in other genes across species.

Taken together, this study demonstrates the role of uORF-mediated translational buffering in mitigating variability in gene translation during species divergence and development, using large-scale comparative transcriptome and translatome data. Our work addresses gaps in understanding the stabilization of gene expression at the translational level and offers new insights into the evolutionary and functional significance of uORFs.

Materials and Methods

Modeling the uORF-mediated buffering effect on CDS translation

We adapted the stochastic ICIER framework developed by Andreev *et al.* [43](#) based on the TASEP model, with several major modifications. First, we concatenated a 500-codon CDS downstream of the uORF. A leaky scanning ribosome from the uORF would initiate translation at the CDS start codon with a probability of I_{CCS} . The probability of elongating ribosomes moving along the CDS ($v_{EC} = 0.5$) in each action was higher than that along uORFs ($v_{Eu} = 0.3$) in the model settings, considering that uORFs usually encode blocking peptides [99](#)[102](#)[130](#) or contain stalling codons [56](#)[96](#)[97](#). Second, we recorded the number of elongating ribosomes that completed translation at the stop codon of a CDS (N_{EC}) or uORF (N_{Eu}) during a given time period and regarded them as proxies for quantifying the CDS or uORF TE. Third, we considered three models for simulating the consequences of scanning ribosomes colliding with elongating ribosomes mentioned above: a downstream dissociation model, an upstream dissociation model, and a double dissociation model.

In detail, the mRNA molecule structure in the simulations consisted of five parts: a 5' leader before the uORF (150 nucleotides), the uORF (ranging from 2 to 100 codons), a segment between the uORF and CDS (150 nucleotides), the CDS (500 codons) and the 3' UTR (150 nucleotides). The R_{in} determined the loading rate of the 40S scanning ribosomes on the mRNA 5'-terminus, and it roughly represented the availability of *trans* translational resources in the cell. We generated two sets of R_{in} values (**Figure S1A**) to simulate the variation in the availability of translational resources (40S scanning ribosomes, etc.): i) 1000 values of R_{in} were generated from a random generator of a uniform distribution, $U(0, 0.1)$; ii) 1000 values were first generated from a random generator of an exponential distribution, $E(1)$, and then each of the 1000 values was divided by 70 to obtain 1000 values of R_{in} . I_{uORF} and I_{CCS} represent the probability of translation at the start codon of a uORF or CDS, respectively. We used different combinations of I_{uORF} and I_{CCS} to test how translational initiation strength influenced the buffering effect of uORFs. v_s determined the scanning rate of the 40S ribosome, and v_{Eu} and v_{EC} determined the elongation rate of the 80S ribosome at a uORF or CDS, respectively. All the parameters used in our simulation are listed in Table S1.

The mRNA molecule was modeled as an array, and the value of each position in the array represented the occupation status of ribosomes along the mRNA molecule. The process of translation was simulated by a series of discrete actions referred to Andreev *et al.* Upon each action, there was a probability that certain events would occur, including the addition of a 40S ribosome to the 5'-terminus, the transformation of a 40S ribosome into an 80S ribosome at a CDS or uORF start codon, the movement of a 40S ribosome or 80S ribosome to the next codon in the CDS or uORF, etc. More specific operations in each action were described in Supplementary Materials and Methods.

After 1,000,000 actions, we recorded the number of 80S ribosomes that completed translation at the uORF (N_{Eu}) and CDS (N_{EC}) for each R_{in} input, which was used to represent the protein production rates (i.e. translation rate) of the uORF and CDS, respectively. To test the effects of different factors, we used various combinations of the parameters in Table S1 in the simulations. For a given distribution of R_{in} input (1000 values) and the dissociation model, we obtained the corresponding 1000 N_{EC} values and their CV values by calculating the ratio of the standard deviation to the mean of N_{EC} .

In the double-uORF simulation, we only adopted a uniform distribution of R_{in} input and the downstream dissociation model. Most parameters and simulation processes were similar to those in the single-uORF simulation, except that an additional uORF was introduced upstream of the CDS with an initiation probability of I_{uORF2} .

Annotation of uORFs in *D. melanogaster* and *D. simulans*

We downloaded the whole-genome sequence alignment (maf) of *D. melanogaster* (dm6) and 26 other insect species from the University of California Santa Cruz (UCSC) genome browser (genome.ucsc.edu)¹³¹. To improve the genome correspondence between *D. melanogaster* and *D. simulans*, we adopted a newly published reference-quality genome of *D. simulans* (NCBI, ASM438218v1)¹³² to replace the original *D. simulans* genome in UCSC maf. We softly masked repetitive sequences in the new genome by RepeatMasker 4.1.1 (<http://www.repeatmasker.org>)¹³³ and aligned it to dm6 with lastz¹³⁴ in runLastzChain.sh following UCSC guidelines. The lastz alignment parameters and the scoring matrix were the same as the parameters of dm6 and droSim1 in UCSC. Chained alignments were processed into nets by the chainNet and netSyntenic programs. The alignment was integrated into the multiple alignments of 27 species with multiz¹³⁴.

Based on the genome annotation of *D. melanogaster* (FlyBase r6.04, <https://flybase.org/>), we used the Galaxy platform to parse the multiple sequence alignments of 5' UTRs in *Drosophila*. The 5' UTR sequences of each annotated transcript from *D. melanogaster* and the corresponding

sequences in *D. simulans* were extracted from the maf. The start codons of putative uORFs were identified by scanning all the ATG triplets (uATGs) within the 5' UTRs of *D. melanogaster* and *D. simulans*. uATGs that overlapped with any annotated CDS region were removed. The presence or absence of each uATG of *D. melanogaster* was determined at orthologous sites in *D. simulans* based on multiple genome alignments, and vice versa. The conserved uORF was defined as a uORF where its uATG was present in both *D. melanogaster* and the corresponding orthologous positions of *D. simulans*. For each protein-coding gene, we only considered the canonical transcript. For transcripts containing multiple uORFs, we defined the uORF that showed the highest TE as the dominant uORF. The branch length score (BLS) of the uATGs was calculated as we previously described ¹⁸.

Fly materials and general raising conditions

The *sim4* strain of *D. simulans* was used to generate all the libraries of *D. simulans* in this study. All flies were raised on standard corn medium and grown in 12 h light: 12 h dark cycles at 25 °C for general conditions or at 29 °C for specific experimental design. The samples of embryos at different stages, larva, pupa, bodies and heads were collected following a previous protocol ²³.

Processing *Drosophila* mRNA-Seq and Ribo-Seq data

Ribo-Seq and matched mRNA-Seq libraries for different developmental stages and tissues of *D. simulans* were constructed as we previously described ²³ and sequenced on an Illumina HiSeq-2500 sequencer (run type: single-end; read length: 50 nucleotides) according to the manufacturer's protocol. The 3' adaptor sequences (TGGAATTCTCGGGTGCCAAGG) were trimmed using Cutadapt 3.0 with default parameters ¹³⁵, and the NGS reads were mapped to the genomes of yeast, *Wolbachia*, *Drosophila* viruses and the sequences of tRNAs, ribosomal RNAs (rRNAs), small nuclear RNAs (snRNAs) or small nucleolar RNAs (snoRNAs) of *D. melanogaster* (FlyBase r6.04) and *D. simulans* (FlyBase r1.3) using Bowtie2 version 2.2.3 ¹³⁶ with the parameters -p8 -local -k1. The mapped reads of these genomes/sequences were further removed in the downstream analysis.

After filtering, the mRNA-Seq and Ribo-Seq reads were mapped to the reference genomes of *D. melanogaster* (FlyBase, r6.04) and *D. simulans* (NCBI, ASM438218v1), respectively, using the Spliced Transcripts Alignment to a Reference (STAR) algorithm ¹³⁷. For Ribo-Seq reads, we assigned a mapped RPF (27-34 nucleotides in length) to its P-site using the psite script from Plastid ¹³⁸. The uniquely mapped reads were extracted and then mapped to the CDS of *D. melanogaster* and *D. simulans* respectively using STAR ¹³⁷. The P-sites of RPF or mRNA-Seq reads that overlapped with a CDS were counted separately. The reads that were not mapped to CDSs were then mapped to the 5' UTRs of *D. melanogaster* and *D. simulans*. The P-sites of RPF or mRNA-Seq reads that overlapped with uORFs were counted separately.

The TE of a given uORF or CDS was calculated as the $\text{RPKM}_{\text{P-site}}/\text{RPKM}_{\text{mRNA}}$ ratio. For a few uORFs with $\text{RPKM}_{\text{mRNA}} = 0$ but $\text{RPKM}_{\text{P-site}} > 0$, a pseudocount of 0.1 was added to both $\text{RPKM}_{\text{mRNA}}$ and $\text{RPKM}_{\text{P-site}}$ to avoid dividing a positive value by zero. In each sample, expressed genes were defined as the genes with a CDS $\text{RPKM}_{\text{mRNA}} > 0.1$ in both species, and translated uORFs were defined as uORFs with a TE > 0.1.

Testing the statistical significance of the difference in the interspecific TE change between a uORF and its downstream CDS

For this analysis, we adopted the methods developed by Zhang *et al.* ²³. Briefly, for the samples of female bodies or male bodies of *D. melanogaster* with biological replicates, we obtained the $\log_2(\text{TE})$ and SE of the $\log_2(\text{TE})$ of a uORF or CDS by contrasting the RPF counts against mRNA-Seq read counts using DESeq2. Then, we fitted the SE values against the normalized mRNA counts and $\log_2(\text{TE})$ values using the gam function in the R package mgcv, with a log link to obtain the SE ~ mRNA counts + $\log_2\text{TE}$ functions. For other samples without biological replicates, we estimated the

SE of the $\log_2(\text{TE})$ for a feature (CDS or uORF) by applying the fitted functions obtained based on the biological replicates of female and male bodies to the observed mRNA counts and $\log_2(\text{TE})$ values. We identified uORFs whose TEs differed significantly between the paired samples of *D. melanogaster* and *D. simulans* by testing whether the value obtained for $\log_2(\beta_u) = \log_2(\text{TE}_{u\text{ORF},\text{sim}}) - \log_2(\text{TE}_{u\text{ORF},\text{mel}})$ was significantly different from 0. Based on the SE of the $\log_2(\text{TE})$ derived as described above, the SE of the $\log_2(\beta_u)$ can be derived as follows:

$$\text{SE}_{\log_2(\beta_u)} = \sqrt{(\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{sim}}})^2 + (\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{mel}}})^2}$$

As the Wald statistic, $\frac{\log_2(\beta_u)}{\text{SE}_{\log_2(\beta_u)}}$, follows a standard normal distribution under the null hypothesis of $\log_2(\beta_u) = 0$, we calculated the *P* value as follows: $2 \cdot (1 - \Phi(|\frac{\log_2(\beta_u)}{\text{SE}_{\log_2(\beta_u)}}|))$. The identification of CDSs whose TEs differed significantly between the paired samples of *D. melanogaster* and *D. simulans* was similar to the procedures above.

Then, we defined $\gamma = \frac{\beta_c}{\beta_u}$ (where $\beta_c = \text{TE}_{\text{CDS},\text{sim}}/\text{TE}_{\text{CDS},\text{mel}}$) and tested whether $\log_2(\gamma)$ was significantly different from 0 to determine whether the magnitude of interspecific TE changes in CDSs and uORFs was significantly different. We obtained $\log_2\text{TE}_{u\text{ORF},\text{sim}}$, $\log_2\text{TE}_{u\text{ORF},\text{mel}}$, $\log_2\text{TE}_{\text{CDS},\text{sim}}$, and $\log_2\text{TE}_{\text{CDS},\text{mel}}$ as described above and estimated $\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{sim}}}$, $\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{mel}}}$, $\text{SE}_{\log_2\text{TE}_{\text{CDS},\text{sim}}}$, and $\text{SE}_{\log_2\text{TE}_{\text{CDS},\text{mel}}}$ based on the biological replicates of female and male bodies of *D. melanogaster*. Finally, $\log_2(\gamma)$ also follows a normal distribution with SE denoted as $\text{SE}_{\log_2(\gamma)}$ =

$$\sqrt{(\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{sim}}})^2 + (\text{SE}_{\log_2\text{TE}_{u\text{ORF},\text{mel}}})^2 + (\text{SE}_{\log_2\text{TE}_{\text{CDS},\text{sim}}})^2 + (\text{SE}_{\log_2\text{TE}_{\text{CDS},\text{mel}}})^2}$$

As the Wald statistic, $\frac{\log_2(\gamma)}{\text{SE}_{\log_2(\gamma)}}$, follows a standard normal distribution under the null hypothesis that $\log_2(\beta_u) = 0$, we calculated the *P* value as follows: $2 \cdot (1 - \Phi(|\frac{\log_2(\gamma)}{\text{SE}_{\log_2(\gamma)}}|))$.

uORFs are shorter than CDSs and have substantially lower read counts, reducing statistical power when comparing TE changes. To mitigate this bias, we focused on highly expressed genes, defined as those with both mRNA and RPF RPKM values above the 50th percentile in *D. melanogaster* and *D. simulans*. uORFs and CDSs of these genes were used in the calculation of significant TE change.

Knocking out a uORF with CRISPR-Cas9 technology in *D. melanogaster*

We searched for possible sgRNA target sites near the uATG start codon of a uORF using the Benchling website (<https://www.benchling.com/crispr/>) to design optimal single guide RNA (sgRNA) sequences with high specificity and low off-target effects. We then synthesized single-stranded complementary DNAs (ssDNAs) and annealed them to obtain double-stranded DNA (dsDNA), which served as the template for sgRNA expression. The template sequences of the sgRNA used for *bcd* uATG-KO are listed in Table S6. The dsDNA was then ligated into the BbsI-digested pU6B vector. The pU6B-sgRNA plasmid was purified and injected into the embryos of transgenic Cas9 flies collected within one hour of laying at the Tsinghua Fly Center as described in Ni *et al.* ¹³⁹. The injected embryos were kept at 25°C and 60% humidity until adulthood (G0). The G0 adult flies that hatched from injected embryos were individually crossed with other flies (*y sc v*) to increase the number of offspring. Then, the F1 progeny were crossed with flies carrying an appropriate balancer (*Dr*, *e/TM3*, *Sb*). After F2 spawning, the F1 individuals were screened for mutations of interest by genotyping. The primers used for genotyping are listed in Table S6. The F2 progeny whose parents showed positive genotyping results were then screened for the *yellow* gene to separate the chromosome carrying *nos-Cas9*. The screened F2 males were crossed with the flies containing the same balancer as above. After F2 genotyping, the progeny (F3) of positive F2 individuals showing the same mutation status were crossed individually to generate homozygous mutants in the F4 generation. The original homozygous mutants were sequentially outcrossed with *w*¹¹¹⁸ flies for 9 generations to purify the genetic background (Figure S25B).

Ribosome fraction analysis by sucrose gradient fractionation and RT-qPCR

The 0-2 h embryos obtained from *w¹¹¹⁸*, *uKO1/uKO1*, and *uKO2/uKO2* mutants raised at 25°C and 29°C were collected and homogenized in a Dounce homogenizer with lysis buffer [50 mM Tris pH 7.5, 150 mM NaCl, 5 mM MgCl₂, 1% Triton X-100, 2 mM dithiothreitol (DTT), 20 U/ml Superscript^{III} (Ambion), 0.5 tablets of proteinase inhibitor (Roche), 100 µg/ml emetine (Sigma Aldrich), and 50 µM guanosine 5'-[β,γ-imido]triphosphate trisodium salt hydrate (GMP-PNP) (Sigma Aldrich)] at 4°C. The lysates were clarified by centrifugation at 4°C and 20,000×g for 8 min, and the supernatants were transferred to new 1.5 ml tubes. 10-45% sucrose gradients were prepared in buffer (250 mM NaCl, 50 mM Tris pH 7.5, 15 mM MgCl₂, 0.5 mM DTT, 12 U/ml RNaseOUT, 0.5 tablets of protease inhibitor, and 20 µg/ml emetine) using a Gradient Master (Biocomp Instruments) in ULTRA-CLEAR Thinwall Tubes (Beckman Coulter). A sample volume of up to 500 µl was applied to the top of each gradient. After ultracentrifugation with a Hitachi P40ST rotor at 35,000 × rpm for 3 h at 4°C, the monosome and polysome fractions were collected, flash-frozen in liquid nitrogen, and stored at -80°C until further use.

The RNA in the monosome and polysome fractions was extracted separately using TRIzol reagent (Life Technologies, Inc.) and chloroform (Beijing Chemical Works) following the manufacturer's instructions and were reverse transcribed into cDNA using the PrimeScriptTM II 1st Strand cDNA Synthesis Kit (Takara). RT-qPCR analysis of *bcd* cDNA and its targets was performed using PowerUpTM SYBRTM Green Master Mix (Thermo Fisher) following the manufacturer's instructions with *rp49* as an internal control. The primer sequences employed for RT-qPCR are listed in Table S6. For each sample, the ratio of *bcd* mRNA abundance in the polysome fraction to that in the monosome fraction was calculated as the P-to-M ratio. Six biological replicates were performed for each sample.

Dual-luciferase reporter assays

The wild-type (WT) 5' UTR of *bcd* was cloned from cDNA by PCR, and uATG mutations were introduced into 5' UTR using specific amplification primers (listed in Table S6). The WT and mutated 5' UTR sequences were ligated into a linearized reporter plasmid (psiCHECK-2 vector, Promega). The whole sequence of all the plasmids was validated by Sanger sequencing.

Drosophila S2 cells were cultured in Schneider's Insect Medium (Sigma) plus 10% (by volume) heat-inactivated fetal bovine serum, 100 U/ml penicillin and 100 µg/ml streptomycin (Thermo Fisher) at 25°C without CO₂ for 24 h to reach 1–2×10⁶ cells/ml before further treatments. Plasmid transfection was conducted with Lipofectamine 3000 (L3000001, Thermo Fisher) according to the supplier's protocol.

The *Renilla* luciferase activity associated with WT or uORF-mutated 5' UTRs was measured according to the manual of the Dual-Luciferase Reporter Assay System (Promega) 32 h after transfection and was normalized to the activity of firefly luciferase.

mRNA-Seq in the embryos of mutant and WT flies

We collected 0-2 h, 2-6 h, 6-12 h, and 12-24 h embryos of *w¹¹¹⁸* and *uKO2/uKO2* mutants raised at 25°C and 29°C and conducted mRNA-Seq. Library construction and sequencing with PE150 were conducted by Annoroad on the Illumina Nova6000 platform. Two biological replicates were sequenced for each sample. The clean data were mapped to the reference genome of *D. melanogaster* (FlyBase, r6.04) using STAR ¹³⁷[137](#). Reads mapped to the exons of each gene were tabulated with htseq-count ¹⁴⁰[140](#). The differentially expressed genes were identified using DESeq2 ¹⁴¹[141](#). The Gene Ontology (GO) analyses were conducted using the “clusterProfiler” package in R ¹⁴²[142](#).

Measurement of embryo hatchability

We collected embryos from WT and mutant flies and manually seeded them in vials containing standard corn medium at a density of 30 embryos per vial and 20 vials per strain. The embryos were cultivated under standard conditions (60% humidity, 12 h light: 12 h dark cycles at 25°C). We counted the number of pupae in each vial after the completion of pupation and calculated the corresponding hatching rate = $\text{number}_{\text{pupa}} / 30$

Quantification of offspring number per female fly

Newly hatched virgins were picked out and allowed to mature for two days in separate vials. They were then mated by placing one virgin female with three male flies for two days. After that, each female parent was transferred to a new vial to count the offspring number produced in each 10-day period at 25°C and 29°C. All assays were performed with 20 females per genotype.

Measurement of starvation resistance in adult flies

We selected 3- to 5-day-old adult males and females and placed them in the starvation medium (1.5% agar), with 10 flies per vial and 10 vials for both males and females from each strain. We made observations every 6 h or 12 h to count the number of deaths under starvation conditions until all flies had starved to death. The survival curves were plotted by the `ggsurvplot` package in R.

mRNA-Seq and Ribo-Seq data analysis in primates

The mRNA-Seq and Ribo-Seq data from the brains, livers, and testes of humans and macaques were downloaded from reference ⁷⁹ with accession number E-MTAB-7247 (ArrayExpress). The mRNA-Seq and Ribo-Seq data of human lymphoblastoid cell lines (LCL) from Yoruba individuals were downloaded from references ^{81,82} under accession numbers GSE61742 and E-GEUV-1. The matched mRNA-Seq and Ribo-Seq libraries of 69 individuals were used.

The uORF annotation and downstream analysis procedures for the human and macaque data were similar to those applied in *Drosophila* as described above. The differential analysis of translational efficiency in humans and macaques was conducted by Xtail ¹⁴³. In each pair of human-macaque samples, expressed genes were defined as the genes with a $\text{CDS RPKM}_{\text{mRNA}} > 0.1$ in both species. The translated uORFs in a sample were defined as uORFs with a $\text{TE} > 0.1$. For the human cell line data, expressed genes were defined as genes with a mean $\text{CDS RPKM}_{\text{mRNA}} > 0.1$ across the cell lines, and translated uORFs were defined as uORFs with a mean $\text{TE} > 0.1$.

Supplementary Materials and Methods

The detailed operations per action in the ICIER simulation

The translation process of an mRNA molecule is modeled as a series of discrete actions. During each action, certain events may occur with certain probabilities for each entity (40S or 80S ribosome). The operations per action are described as follows:

1. If the first 10 triplets were vacant of the mRNA, a 40S ribosome would be added there with a probability of R_{in} .
2. If a 40S ribosome was located at the start codon of the uORF, it would transform into an 80S ribosome with a probability of I_{uORF} . If a 40S ribosome was located at the start codon of the CDS, it would transform into an 80S ribosome with a probability of I_{CCS} .

3. If an 80S ribosome was located at the stop codon of the uORF or CDS, it would dissociate from the mRNA. The dissociation of an 80S ribosome from the uORF or CDS stop codon was counted as a completed translation event, and the values of N_{EC} and N_{Eu} were incremented by one, respectively.
4. If a 40S ribosome arrived at the 3' end of the mRNA, it would dissociate from the mRNA.
5. For each 40S ribosome on the mRNA, if its next position was vacant (in the downstream-dissociation model) or not occupied by another 40S ribosome (in the upstream-dissociation model and double-dissociation model), it would move to the next position with a probability of v_s .
6. For each 80S ribosome in the uORF, if its next position was vacant (in the upstream-dissociation model) or not occupied by another 80S ribosome (in the downstream-dissociation model and double-dissociation model), it would move to the next position with a probability of v_{Eu} . Similarly, for each 80S ribosome in the CDS, if its next position was vacant (in the upstream-dissociation model) or not occupied by an 80S ribosome (in the downstream-dissociation model and double-dissociation model), it would move to the next position with a probability of v_{EC} .
7. It was checked whether any 80S ribosome collided with a 40S ribosome. If such a collision occurred, the 40S ribosome involved would dissociate from the mRNA with a probability of K_{up} (upstream dissociation or double-dissociation model) or K_{down} (downstream dissociation or double-dissociation model).

These above operations are executed repeatedly until the action count reaches the limit given by the user (1,000,000 in this study).

Supplementary Figures and Tables

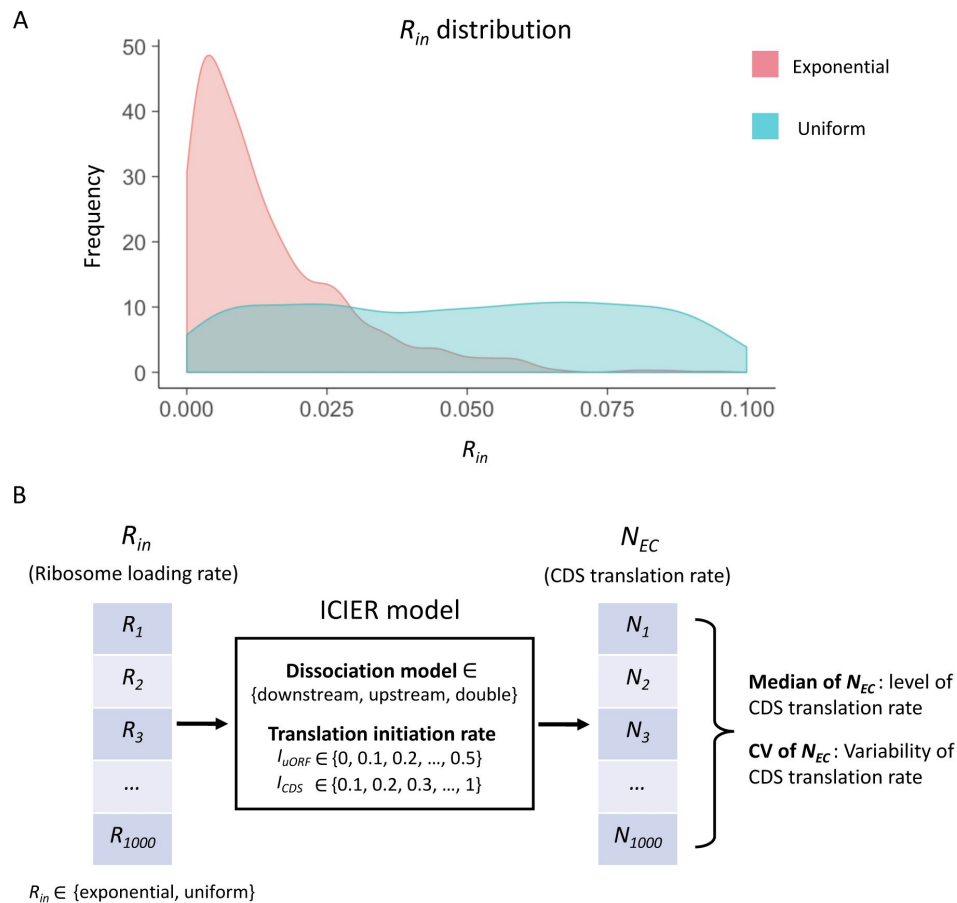


Figure S1.

The input distribution and model simulation flow.

(A) Two distributions of the R_{in} input were used in our simulation: an exponential distribution and a uniform distribution. A total of 1,000 R_{in} values (ranging from 0 to 0.1) were randomly generated following either distribution. (B) Under a fixed combination of parameters (R_{in} distribution, dissociation model, I_{uORF} and I_{CDS}), the model simulation produces 1000 N_{EC} values from 1000 R_{in} input. The median of these 1000 N_{EC} represented the median of translation rate of CDS under the 1000 varying R_{in} inputs and the other fixed parameters. The the coefficient of variation (CV) of these 1000 N_{EC} reflected the variability of translation rate of CDS under the 1000 varying R_{in} inputs and the other fixed parameters.

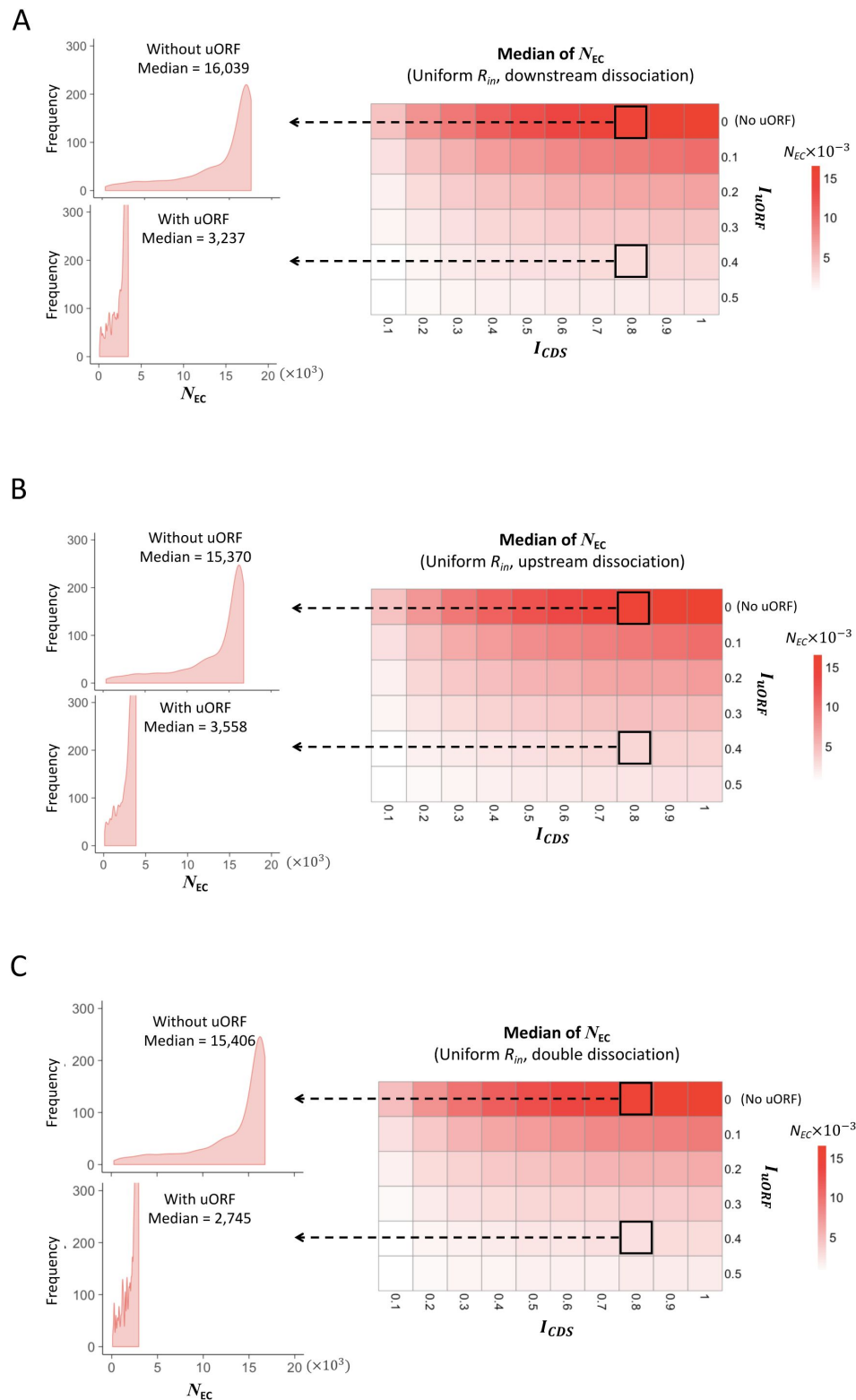


Figure S2.

Heatmaps showing the median CDS translation rate (N_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model (A), a uniform distribution of R_{in} input and the upstream dissociation model (B), a uniform distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $I_{CDS} = 0.8$ & $I_{uORF} = 0$ (top panel, without uORF) and $I_{CDS} = 0.8$ & $I_{uORF} = 0.4$ (bottom panel, with uORF).

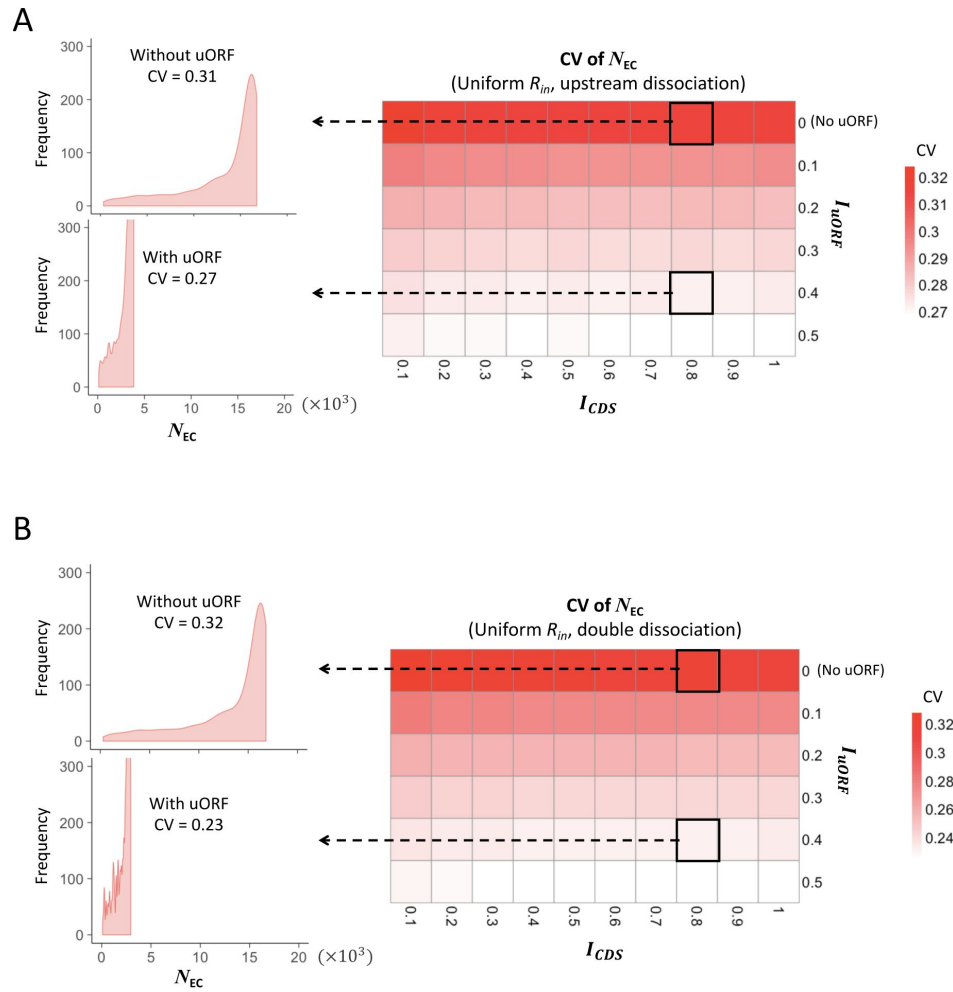


Figure S4.

Heatmaps showing the CVs of CDS translation rate (N_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the upstream dissociation model (A), a uniform distribution of R_{in} input and the double dissociation model (B). For each heatmap, the left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $I_{CDS} = 0.8$ & $I_{uORF} = 0$ (top panel, without uORF) and $I_{CDS} = 0.8$ & $I_{uORF} = 0.4$ (bottom panel, with uORF).

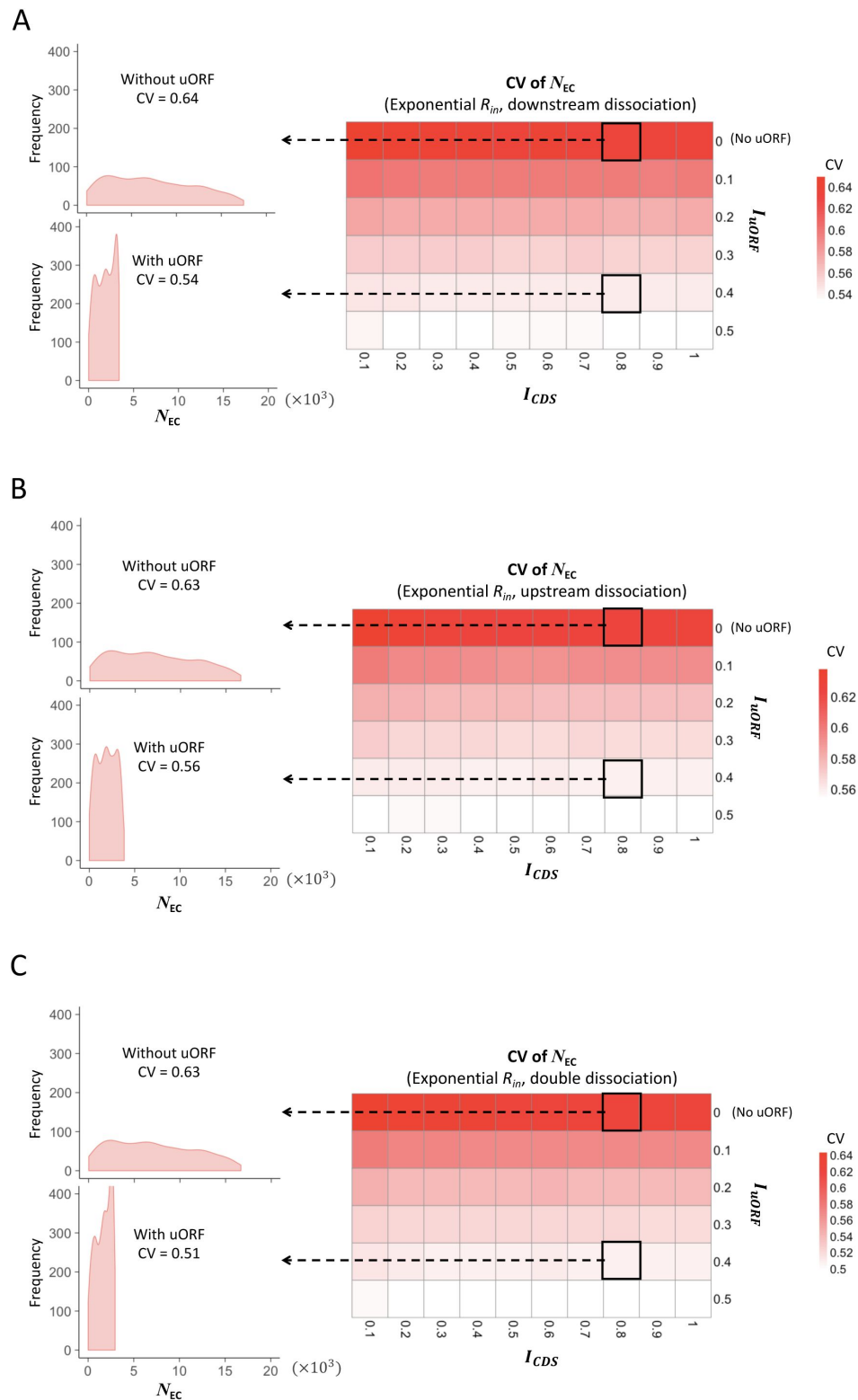


Figure S5.

Heatmaps showing the CVs of CDS translation rate (N_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with an exponential distribution of R_{in} input and the downstream dissociation model (A), an exponential distribution of R_{in} input and the upstream dissociation model (B), an exponential distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $I_{CDS} = 0.8$ & $I_{uORF} = 0$ (top panel, without uORF) and $I_{CDS} = 0.8$ & $I_{uORF} = 0.4$ (bottom panel, with uORF).

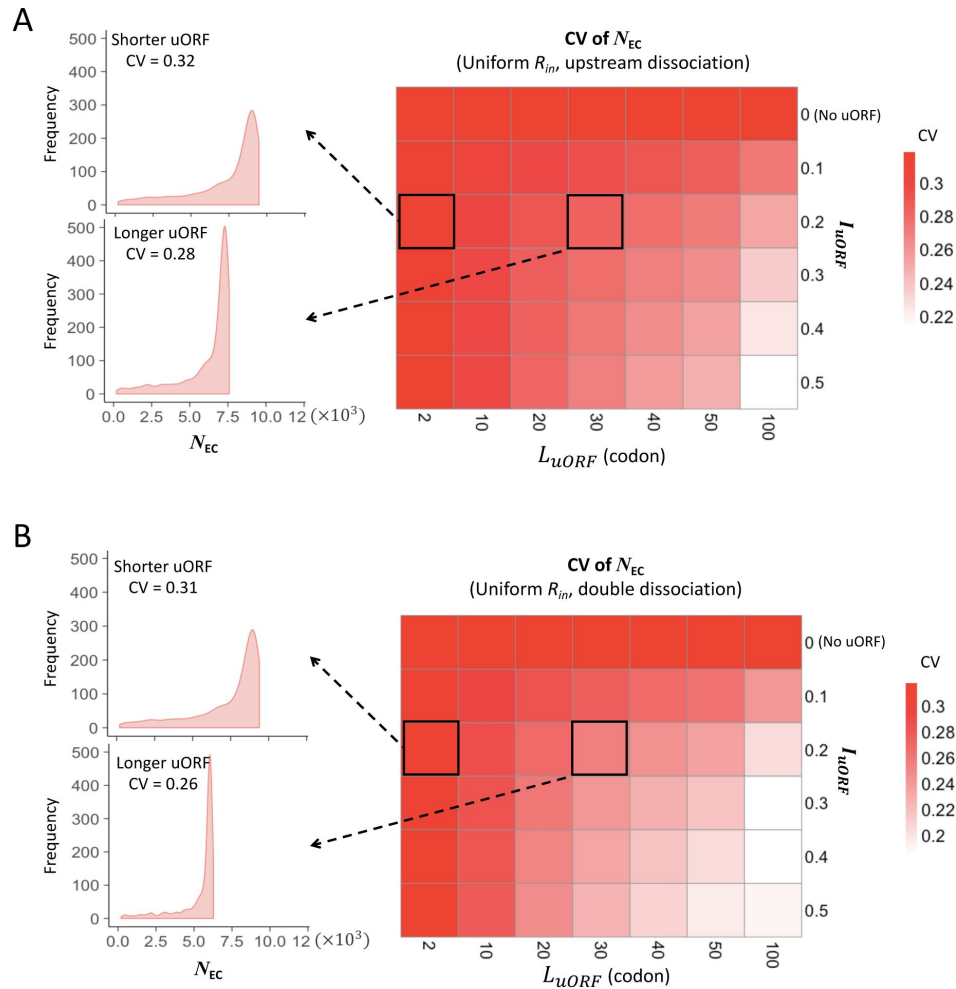


Figure S6.

Heatmaps showing the CVs of CDS translation rate (N_{EC}) under different L_{uORF} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the upstream dissociation model (A), a uniform distribution of R_{in} input and the double dissociation model (B). For each heatmap, the left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $L_{uORF} = 2$ & $I_{uORF} = 0.2$ (top panel, shorter uORF) and $L_{uORF} = 30$ & $I_{uORF} = 0.2$ (bottom panel, longer uORF).

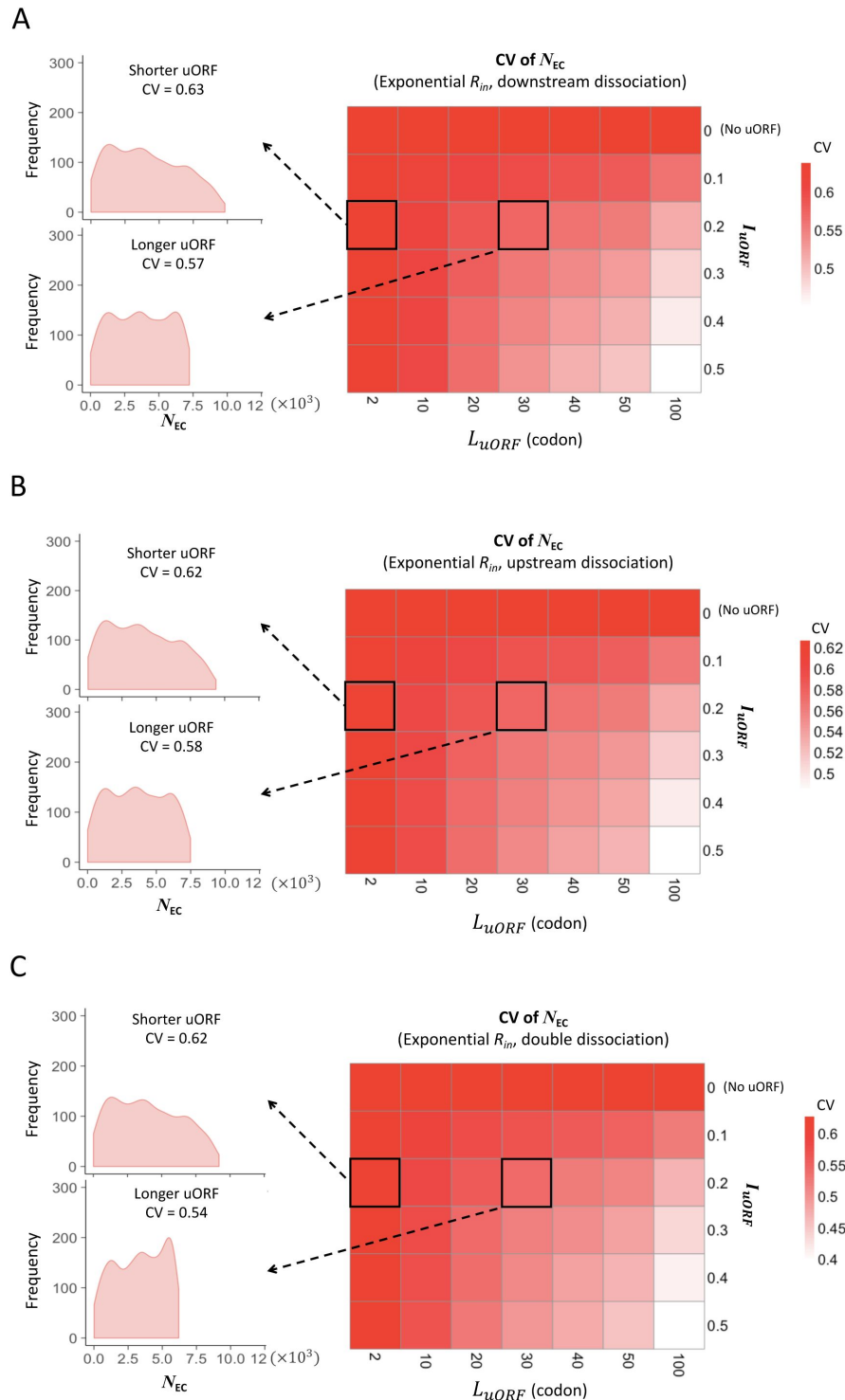


Figure S7.

Heatmaps showing the CVs of CDS translation rate (N_{EC}) under different L_{uORF} (x-axis) and I_{uORF} (y-axis) combinations with an exponential distribution of R_{in} input and the downstream dissociation model (A), an exponential distribution of R_{in} input and the upstream dissociation model (B), an exponential distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $L_{uORF} = 2$ & $I_{uORF} = 0.2$ (top panel, shorter uORF) and $L_{uORF} = 30$ & $I_{uORF} = 0.2$ (bottom panel, longer uORF).

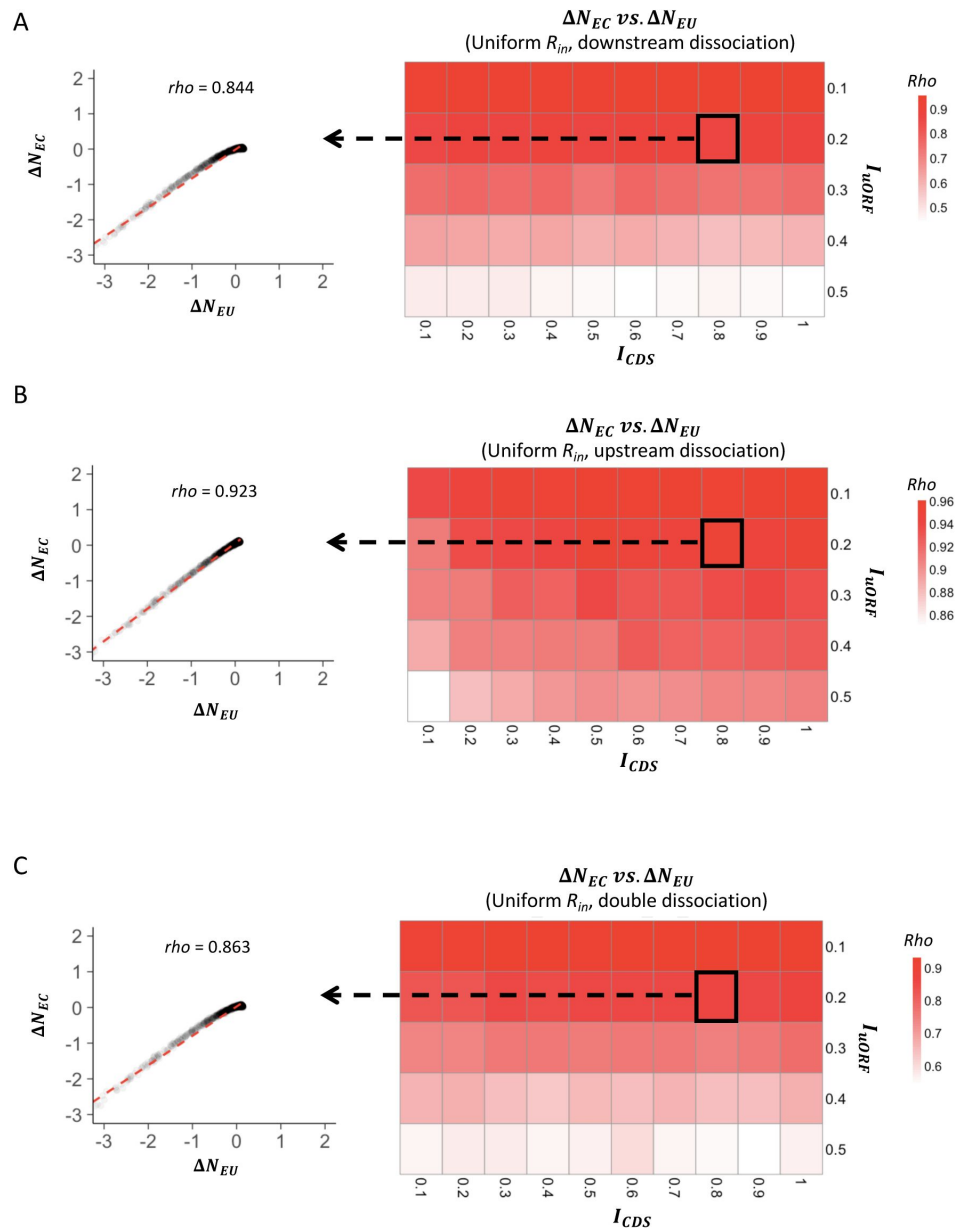


Figure S8.

Heatmaps showing the Spearman's correlations between changes of uORF translation rate (ΔN_{EU}) and downstream CDS translation rate (ΔN_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model (A), a uniform distribution of R_{in} input and the upstream dissociation model (B), a uniform distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panel elicited by the dotted line from the specific square of right heatmap was an example showing the Spearman's correlations between changes of uORF translation rate (ΔN_{EU}) and downstream CDS translation rate (ΔN_{EC}) under $I_{CDS} = 0.8$ & $I_{uORF} = 0.2$. All P values < 0.001.

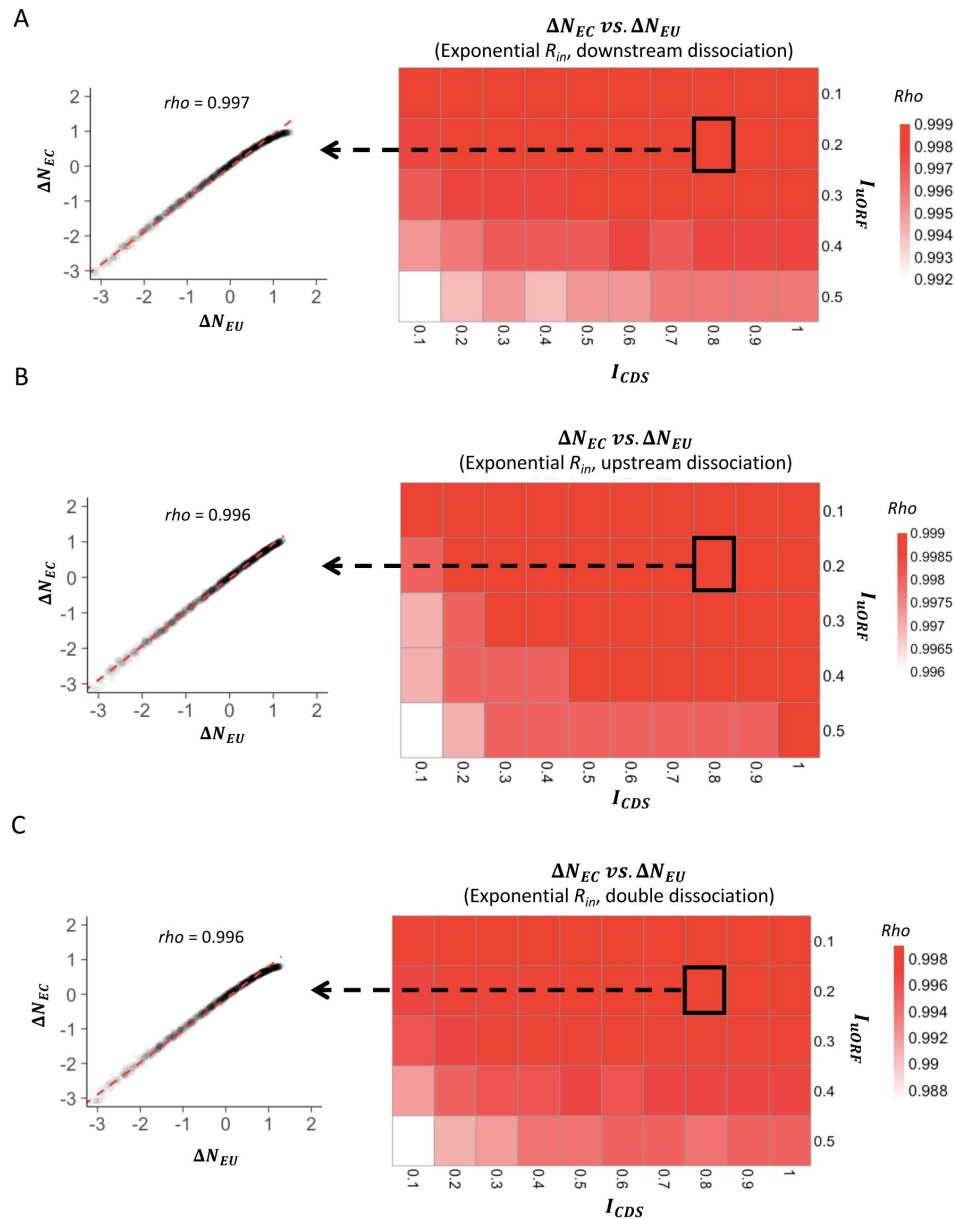


Figure S9.

Heatmaps showing the Spearman's correlations between changes of uORF translation rate (ΔN_{EU}) and downstream CDS translation rate (ΔN_{EC}) under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with an exponential distribution of R_{in} input and the downstream dissociation model (A), an exponential distribution of R_{in} input and the upstream dissociation model (B), an exponential distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panel elicited by the dotted line from the specific square of right heatmap was an example showing the Spearman's correlations between changes of uORF translation rate (ΔN_{EU}) and downstream CDS translation rate (ΔN_{EC}) under $I_{CDS} = 0.8$ & $I_{uORF} = 0.2$. All P values < 0.001.

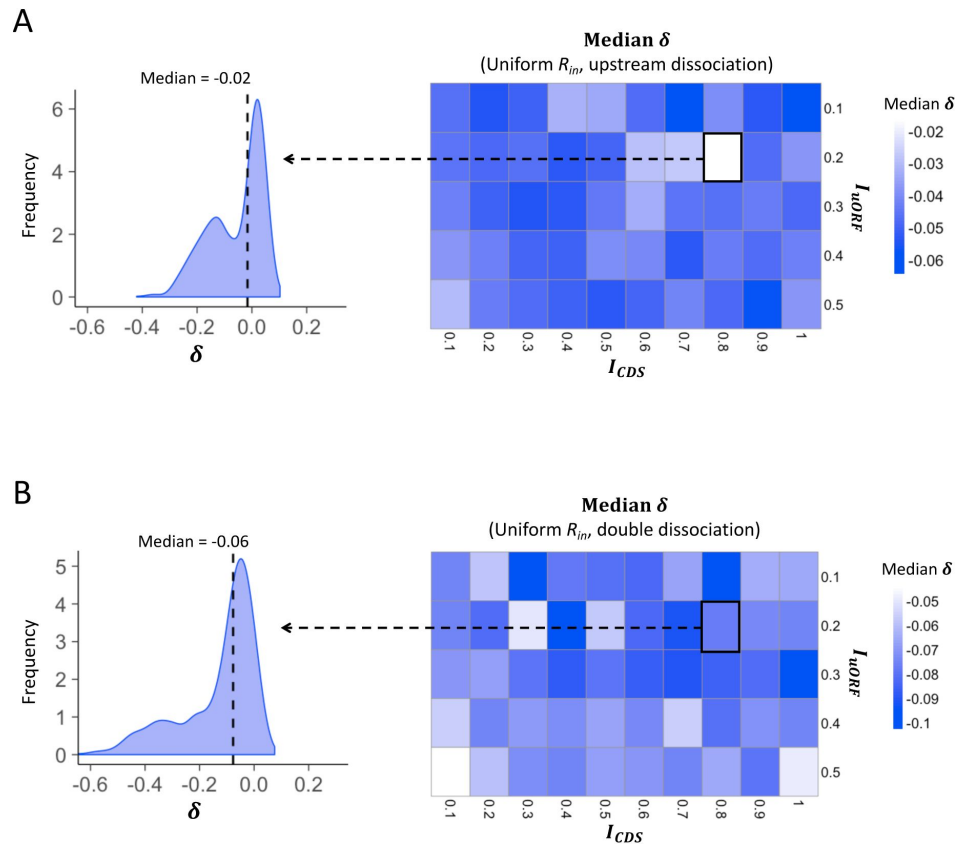


Figure S10.

Heatmaps showing the median δ under different I_{CDS} (x-axis) and I_{uORF} (y-axis) combinations with a uniform distribution of R_{in} input and the upstream dissociation model (A), a uniform distribution of R_{in} input, and the double dissociation model (B). For each heatmap, the left panel elicited by the dotted line from the specific square of right heatmap was an example showing the distribution of δ under $I_{CDS} = 0.8$ & $I_{uORF} = 0.2$. The vertical dashed line indicated the median value of δ .

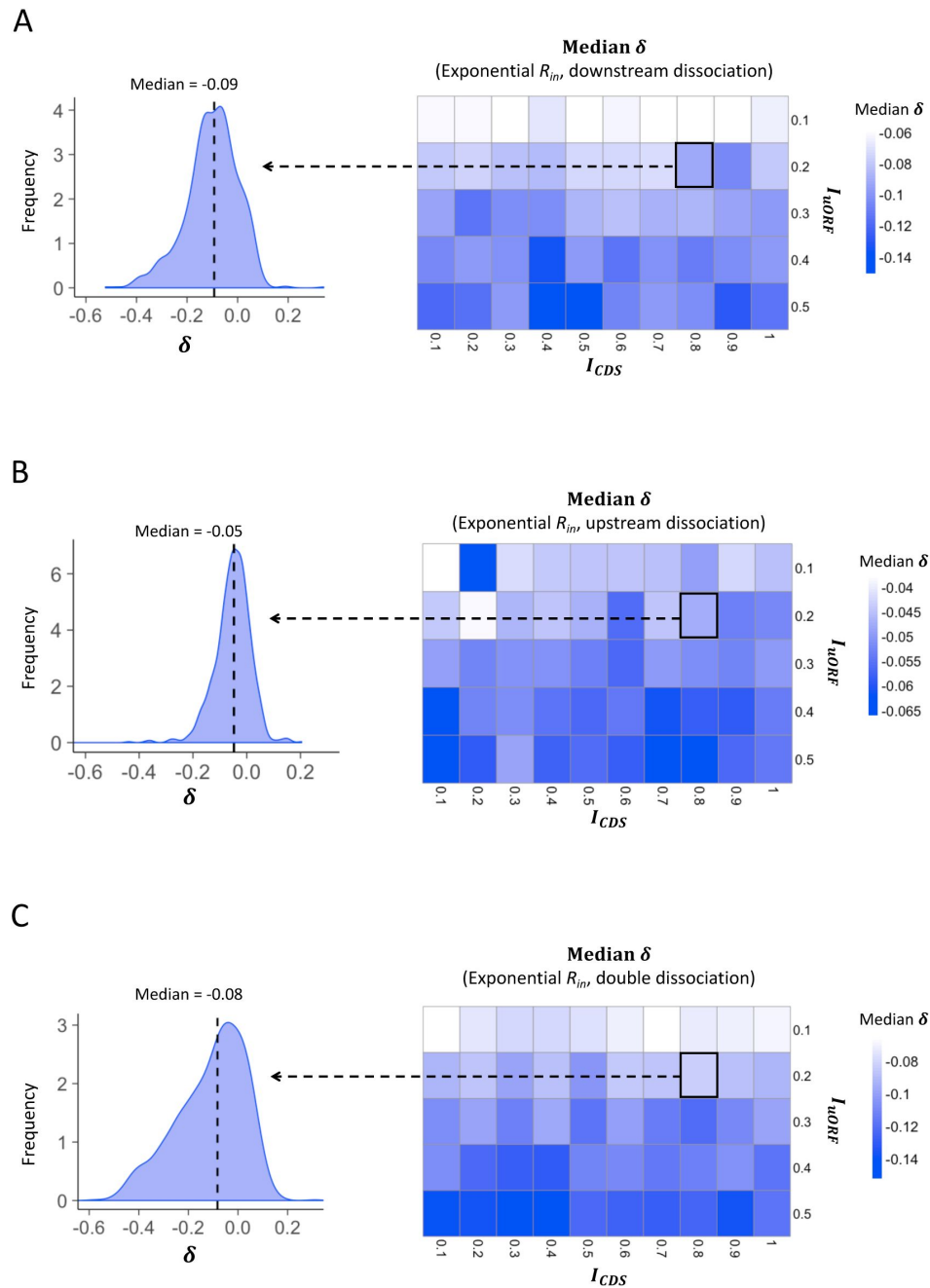


Figure S11.

Heatmaps showing the median δ under different I_{CDS} (x-axis) and I_{URF} (y-axis) combinations with an exponential distribution of R_{in} input and the downstream dissociation model (A), an exponential distribution of R_{in} input and the upstream dissociation model (B), an exponential distribution of R_{in} input and the double dissociation model (C). For each heatmap, the left panel elicited by the dotted line from the specific square of right heatmap was an example showing the distribution of δ under $I_{CDS} = 0.8$ & $I_{URF} = 0.2$. The vertical dashed line indicated the median value of δ .

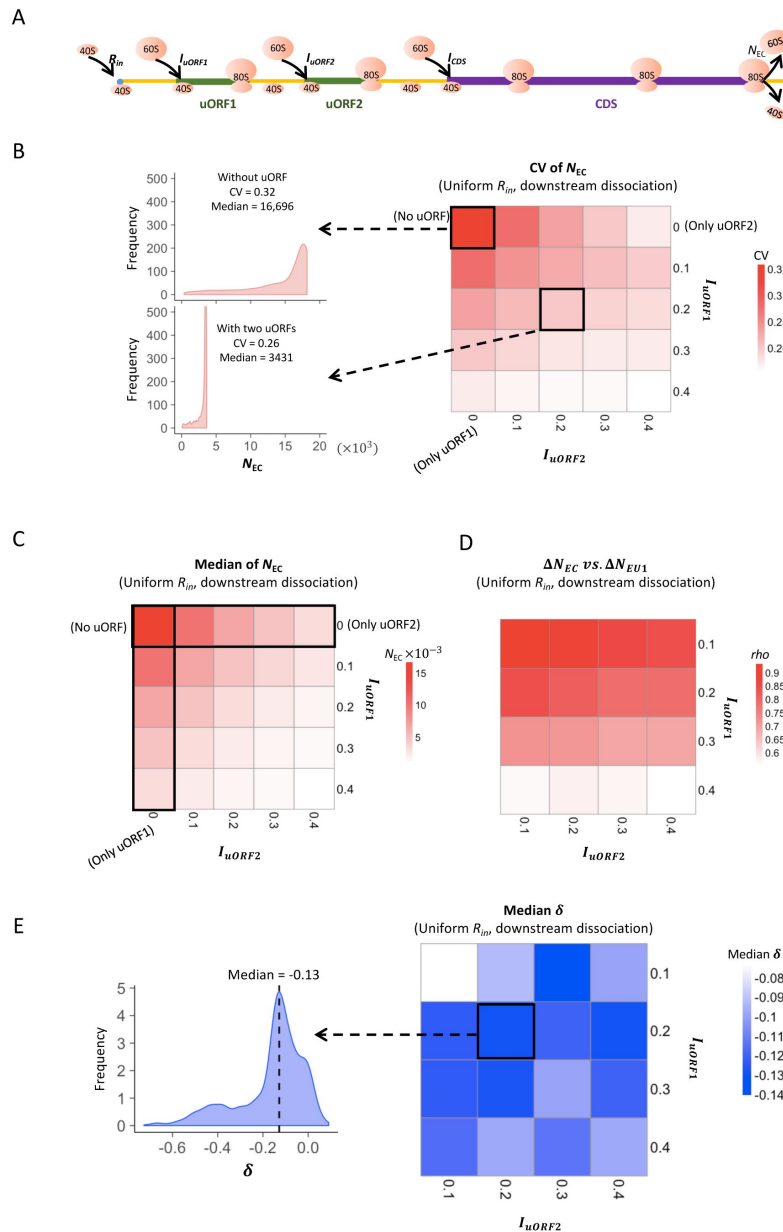


Figure S12.

Two-uORF model simulation measurements.

(A) The two-uORF model schema. The factors and parameters were the same as those illustrated in **Fig. 1A** except where specifically indicated. (B) Heatmap showing the CVs of CDS translation rate (N_{EC}) under different I_{uORF2} (x-axis) and I_{uORF1} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. The left panels elicited by the dotted lines from specific squares of right heatmap were two examples showing the distribution of N_{EC} under $I_{uORF1} = 0$ & $I_{uORF2} = 0$ (top panel, without uORF) and $I_{uORF1} = 0.2$ & $I_{uORF2} = 0.2$ (bottom panel, with two uORFs). (C) Heatmap showing the median CDS translation rate (N_{EC}) under different I_{uORF2} (x-axis) and I_{uORF1} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. (D) Spearman's correlations between changes of uORF1 translation rate (ΔN_{EC}) and downstream CDS translation rate (ΔN_{EC}) under different I_{uORF2} (x-axis) and I_{uORF1} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. (E) Heatmap showing the median δ [$\log_2(\Delta N_{EC}/\Delta N_{EC})$] under different I_{uORF2} (x-axis) and I_{uORF1} (y-axis) combinations with a uniform distribution of R_{in} input and the downstream dissociation model. The left panel elicited by the dotted line from the specific square of right heatmap was an example showing the distribution of δ under $I_{uORF1} = 0.2$ & $I_{uORF2} = 0.2$. The vertical dashed line indicated the median value of δ .

Figure S13.

Comparison of the buffering effects between single uORF and two uORFs.

In each panel, the I_{uORF} of single-uORF model equals to I_{uORF1} of two-uORF model, with both values ranging from 0 to 0.5. The x-axis in each panel denotes the values of different I_{uORF2} in the two-uORF model, ranging from 0 to 0.5. The y-axis in each panel represents the ratio of the CV of N_{EC} with two uORFs to that with a single uORF.

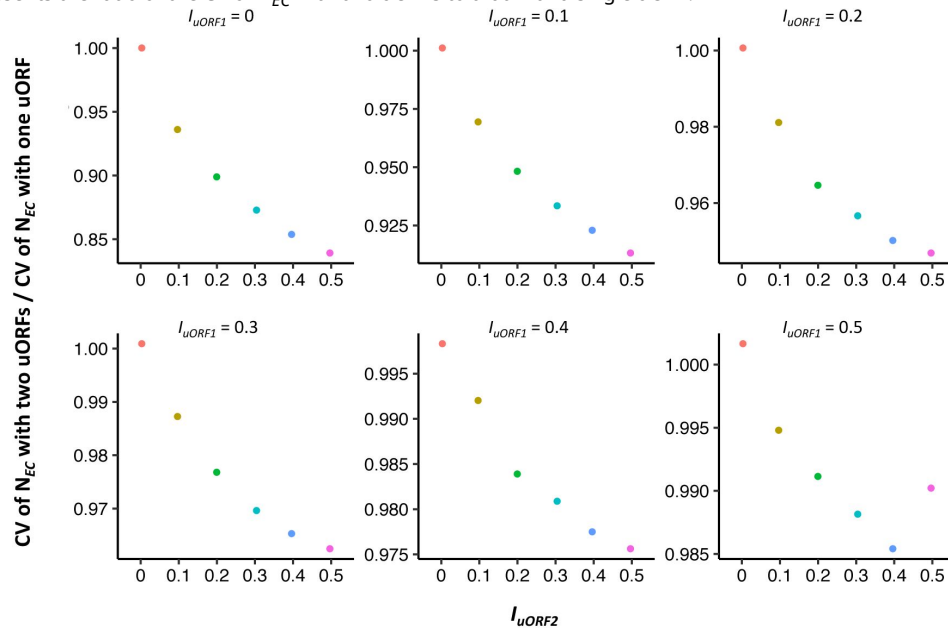


Figure S14.

The dominant uORF showed highest TE than other uORFs with a same gene.

uORFs were ranked by decreasing TEs within each gene. The uORF with the highest TE within each gene was defined as the dominant uORF (#1). “#2” represents the second highest uORF TE and the same goes for “#3” and “>3”. Each dot represents the median TE of a sample.

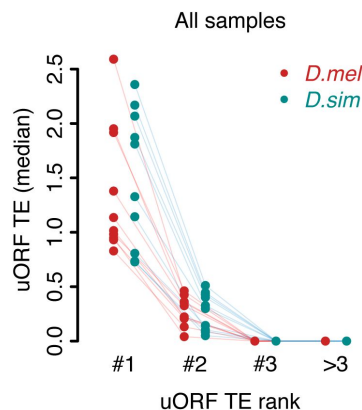


Figure S15.

Observed and expected numbers of genes sharing dominant uORFs in all samples.

Among the genes of *D. melanogaster* (where only the longest transcripts were considered), 7,259 (52.2%) genes had no uORF ("no-uORF" genes), 2,687 (19.3%) genes had one uORF ("one-uORF" genes), and 3,961 (28.5%) genes had multiple (≥ 2) uORFs ("multiple-uORF" genes). The numbers in the header of each panel represent genes with translated uORFs (TE > 0.1) in $\geq N$ samples ($2 \leq N \leq 10$). The TEs of all uORFs were shuffled 1,000 times. P values were obtained from randomization tests.

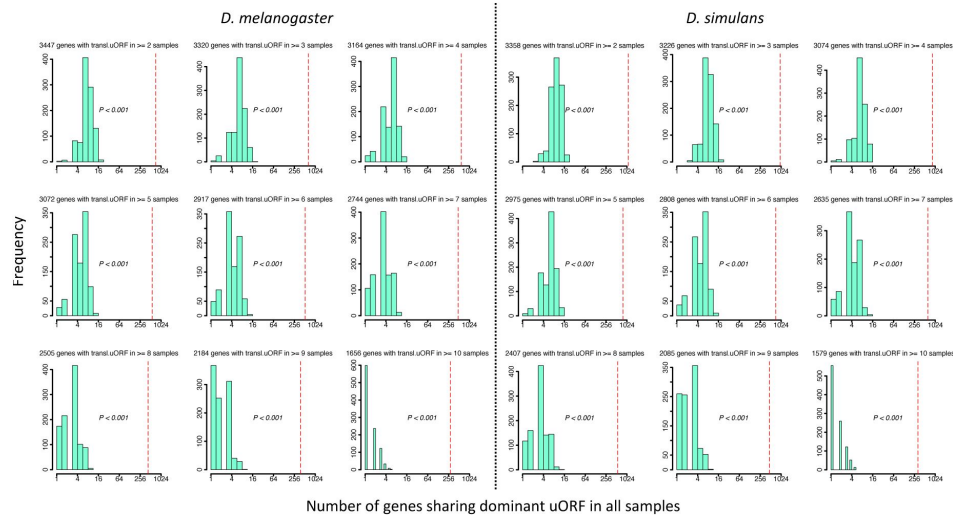
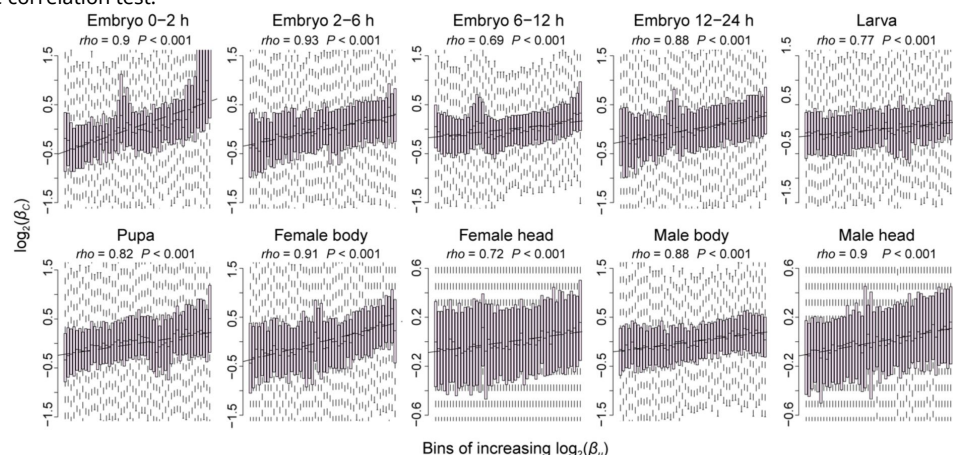


Figure S16.

The positive correlation of interspecific TE changes between uORFs and CDSs.

Correlations between interspecific uORF TE changes ($\log_2 \beta_u$) and CDS TE changes ($\log_2 \beta_c$) in 10 samples. The x-axis was divided into 50 equal bins with increasing β_u . Spearman's correlation coefficients (ρ) are shown at the top left. ***, $P < 0.001$ in the correlation test.



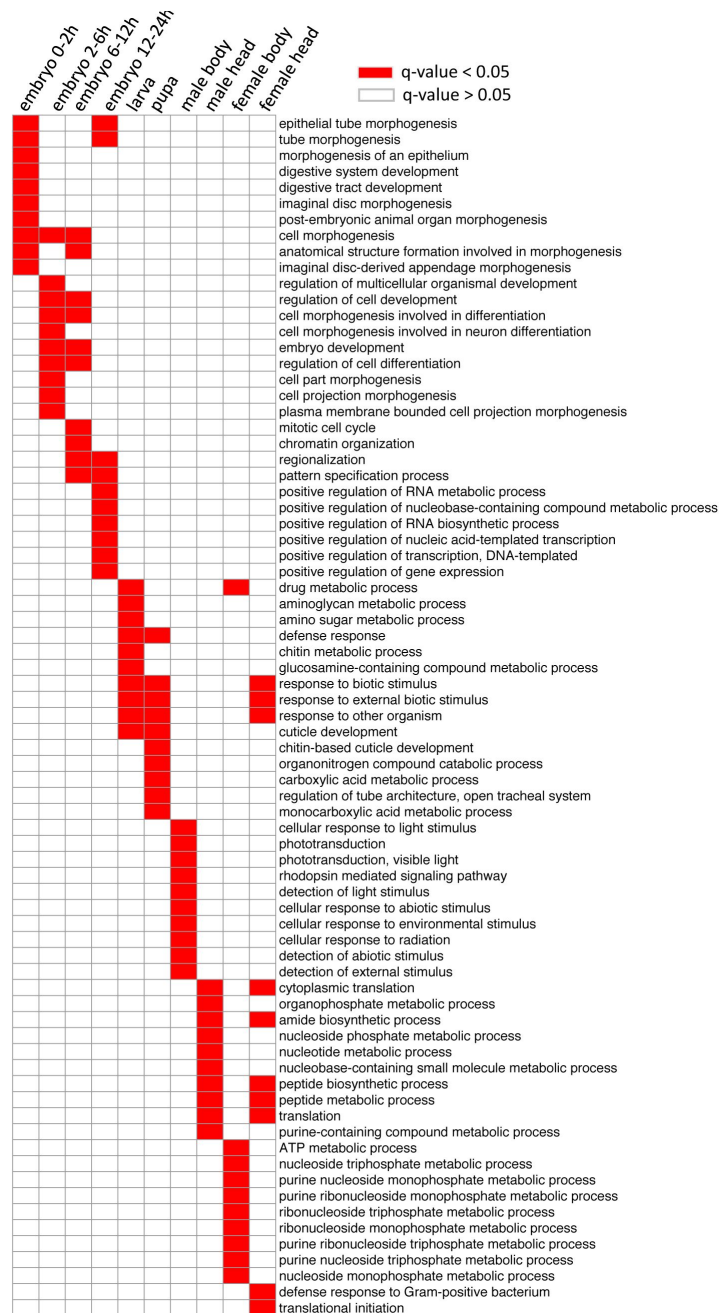


Figure S17.

Gene ontology analysis of the genes with $\log_2(\beta_c) \neq 0$ in each stage and tissue.

The biological process (BP) terms with q-values < 0.05 in each sample type are indicated with red, and others are indicated with white.

Figure S18.

Reads count distribution of uORFs and CDSs.

Distribution of mRNA reads counts and RPF counts mapped to uORFs and CDSs for all expressed uORFs (A) or only highly expressed genes (B) in female head sample. The distribution patterns were similar across other samples, and the data are not shown here. *Dm*, *D. melanogaster*; *Ds*, *D. simulans*. The reads counts were $\log_2$ transformed.

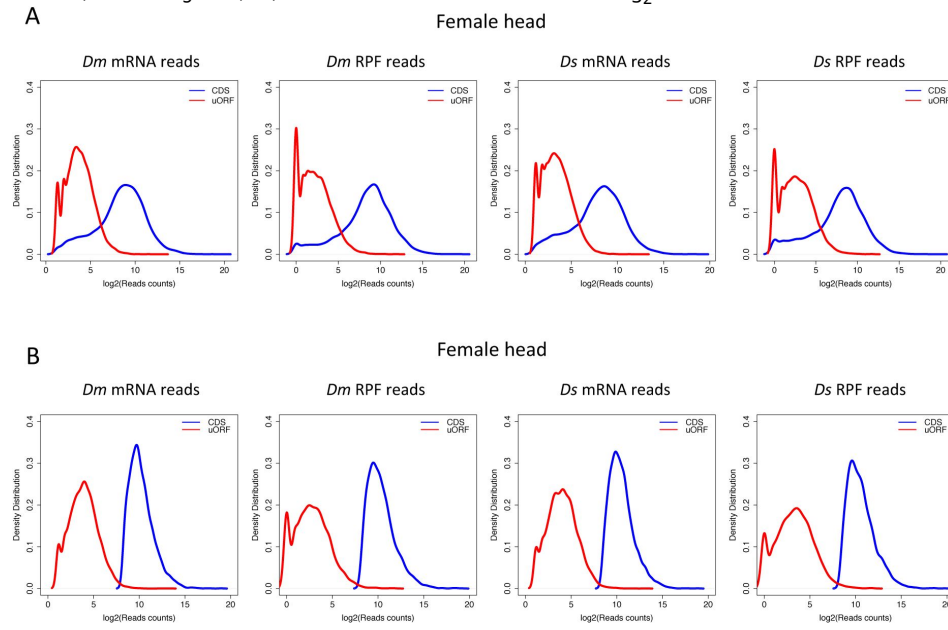
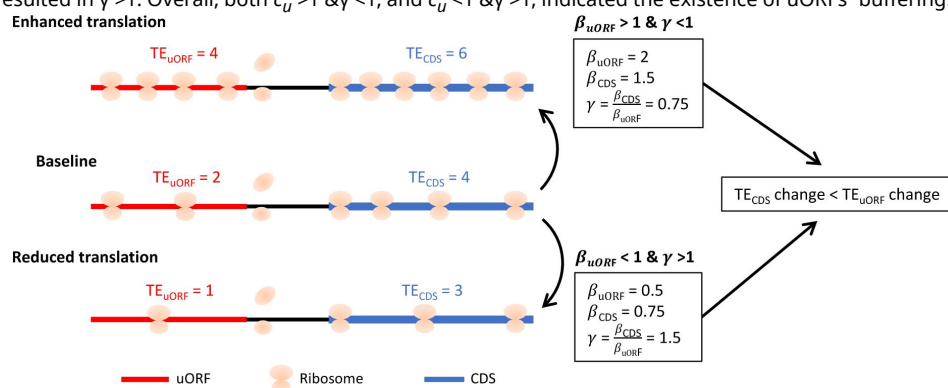


Figure S19.

The scheme illustrating the calculation of β_u , c_c and γ .

When the cellular environment causes mRNA translation to be enhanced from baseline level, the TE_{uORF} changed from 2 to 4 ($c_u = 2$), while the TE_{uORF} changed with a smaller degree due to uORF's buffering, from 4 to 6 ($c_c = 1.5$). This resulted in $\gamma < 1$. Conversely, When the cellular environment causes mRNA translation to be reduced from the baseline level, the TE_{uORF} changed from 2 to 1 ($c_u = 0.5$), while the TE_{uORF} changed with a smaller degree due to uORF's buffering, from 4 to 3 ($c_c = 0.75$). This resulted in $\gamma > 1$. Overall, both $c_u > 1$ & $\gamma < 1$, and $c_u < 1$ & $\gamma > 1$, indicated the existence of uORFs' buffering.



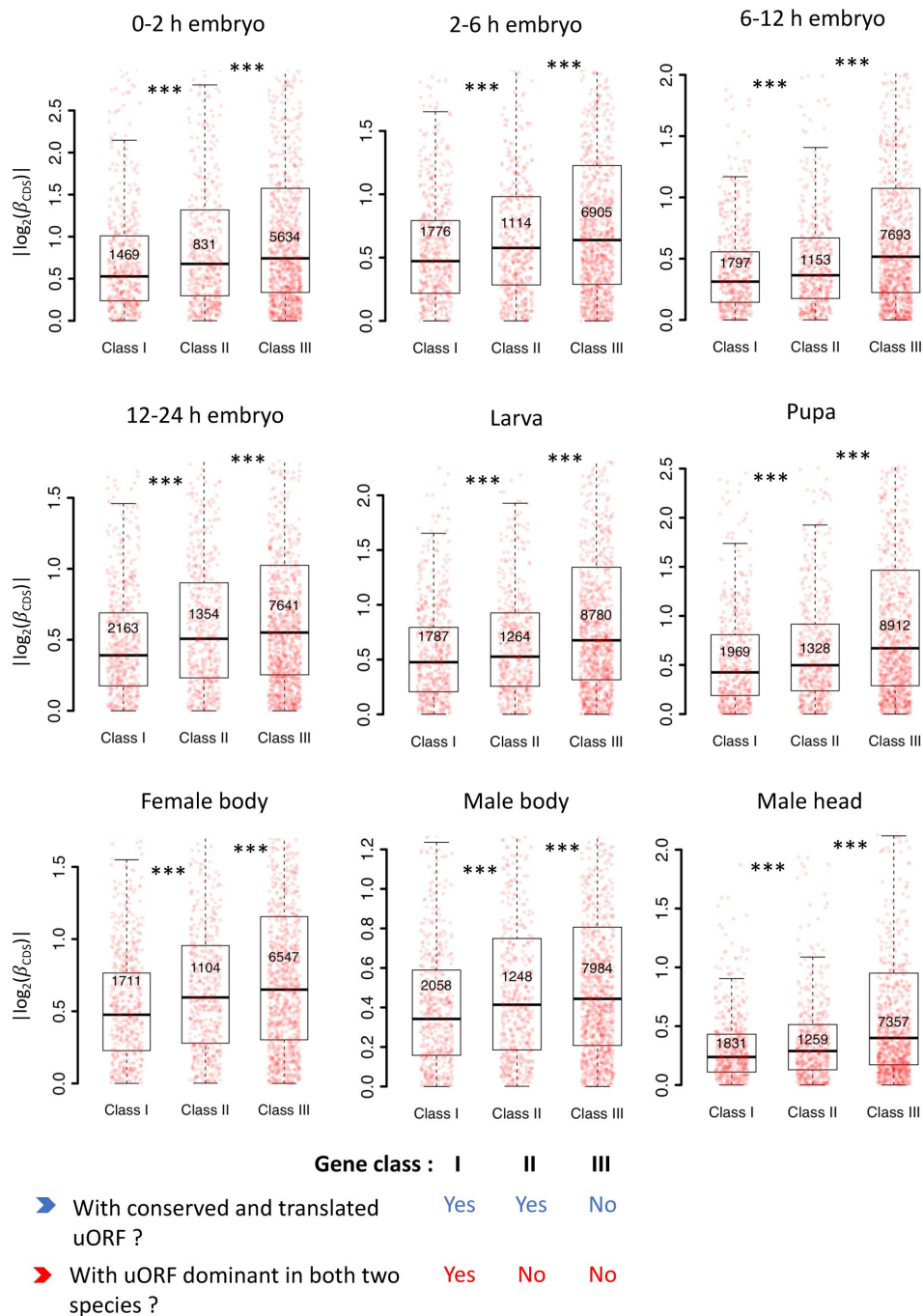


Figure S20.

Conserved and dominantly translated uORFs showed the stronger buffering effect.

Genes expressed in different stages/tissues (mRNA RPKM > 0.1 in both species) were classified into three classes according to whether a gene had a conserved and dominantly translated uORF or not. Boxplots showing interspecific CDS TE variability $|\log_2(c_c)|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$.

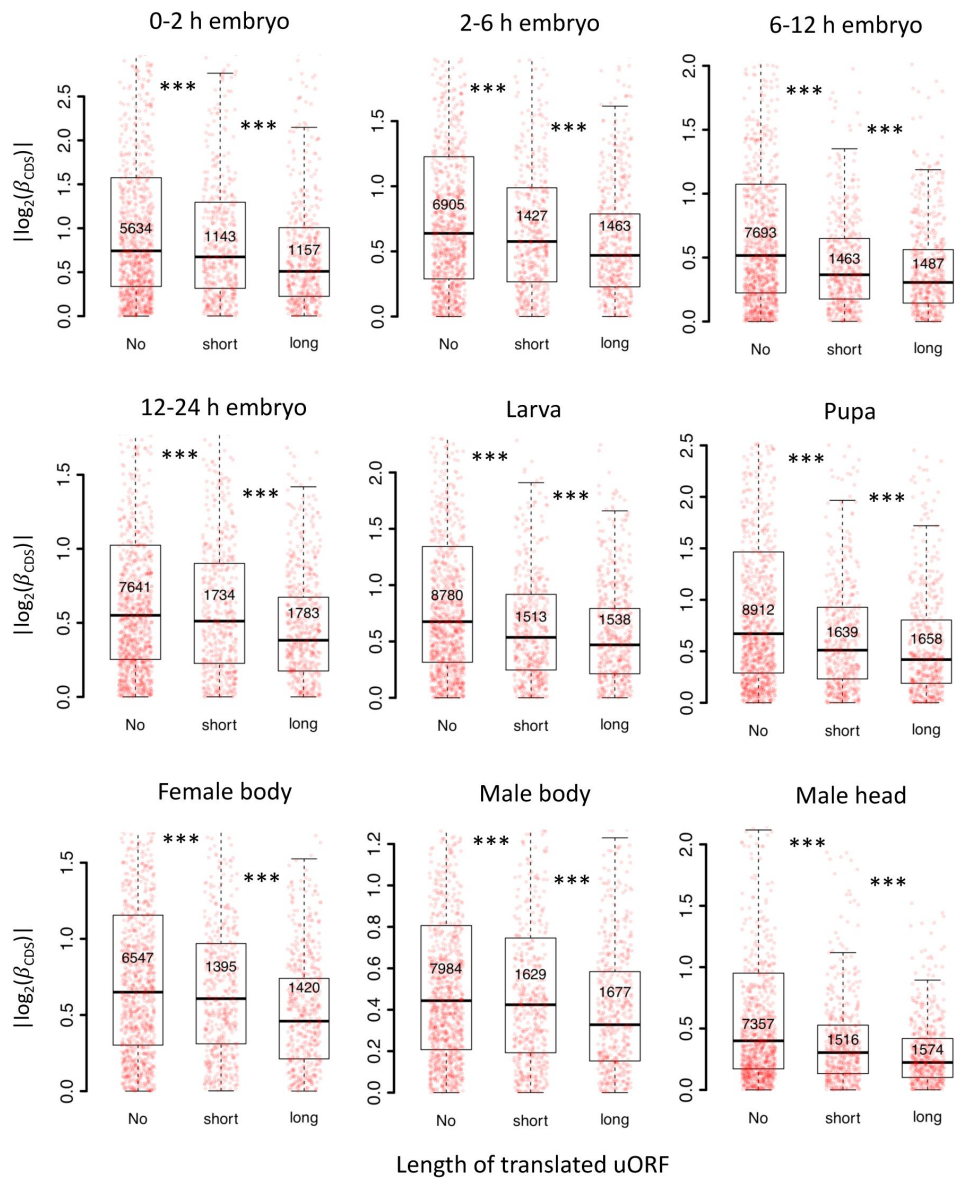


Figure S21.

Longer uORFs showed the stronger buffering effect.

Genes expressed in different stages/tissues (mRNA RPKM > 0.1 in both species) were classified into three classes according to the length of translated uORFs. Boxplots showing interspecific CDS TE variability $|\log_2(c_c)|$ of different gene classes. P values were calculated using Wilcoxon rank-sum tests between the neighboring groups. ***, $P < 0.001$.

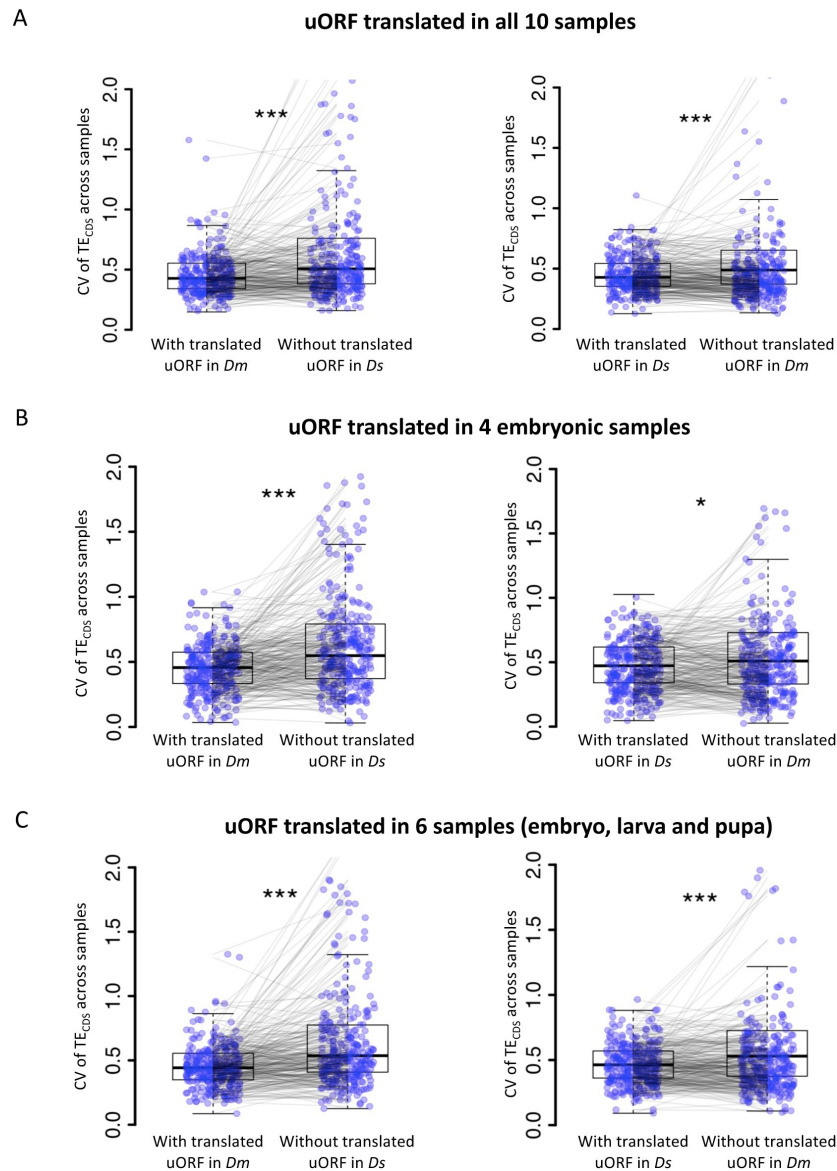


Figure S22.

uORFs reduce CDS translational fluctuation during *Drosophila* development under different cutoffs on defining “translated uORFs”.

The CV of TE_{CDS} across *Dm* (*D. melanogaster*) samples and *Ds* (*D. simulans*) samples was shown as boxplots. The selected genes harbor the translated uORFs in one of *Drosophila* species but its homologous gene without translated uORFs in another *Drosophila* species. Each pair of dots linked by a gray line represents a pair of homologous genes in *Dm* and *Ds*. (A) Translated uORF was defined as uORF translated ($TE > 0.1$) in all 10 samples. (B) Translated uORF was defined as uORF translated ($TE > 0.1$) in 4 embryonic stages. The CV of TE_{CDS} across these 4 stages was calculated. (C) Translated uORF was defined as uORF translated ($TE > 0.1$) in the 6 developmental stages including 4 embryonic stages, larva, and pupa. The CV of TE_{CDS} across these 6 stages was calculated. Wilcoxon signed-rank sum test. *, $P < 0.05$; ***, $P < 0.001$.

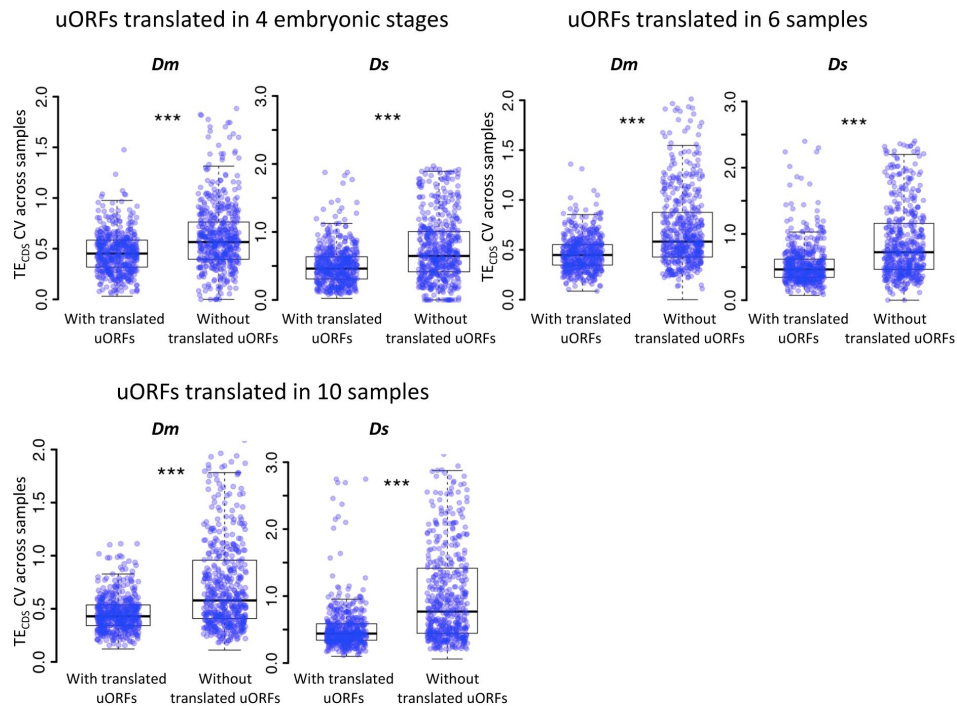


Figure S23.

uORFs could reduce CDS translational variation during *Drosophila* development under different cutoffs on defining “translated uORFs”.

Within each species, the CV of TE_{CDS} across different samples was measured for genes with translated uORFs and genes without translated uORFs, respectively. Each dot in the boxplot represents one gene. Translated uORF was defined as uORF translated (TE > 0.1) in at least 4 embryonic stages, six developmental stages (left) or translated or in all 10 samples. Wilcoxon rank-sum tests. ***, $P < 0.001$. *Dm*, *D. melanogaster*; *Ds*, *D. simulans*.

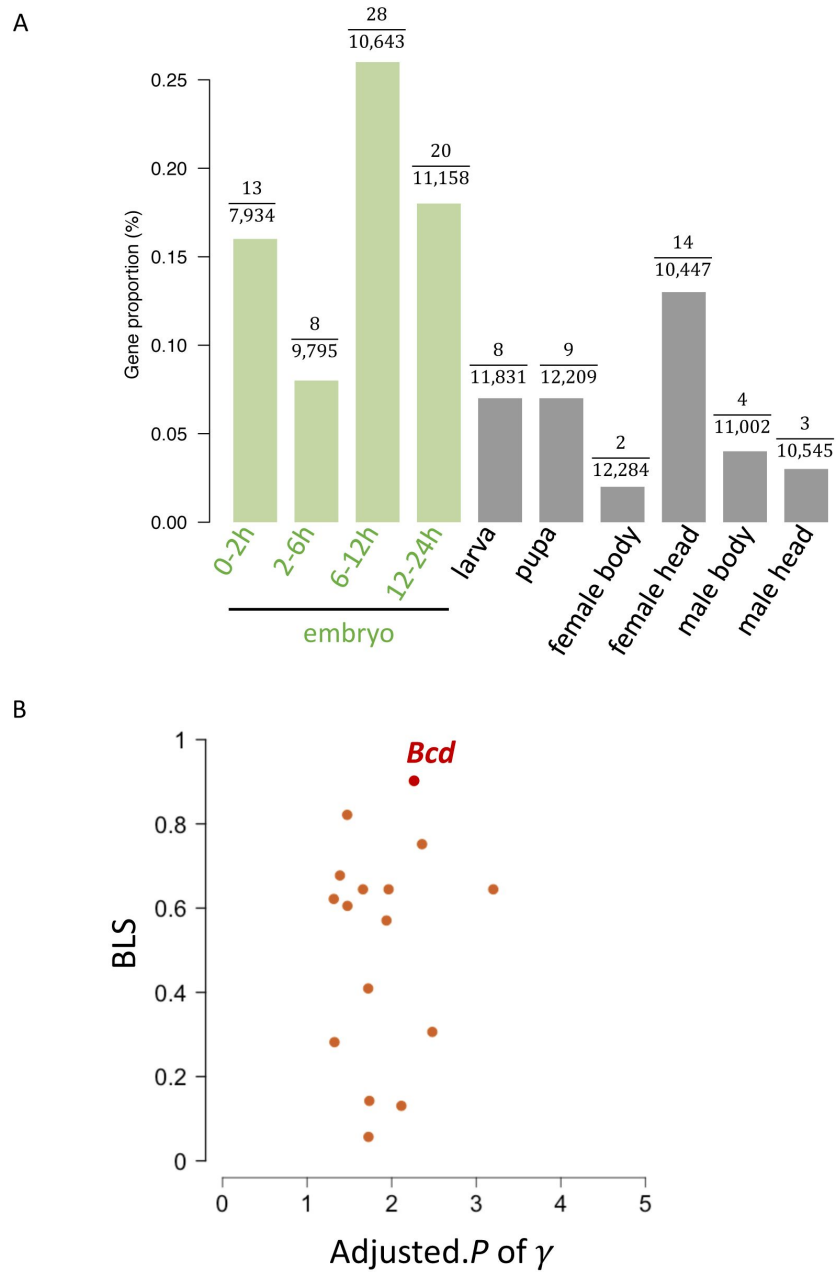


Figure S24.

(A) The proportion (%) of genes with buffering uORFs in all expressed genes across developmental stages/tissues. (B) The distribution of adjusted. P of γ (x-axis) and BLS (y-axis) of dominant uORFs in embryonic stages. The *bcd* was highlighted by red color.

Figure S25.

Overview of homozygous uORF-KO mutant screening.

(A) Procedures for the primary screening of CRISPR/Cas9-induced mutants. Red “-” denotes the *bcd*-uORF-KO allele. (B) Purification of the genetic background of homozygous mutants by back-crossing with WT lines for nine generations.

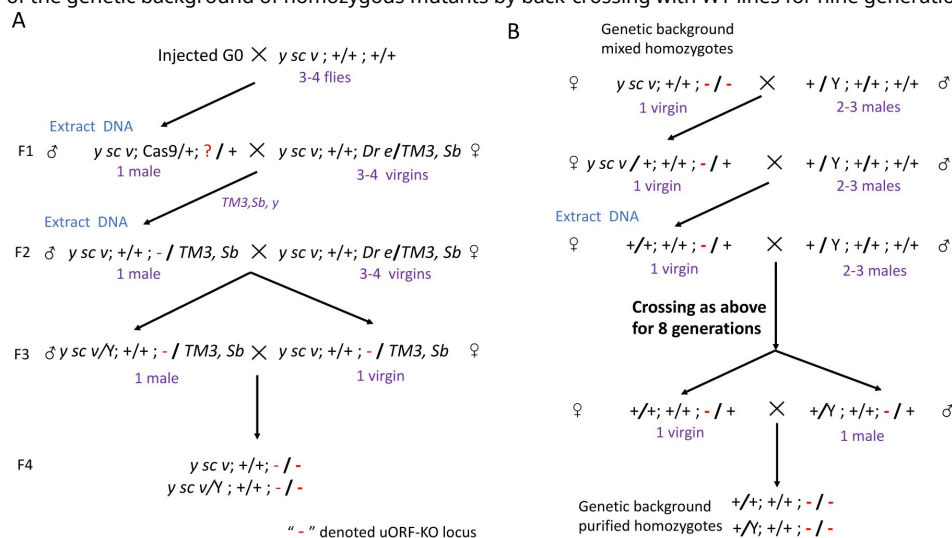


Figure S26.

Heatmap of sample-to-sample distances of RNA-Seq libraries.

Mut, mutant; WT, wild-type; rep1/rep2, biological replicate 1/biological replicate 2.

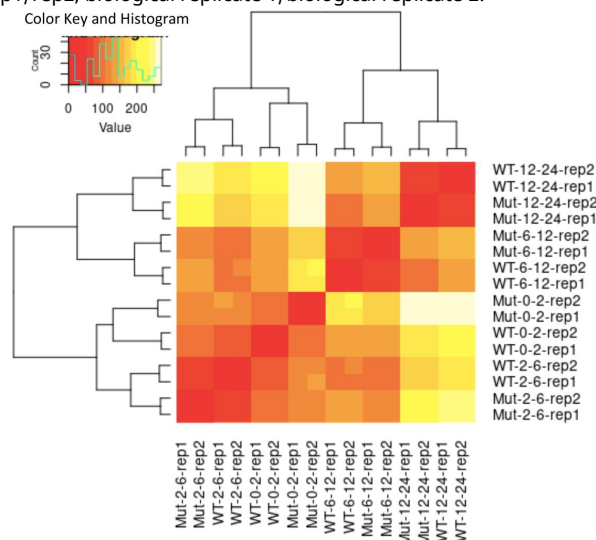


Figure S27.

(A) The enrichment of Bcd targets in each embryo stage at 29°C. Enrichment was calculated as the ratio of the Bcd target proportion in DEGs to the Bcd target proportion in non-DEGs. χ^2 tests were performed to compare the difference in Bcd target distribution in DEGs and non-DEGs. The number of DEGs and the number of Bcd targets in DEGs for each stage are provided above the bar. Asterisks indicate statistical significance (χ^2 tests; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; n.s., $P > 0.05$). (B) The expression [$\log_2(\text{fold change})$] levels of *bcd* and 3 of its targets, *Kr*, *btd*, and *otd*, in mutants compared to those in the WT in 4 embryonic stages. Asterisks are present at the top of bars only for $|\text{fold-changes}| > 1.5$ and adjusted P values < 0.05 (*, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$). (C) Expression of *bcd* and its targets (*Kr*, *btd*, *otd*) measured by RNA-Seq in uKO2/uKO2 mutant (red line) and WT (blue line) of *D. melanogaster*. The expression of each gene in each stage is shown as the RPKM value.

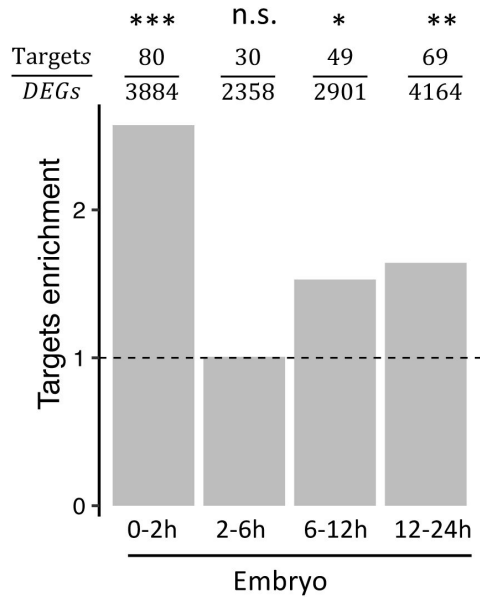


Figure S28.

The correlation of expression changes ($\log_2\text{FoldChange}$) between RNA-Seq (y-axis) and RT-qPCR (x-axis) of 20 *bcd* targets in four embryo stages in uKO2/uKO2 mutant compared to WT.

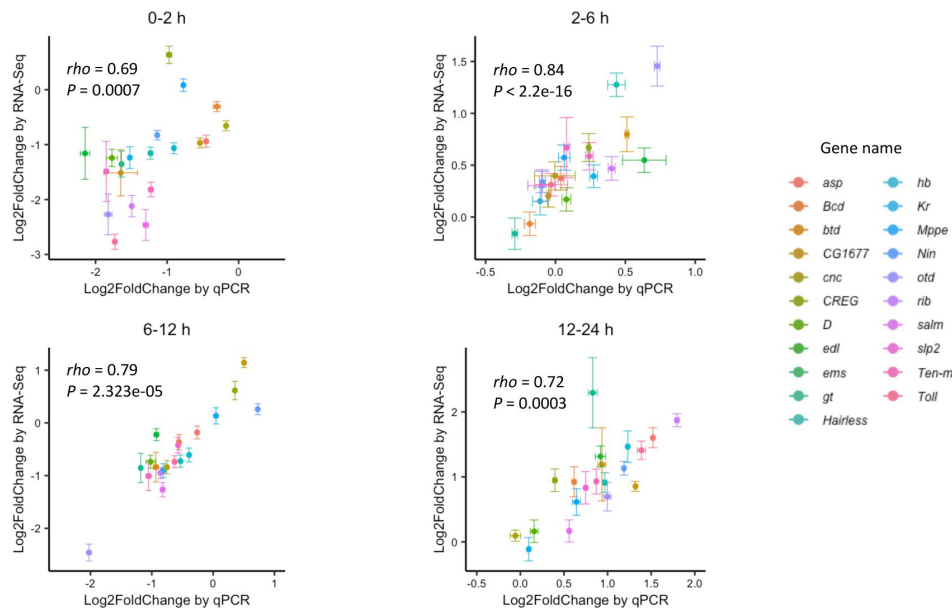


Figure S29.

Boxplots showing the TEs of conserved and species-specific uORFs in the liver and testis of *H. sapiens* and *M. mulatta*.

Wilcoxon rank-sum tests. ***, $P < 0.001$.

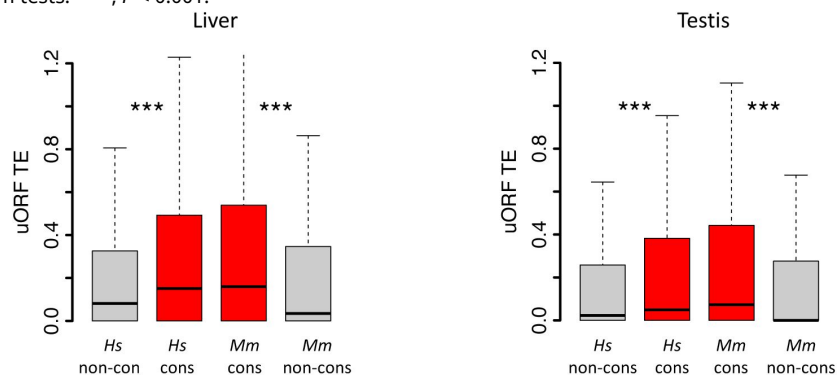


Figure S30.

Correlations of uORF TE and corresponding CDS TE in 3 tissues of *H. sapiens* and *M. mulatta*.

The bars represent Spearman's correlation coefficient (ρ). In all samples, we obtained P values < 0.001 . Data from brains are shown as examples in the right panel.

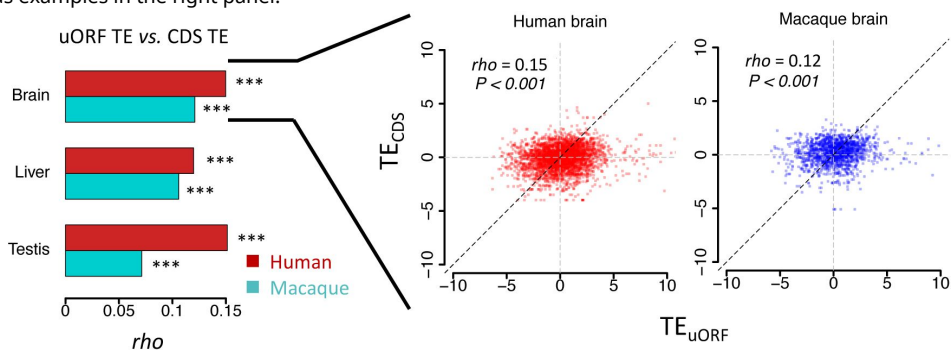


Figure S31.

Boxplots showing interspecific CDS TE variability $|\log_2(c_{CCS})|$ for three gene groups according to the length of their translated uORFs (No, short, long) in the liver (left) and testis (testis).

Wilcoxon rank-sum tests. ***, $P < 0.001$.

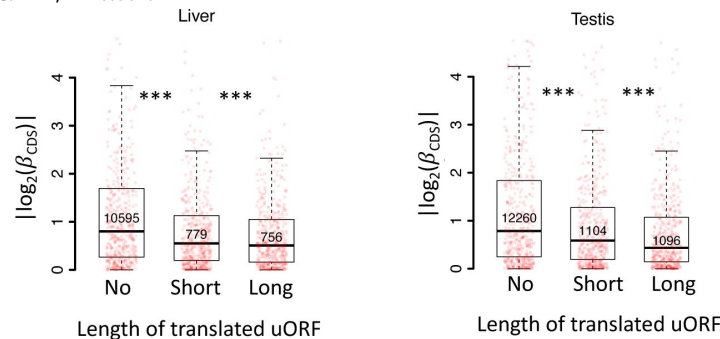
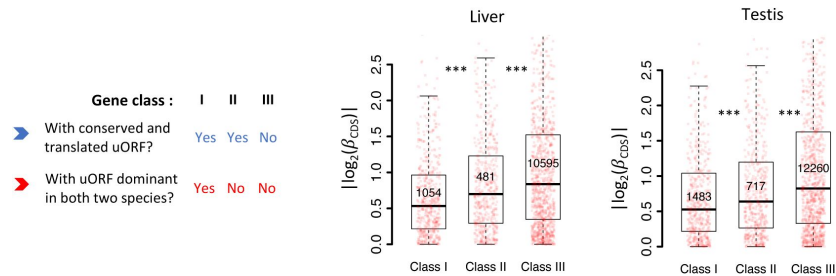


Figure S32.

Boxplots showing interspecific CDS TE variability $|\log_2(c_{CCS})|$ for different gene classes in the liver and testis.

Genes expressed in liver or testis (mRNA RPKM > 0.1 in both species) were classified into three classes according to whether a gene had a conserved and dominant translated uORF (TE > 0.1) in both species or not. Wilcoxon rank-sum tests. ***, $P < 0.001$.



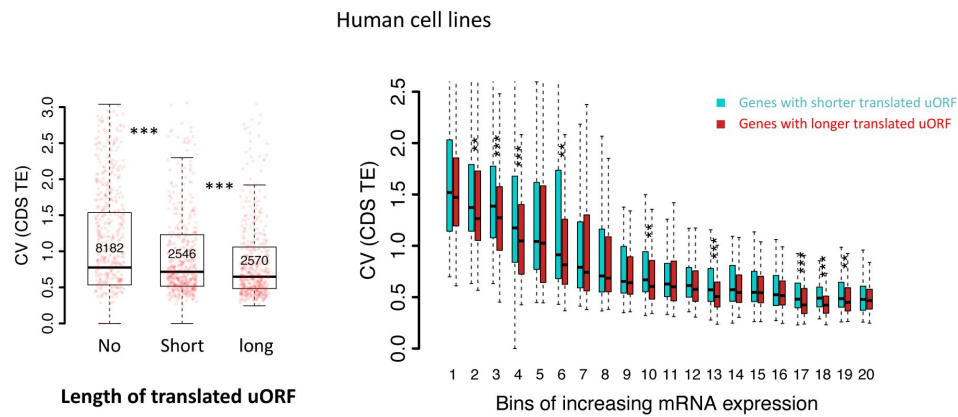


Figure S33.

Left panel: Boxplot showing the coefficients of variation (CVs) of CDS TEs among 69 human lymphoblastoid cell lines (LCLs) for three gene groups. Genes were classified into 3 groups according to the total length of translated uORFs. Right panel: boxplot showing the coefficients of variation (CVs) of CDS TEs among the human cell lines. Genes with translated uORFs were divided into 20 bins with increasing mRNA expression levels. In each bin, genes were divided into two fractions according to the total length of translated uORFs.

Symbol	Description	Value
Len_a	Length of 5'-leader before uORF	50 triplets
Len_u	Length of uORF	Default: 30 triplets Variable: 2, 10, 20, 30, 40, 50, 100 triplets
Len_b	Distance between uORF and CDS	50 triplets
Len_m	Length of CDS	500 triplets
Len_c	Length of 3'UTR	50 triplets
v_s	Probability of movement of a 40S ribosome to the next position in a single action	0.3*
v_{Eu}	Probability of movement of an 80S ribosome to the next position in uORF in a single action	0.3**
v_{EC}	Probability of movement of an 80S ribosome to the next position in CDS in a single action	0.5***
R_{in}	Probability of loading a new 40S ribosome at the 5'-terminus of the mRNA in a single action	1000 values generated from uniform distribution or exponential distribution
K_{up}	Dissociation probability of upstream 40S ribosome colliding with downstream 80S ribosome	0 for downstream dissociation; 1 for upstream or double dissociation
K_{down}	Dissociation probability of downstream 40S ribosome colliding with upstream 80S ribosome	0 for upstream dissociation; 1 for downstream or double dissociation
I_{uORF}	Probability of translation initiation at the uORF start codon in a single action	Single-uORF model: 0, 0.1, 0.2, 0.3, 0.4, 0.5 Double-uORF model: 0, 0.1, 0.2, 0.3, 0.4
I_{CDS}	Probability of translation initiation at the CDS start codon in a single action	Single-uORF model: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0 Double-uORF model: 0.9

*Adopted from Andreev *et al.*'s original ICIER model¹, corresponding to a movement rate of 5 triplets/s.

**Adopted from Andreev *et al.*'s original ICIER model¹, corresponding to a movement rate of 5 codons/s.

*** Considering that uORFs usually encode blocking peptides²⁻⁶ or contain stalling codons⁷⁻⁹, we set v_{Eu} slightly lower than v_{EC} .

Table S1.

Parameters used in our simulation.

Sample	Species*	Library type	Total reads (M)	After quality control (M) **	Unique mapping rate (%)	Multiple mapping rate (%)
0-2 h embryos	<i>D. melanogaster</i>	mRNA-Seq	61.01	52.74	93.14	2.84
2-6 h embryos	<i>D. melanogaster</i>	mRNA-Seq	59.44	52.07	93.26	3.24
6-12 h embryos	<i>D. melanogaster</i>	mRNA-Seq	53.56	45.11	92.33	3.09
12-24 h embryos	<i>D. melanogaster</i>	mRNA-Seq	61.18	50.64	91.89	4.31
Third-instar larvae	<i>D. melanogaster</i>	mRNA-Seq	58.28	45.46	88.53	6.09
P7-8 pupae	<i>D. melanogaster</i>	mRNA-Seq	52.99	41.16	92.65	4.46
Female adult heads	<i>D. melanogaster</i>	mRNA-Seq	21.48	16.87	93.00	3.27
Male adult heads	<i>D. melanogaster</i>	mRNA-Seq	16.32	14.18	91.17	4.24
Female adult bodies rep1	<i>D. melanogaster</i>	mRNA-Seq	16.84	11.20	88.83	5.76
Female adult bodies rep2	<i>D. melanogaster</i>	mRNA-Seq	24.94	19.37	88.73	6.00
Male adult bodies rep1	<i>D. melanogaster</i>	mRNA-Seq	19.79	16.01	91.62	6.39
Male adult bodies rep2	<i>D. melanogaster</i>	mRNA-Seq	11.24	7.64	89.86	6.69
0-2 h embryos	<i>D. melanogaster</i>	Ribo-Seq	64.81	51.96	90.36	6.31
2-6 h embryos	<i>D. melanogaster</i>	Ribo-Seq	53.14	41.19	69.47	16.92
6-12 h embryos	<i>D. melanogaster</i>	Ribo-Seq	66.24	46.76	86.02	8.94
12-24 h embryos	<i>D. melanogaster</i>	Ribo-Seq	63.7	47.45	88.96	6.30
Third-instar larvae	<i>D. melanogaster</i>	Ribo-Seq	39.53	22.13	58.83	6.82
P7-8 pupae	<i>D. melanogaster</i>	Ribo-Seq	41.53	20.33	80.92	5.51
Female adult heads	<i>D. melanogaster</i>	Ribo-Seq	63.94	39.90	94.39	2.92
Male adult heads	<i>D. melanogaster</i>	Ribo-Seq	58.22	23.87	93.36	3.24
Female adult bodies rep1	<i>D. melanogaster</i>	Ribo-Seq	44.11	19.85	84.20	6.36
Female adult bodies rep2	<i>D. melanogaster</i>	Ribo-Seq	41.95	18.73	70.74	11.33
Male adult bodies rep1	<i>D. melanogaster</i>	Ribo-Seq	42.56	18.98	79.39	8.65
Male adult bodies rep2	<i>D. melanogaster</i>	Ribo-Seq	39.52	28.66	81.19	6.93
0-2 h embryos	<i>D. simulans</i>	mRNA-Seq	19.92	18.33	90.85	5.82
2-6 h embryos	<i>D. simulans</i>	mRNA-Seq	13.88	7.47	88.32	7.46
6-12 h embryos	<i>D. simulans</i>	mRNA-Seq	21.13	18.64	89.26	7.06
12-24 h embryos	<i>D. simulans</i>	mRNA-Seq	16.17	13.96	87.77	8.47
Third-instar larvae	<i>D. simulans</i>	mRNA-Seq	17.56	12.69	89.88	6.57
P7-8 pupae	<i>D. simulans</i>	mRNA-Seq	23.87	19.04	89.92	5.62
Female adult heads	<i>D. simulans</i>	mRNA-Seq	17.13	15.11	89.70	6.39
Male adult heads	<i>D. simulans</i>	mRNA-Seq	17.72	15.96	90.73	6.15
Female adult bodies	<i>D. simulans</i>	mRNA-Seq	21.64	15.37	85.24	9.20
Male adult bodies	<i>D. simulans</i>	mRNA-Seq	21.54	17.34	87.45	8.84
0-2 h embryos	<i>D. simulans</i>	Ribo-Seq	63.38	49.48	80.09	10.81
2-6 h embryos	<i>D. simulans</i>	Ribo-Seq	53.46	38.09	77.27	13.72
6-12 h embryos	<i>D. simulans</i>	Ribo-Seq	57.76	43.84	81.58	10.81
12-24 h embryos	<i>D. simulans</i>	Ribo-Seq	50.55	35.05	83.55	8.21
Third-instar larvae	<i>D. simulans</i>	Ribo-Seq	67.69	35.40	86.44	7.10
P7-8 pupae	<i>D. simulans</i>	Ribo-Seq	61.74	40.45	87.45	4.76
Female adult heads	<i>D. simulans</i>	Ribo-Seq	62.52	26.18	86.51	4.58
Male adult heads	<i>D. simulans</i>	Ribo-Seq	65.07	28.99	89.62	3.96
Female adult bodies	<i>D. simulans</i>	Ribo-Seq	59.47	12.91	83.01	5.72
Male adult bodies	<i>D. simulans</i>	Ribo-Seq	53.55	18.35	87.44	5.61

* The mRNA-seq and Ribo-Seq datasets for *D. melanogaster* were previously generated from Zhang *et al.* ¹⁰. The datasets for *D. simulans* were generated in this study.

** After removing the reads mapped to rRNA, miscRNA (snoRNA, snRNA, rRNA, tRNA), the yeast genome, and the Wolbachia genome.

Table S2.

Mapping statistics of Ribo-Seq and matched mRNA-Seq libraries.

Sample	Condition	Stain	Stage	Total read-pairs (M)	Unique mapping reads (M)	Unique mapping rate (%)	Multiple mapping rate (%)
uKO2-0-2 h-embryo-29-rep1	29°C	Mutant	0-2 h	19.39	16.73	86.31	9.13
uKO2-0-2 h-embryo-29-rep2	29°C	Mutant	0-2 h	18.72	12.72	67.94	26.64
uKO2-2-6 h-embryo-29-rep1	29°C	Mutant	2-6 h	19.60	16.76	85.51	11.60
uKO2-2-6 h-embryo-29-rep2	29°C	Mutant	2-6 h	18.01	15.40	85.52	11.53
uKO2-6-12 h-embryo-29-rep1	29°C	Mutant	6-12 h	18.55	16.26	87.64	7.52
uKO2-6-12 h-embryo-29-rep2	29°C	Mutant	6-12 h	19.76	17.31	87.58	7.79
uKO2-12-24 h-embryo-29-rep1	29°C	Mutant	12-24 h	17.48	15.23	87.14	8.36
uKO2-12-24 h-embryo-29-rep2	29°C	Mutant	12-24 h	18.78	15.98	85.09	10.05
w1118-0-2 h-embryo-29-rep1	29°C	WT	0-2 h	19.10	17.24	90.25	6.87
w1118-0-2 h-embryo-29-rep2	29°C	WT	0-2 h	19.02	17.12	90.01	7.38
w1118-2-6 h-embryo-29-rep1	29°C	WT	2-6 h	18.54	15.60	84.11	12.40
w1118-2-6 h-embryo-29-rep2	29°C	WT	2-6 h	19.24	16.34	84.93	11.84
w1118-6-12 h-embryo-29-rep1	29°C	WT	6-12 h	18.61	15.87	85.28	10.50
w1118-6-12 h-embryo-29-rep2	29°C	WT	6-12 h	19.48	16.83	86.39	9.45
w1118-12-24 h-embryo-29-rep1	29°C	WT	12-24 h	18.93	16.44	86.85	6.96
w1118-12-24 h-embryo-29-rep2	29°C	WT	12-24 h	19.70	16.95	86.03	7.69
uKO2-0-2 h-embryo-25-rep1	25°C	Mutant	0-2 h	16.22	14.94	92.12	4.27
uKO2-0-2 h-embryo-25-rep2	25°C	Mutant	0-2 h	16.23	14.91	91.91	4.15
uKO2-2-6 h-embryo-25-rep1	25°C	Mutant	2-6 h	16.80	14.50	86.31	7.53
uKO2-2-6 h-embryo-25-rep2	25°C	Mutant	2-6 h	19.34	16.58	85.71	7.61
uKO2-6-12 h-embryo-25-rep1	25°C	Mutant	6-12 h	16.17	14.20	87.80	8.49
uKO2-6-12 h-embryo-25-rep2	25°C	Mutant	6-12 h	16.34	14.36	87.90	8.11
uKO2-12-24 h-embryo-25-rep1	25°C	Mutant	12-24 h	16.85	14.35	85.20	11.33
uKO2-12-24 h-embryo-25-rep2	25°C	Mutant	12-24 h	16.88	14.25	84.44	12.32
w1118-0-2 h-embryo-25-rep1	25°C	WT	0-2 h	17.29	15.74	91.06	5.27
w1118-0-2 h-embryo-25-rep2	25°C	WT	0-2 h	19.53	17.83	91.26	5.06
w1118-2-6 h-embryo-25-rep1	25°C	WT	2-6 h	19.23	16.85	87.62	7.76
w1118-2-6 h-embryo-25-rep2	25°C	WT	2-6 h	17.22	14.86	86.27	9.67
w1118-6-12 h-embryo-25-rep1	25°C	WT	6-12 h	19.04	16.02	84.15	11.76
w1118-6-12 h-embryo-25-rep2	25°C	WT	6-12 h	17.59	15.00	85.31	10.71
w1118-12-24 h-embryo-25-rep1	25°C	WT	12-24 h	16.25	14.03	86.35	10.29
w1118-12-24 h-embryo-25-rep2	25°C	WT	12-24 h	16.23	14.08	86.73	9.37

Table S4.

Mapping statistics of mRNA-Seq libraries for *bcd* uKO2/uKO2 mutant and WT flies.

Tissues	# of expressed uORFs	$\beta_u \neq 1$ (%)	# of expressed CDSs	$\beta_c \neq 1$ (%)	uORF-CDS pairs with $\beta_u > 1$			uORF-CDS pairs with $\beta_u < 1$		
					Total	$\gamma > 1$	$\gamma < 1$	Total	$\gamma > 1$	$\gamma < 1$
Brain	7,380	80 (1.08)	15,086	507 (3.36)	51	0	27	29	14	0
Liver	4,429	28 (0.63)	13,246	149 (1.24)	10	0	4	18	10	0
Testis	7,384	53 (0.72)	15,134	272 (1.80)	33	0	9	20	5	0

Only uORFs and CDSs with an mRNA RPKM > 0.1 in both *H. sapiens* and *M. mulatta* were considered in each sample pair in the analysis. $\beta_u = TE_{uORF,macaque}/TE_{uORF,human}$, is the fold change in TE_{uORF} in *M. mulatta* relative to *H. sapiens* for each sample. $\beta_c = TE_{CDS,macaque}/TE_{CDS,human}$, is the fold change in TE_{CDS} in *M. mulatta* relative to *H. sapiens* for each sample. $\gamma = \beta_c/\beta_u$. For each CDS-uORF pair, $\beta_u > 1$ and $\gamma < 1$ or $\beta_u < 1$ and $\gamma > 1$ means that the magnitude of TE change is lower for a CDS than a uORF. The statistical significance of β_c , β_u , and γ were all determined at an FDR < 0.05.

Table S5.

Numbers of genes showing different magnitudes of TE changes between uORFs and CDS at the interspecific level, *H. sapiens* and *M. mulatta*.

Primer	Sequence (5'→3')	Application
Bcd-uORF-sgRNA-F	ATCGCAAAAACGCAAAATGT	sgRNA for Bcd uORF knock-out
Bcd-uORF-sgRNA-R	ACATTTTGCGTTTTTGCGAT	
Bcd-genotyping-F	GCTTTGCCGTACTGTTTCGAT	Primers used for Bcd genotyping
Bcd-genotyping-R	AACTGAAGCTGCGGATGTTG	
Bcd-qPCR-F	GATGTATCTGGGTGGCTGCT	RT-qPCR primers used for Bcd mRNA P-to-M ratio quantification
Bcd-qPCR-R	CCGAAATGTGGGACGATAAC	
Rp49-qPCR-F	CACTTCATCCGCCACCAGTC	RT-qPCR primers used for rp49 as inter reference
Rp49-qPCR-R	CGCTTGTTTCGATCCGTAACC	
Bcd-5utr-PC1-wF1	CATCCCTAAATAACGGCACT	Used for WT and uATG-mutated <i>bcd</i> 5'UTR amplification
Bcd-5utr-PC1-wR	CGCGAACTTATTTGCGCCACATTTTGCGTTTTTG	
Bcd-5utr-PC1-mR1	CGCGAACTTATTTGCGCCACTTATTTTGCGTTTTTG	
Bcd-5utr-PC1-mR2	CGCGAACTTATTTGCGAACTTATTTGCGATCCACTGCT	
Bcd-5utr-PC2-wF1	CGAAATAAGTTCGCGAGCGT	
Bcd-5utr-PC2-wR1	GTACACCTTGGAAGCCGGCGGTTGCGCCATTTTCC	
btd_qP_F	CACCAATTACCGCCCGTATG	RT-qPCR Primers for Bcd target genes
btd_qP_R	CGGCCAATACCTTCTTGCTC	
kr_qP_F	CCAGTGCGAGCTCTCCTAAT	
kr_qP_R	GCGAGATGGATCCTTGAGAG	
otd_qP_F	CTACAACATGGGCCATTTCGG	
otd_qP_R	GCGACATGTAATCCAGGCTG	
Toll_qP_F	TTACTTCAGTGCCTCGCTCA	
Toll_qP_R	CCAGAGCCGATAACATTGCC	
hb_qP_F	GGAGAAGGAGCACGATCAGA	
hb_qP_R	TGCCGTGCGAATTGTAGATG	
Hairless_qP_F	GCATACATTCCCTCGCCTTC	

Table S6.

The primer sequences used in this study.

Hairless_qP_R	CGCTCCTGTCTGATAGTGGT
salm_qP_F	GCCCCATTCAAGTGCAGGATC
salm_qP_R	GAAGTTTCTCATCGGCGGAC
D_qP_F	CAAGAGGGTCACATCAAGCG
D_qP_R	CAGAGTTGTGCATCTTGGGG
Ten-m_qP_F	CCACCGCATCCTAAAGCATG
Ten-m_qP_R	CTCTTTCTCCTCCTCGGTCC
rib_qP_F	TTCGCCCTACACAGATCGTT
rib_qP_R	CGCACATCATATCCGCTCTG
CG1677_qP_F	TTGGCCGGGGATCATATGAA
CG1677_qP_R	GGTGGAATACGACGAGTTGC
Nin_qP_F	TAATGACCGCGATCACTGGA
Nin_qP_R	GCACCAGTTCAAGTTCCGTT
gt_qP_F	CCCTCGCCGTATCAAACAAG
gt_qP_R	GGGCGGCATACAGAAGATTG
asp_qP_F	AGCTACCATTTTCCCGGACA
asp_qP_R	CATCTGGAGTGATTCGACGC
CREG_qP_F	CTGATTTTGGCCCTCGTGAG
CREG_qP_R	GGCGATCTTGGCGTGATTA
slp2_qP_F	GCAAGCCGGTGAAGGATAAG
slp2_qP_R	ACTCTGACGAATGGCCATCA
edl_qP_F	TCAAATGCTGGACAAGTGCC
edl_qP_R	GCGCTCATCAGTCGATTGTT
Mppe_qP_F	TGGACTTTAGCAACCCGGAT
Mppe_qP_R	CGCTGATGATGGCTCTCAAC
cnc_qP_F	GGAAACAGCAGCAACTTGGA
cnc_qP_R	CCGGCACTGAAATGGGTATG
ems_qP_F	ATCTATCCGGTCCCCAGAGT
ems_qP_R	CCTGGGATCAAGGTGAGTGT

Table S6. (continued)

Data Availability Statement

All deep-sequencing data generated in this study, including single-ended mRNA-Seq and Ribo-Seq data of 10 developmental stages and tissues of *Drosophila simulans* and paired-end mRNA-Seq data of 0–2 h, 2–6 h, 6–12 h, and 12–24 h *Drosophila melanogaster* embryos, were deposited in the China National Genomics Data Center Genome Sequence Archive (GSA) under accession numbers CRA003198, CRA007425, and CRA007426. The mRNA-Seq and Ribo-Seq data for the different developmental stages and tissues of *Drosophila melanogaster* were published in our previous paper ²³ and were deposited in the Sequence Read Archive (SRA) under accession number SRP067542. All original code has been deposited on GitHub: https://github.com/lujlab/uORF_buffer; https://github.com/lujlab/Buffer_eLife2025.

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Additional files

Supplemental figures and tables

Table S3. Genes with uORFs showing strong evidence of translational buffering.

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Author information

Yuanqiang Sun[#]

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China
ORCID iD: [0009-0007-6967-2234](https://orcid.org/0009-0007-6967-2234)

[#]Equal contribution

Yuange Duan[#]

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China
ORCID iD: [0000-0003-2311-9859](https://orcid.org/0000-0003-2311-9859)

#Equal contribution

Peixiang Gao

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China

Chenlu Liu

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China
ORCID iD: [0000-0002-8993-0145](https://orcid.org/0000-0002-8993-0145)

Kaichun Jin

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China

Shengqian Dou

Eye Institute of Shandong First Medical University, State Key Laboratory Cultivation Base, Shandong Provincial Key Laboratory of Ophthalmology, Qingdao, China

Wenxiong Tang

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China

Hong Zhang

College of Ecology, Lanzhou University, Lanzhou, China

Jian Lu

State Key Laboratory of Gene Function and Modulation Research, Center for Bioinformatics, School of Life Sciences, Peking University, Beijing, China, Beijing Advanced Center of RNA Biology (BEACON), Peking University, Beijing, China, Southwest United Graduate School, Kunming, China
ORCID iD: [0000-0002-4409-1667](https://orcid.org/0000-0002-4409-1667)

For correspondence: luj@pku.edu.cn

Editors

Reviewing Editor

Wenfeng Qian

Chinese Academy of Sciences, Beijing, China

Senior Editor

Alan Moses

University of Toronto, Toronto, Canada

Reviewer #1 (Public review):

Summary:

The authors set out to explore the role of upstream open reading frames (uORFs) in stabilizing protein levels during *Drosophila* development and evolution. By utilizing a modified ICIER model for ribosome translation simulations and conducting experimental

validations in *Drosophila* species, the study investigates how uORFs buffer translational variability of downstream coding sequences. The findings reveal that uORFs significantly reduce translational variability, which contributes to gene expression stability across different biological contexts and evolutionary timeframes.

Strengths:

- (1) The study introduces a sophisticated adaptation of the ICIER model, enabling detailed simulation of ribosomal traffic and its implications for translation efficiency.
- (2) The integration of computational predictions with empirical data through knockout experiments and transcriptome analysis in *Drosophila* provides a compelling validation of the model's predictions.
- (3) By demonstrating the evolutionary conservation of uORFs' buffering effects, the study provides insights that are likely applicable to a wide range of eukaryotes.

Weaknesses:

- (1) Although the study is technically sound, it does not clearly articulate the mechanisms through which uORFs buffer translational variability. A clearer hypothesis detailing the potential molecular interactions or regulatory pathways by which uORFs influence translational stability would enhance the comprehension and impact of the findings.
- (2) The study could be further improved by a discussion regarding the evolutionary selection of uORFs. Specifically, it would be beneficial to explore whether uORFs are favored evolutionarily primarily for their role in reducing translation efficiency or for their capability to stabilize translation variability. Such a discussion would provide deeper insights into the evolutionary dynamics and functional significance of uORFs in genetic regulation.

Comments on revisions:

The authors have adequately addressed my previous concerns.

<https://doi.org/10.7554/eLife.104074.2.sa2>

Reviewer #2 (Public review):

uORFs, short open reading frames located in the 5' UTR, are pervasive in genomes. However, their roles in maintaining protein abundance are not clear. In this study, the authors propose that uORFs act as "molecular dam", limiting the fluctuation of the translation of downstream coding sequences. First, they performed *in silico* simulations using an improved ICIER model, and demonstrated that uORF translation reduces CDS translational variability, with buffering capacity increasing in proportion to uORF efficiency, length, and number. Next, they analysed the transcriptome between two related *Drosophila* species, revealing that genes with uORFs exhibit smaller fluctuations in translation between the two species and across different developmental stages within the same species. Moreover, they identified that *bicoid*, a critical gene for *Drosophila* development, contains a uORF with substantial changes in translation efficiency. Deleting this uORF in *Drosophila melanogaster* significantly affected its gene expression, hatching rates, and survival under stress conditions. Lastly, by leveraging public Ribo-seq data, the authors showed that the buffering effect of uORFs is also evident between primates and within human populations. Collectively, the study significantly advances our understanding of how uORFs regulate the translation of downstream coding sequences at the genome-wide scale, as well as during development and evolution. It would be particularly interesting to explore whether similar buffering functions are conserved in other organisms, and whether their regulatory effects could be harnessed for practical applications, such as improving crop traits or benefiting human health.

Comments on revisions:

The authors have fully addressed all of my concerns, and the revisions have substantially improved the manuscript. I have no further comments.

<https://doi.org/10.7554/eLife.104074.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors set out to explore the role of upstream open reading frames (uORFs) in stabilizing protein levels during Drosophila development and evolution. By utilizing a modified ICIER model for ribosome translation simulations and conducting experimental validations in Drosophila species, the study investigates how uORFs buffer translational variability of downstream coding sequences. The findings reveal that uORFs significantly reduce translational variability, which contributes to gene expression stability across different biological contexts and evolutionary timeframes.

We thank the reviewer for carefully reading our manuscript and providing thoughtful and constructive feedback. We believe the manuscript has been significantly improved by incorporating your suggestions. Please find our detailed responses and corresponding revisions below.

Strengths:

- (1) The study introduces a sophisticated adaptation of the ICIER model, enabling detailed simulation of ribosomal traffic and its implications for translation efficiency.*
- (2) The integration of computational predictions with empirical data through knockout experiments and transcriptome analysis in Drosophila provides a compelling validation of the model's predictions.*
- (3) By demonstrating the evolutionary conservation of uORFs' buffering effects, the study provides insights that are likely applicable to a wide range of eukaryotes.*

We appreciate your positive feedback and thoughtful summary of the strengths of our study.

Weaknesses:

- (1) Although the study is technically sound, it does not clearly articulate the mechanisms through which uORFs buffer translational variability. A clearer hypothesis detailing the potential molecular interactions or regulatory pathways by which uORFs influence translational stability would enhance the comprehension and impact of the findings.*

Thanks for your constructive comments. In the Discussion section of our previous submission (Original Lines 470-489), we proposed that uORFs function as “molecular dams” to smooth out fluctuations in ribosomal flow toward downstream CDS regions, primarily via mechanisms involving ribosome collision and dissociation. To further address your concern, we have expanded the Discussion and included a new model figure (Fig. 9) to more clearly

articulate the potential biological and mechanistic basis by which translating 80S ribosomes may induce the dissociation of 40S ribosomes. The revised section (Lines 540–557) now reads:

“Ribosome slowdown or stalling on mRNA due to rare codons [56,96-98] or nascent blocking peptides [99-102] frequently triggers ribosome collisions genome-wide [103-105]. Such collisions, especially among elongating 80S ribosomes, often activate ribosome quality control (RQC) pathways that recognize collision interfaces on the 40S subunit, leading to ribosomal subunit dissociation and degradation [106-108]. In mammals, ZNF598 specifically identifies collided ribosomes to initiate ubiquitin-dependent protein and mRNA quality control pathways [109-113]. Analogously, yeast employs Hel2-mediated ubiquitination of uS10, initiating dissociation via the RQC-trigger complex (RQT) [114]. Furthermore, the human RQT (hRQT) complex recognizes ubiquitinated ribosomes and induces subunit dissociation similarly to yeast RQT [115]. However, transient ribosome collisions can evade RQC by promoting resumed elongation through mechanical force provided by trailing ribosomes, thereby mitigating stalling [116]. Beyond 80S collisions, evidence increasingly highlights a distinct collision type involving scanning 40S subunits or pre-initiation (43S) complexes. Recently, an initiation RQC pathway (iRQC) targeting the small ribosomal subunit (40S) has been described, particularly involving collisions between scanning 43S complexes or between stalled 43S and elongating 80S ribosomes (Figure 9B) [117,118]. During iRQC, E3 ubiquitin ligase RNF10 ubiquitinates uS3 and uS5 proteins, resulting in 40S degradation [118]. This mechanism aligns closely with our ICIER model, proposing collision-driven 43S dissociation in the 5' UTRs. Future studies exploring these mechanisms in greater detail will clarify how uORFs modulate translational regulation through buffering effects.”

(2) The study could be further improved by a discussion regarding the evolutionary selection of uORFs. Specifically, it would be beneficial to explore whether uORFs are favored evolutionarily primarily for their role in reducing translation efficiency or for their capability to stabilize translation variability. Such a discussion would provide deeper insights into the evolutionary dynamics and functional significance of uORFs in genetic regulation.

Thank you for this insightful suggestion. We agree that understanding whether uORFs are evolutionarily favored for their role in translational repression or for their capacity to buffer translational variability is a compelling and unresolved question. Our study suggests that translational buffering, rather than translational repression alone, can also drive evolutionary selection favoring uORFs, although it remains challenging to empirically disentangle these functions due to their inherent linkage. We have expanded the discussion in the revised manuscript to address this point in more detail (Lines 494-513), which is reproduced as follows:

“Previous studies have shown that a significant fraction of fixed uORFs in the populations of *D. melanogaster* and humans were driven by positive Darwinian selection [63,67], suggesting active maintenance through adaptive evolution rather than purely neutral or deleterious processes. While uORFs have traditionally been recognized for their capacity to attenuate translation of downstream CDSs, accumulating evidence now underscores their critical role in stabilizing gene expression under fluctuating cellular and environmental conditions [43,55,56]. Whether the favored evolutionary selection of uORFs acts primarily through their role in translational repression or translational buffering remains a compelling yet unresolved question, as these two functions are inherently linked. Indeed, highly conserved uORFs tend to be translated at higher levels, resulting not only in stronger inhibition of CDS translation [34,45,67] but also in a more pronounced buffering effect, as demonstrated in this study. This buffering capacity of uORFs potentially provides selective advantages by reducing fluctuations in protein synthesis, thus minimizing gene-expression noise and enhancing cellular homeostasis. This suggests that selection may favor uORFs that contribute to translational robustness, a hypothesis supported by findings in yeast and mammals showing

that uORFs are significantly enriched in stressresponse genes and control the translation of certain master regulators of stress responses [41,42,94,95]. Our study suggests that translational buffering, rather than translational repression alone, can also drive evolutionary selection favoring uORFs, although it remains challenging to empirically disentangle these functions. Future comparative genomic analyses, coupled with experimental approaches such as ribosome profiling and functional mutagenesis, will be crucial in elucidating the precise evolutionary forces driving uORF conservation and adaptation.”

Reviewer #2 (Public review):

uORFs, short open reading frames located in the 5' UTR, are pervasive in genomes. However, their roles in maintaining protein abundance are not clear. In this study, the authors propose that uORFs act as "molecular dam", limiting the fluctuation of the translation of downstream coding sequences. First, they performed in silico simulations using an improved ICIER model, and demonstrated that uORF translation reduces CDS translational variability, with buffering capacity increasing in proportion to uORF efficiency, length, and number. Next, they analyzed the translome between two related Drosophila species, revealing that genes with uORFs exhibit smaller fluctuations in translation between the two species and across different developmental stages within the same species. Moreover, they identified that bicoid, a critical gene for Drosophila development, contains a uORF with substantial changes in translation efficiency. Deleting this uORF in Drosophila melanogaster significantly affected its gene expression, hatching rates, and survival under stress condition. Lastly, by leveraging public Ribo-seq data, the authors showed that the buffering effect of uORFs is also evident between primates and within human populations. Collectively, the study advances our understanding of how uORFs regulate the translation of downstream coding sequences at the genome-wide scale, as well as during development and evolution.

The conclusions of this paper are mostly well supported by data, but some definitions and data analysis need to be clarified and extended.

We thank the reviewer for the thoughtful and constructive review. Your summary accurately captures the key findings of our study. We have carefully addressed all your concerns in the revised manuscript, and we believe it has been significantly improved based on your valuable input.

(1) There are two definitions of translation efficiency (TE) in the manuscript: one refers to the number of 80S ribosomes that complete translation at the stop codon of a CDS within a given time interval, while the other is calculated based on Ribo-seq and mRNA-seq data (as described on Page 7, line 209). To avoid potential misunderstandings, please use distinct terms to differentiate these two definitions.

Thank you for highlighting this important point, and we apologize for the confusion. The two definitions of translation efficiency (TE) in our manuscript arise from methodological differences between simulation and experimental analyses. To clarify, in the revised manuscript, we use “translation rate” in the context of simulations to describe the number of 80S ribosomes completing translation at the CDS stop codon per unit time. We retain the conventional “translation efficiency (TE)” for Ribo-seq–based measurements.

In this revised manuscript, we have added a more detailed explanation of TE in the revised manuscript (Lines 202–206), which now reads:

“For each sample, we followed established procedures [62–66] to calculate the translational efficiency (TE) for each feature (CDS or uORF). TE serves as a proxy for the translation rate at

which ribosomes translate mRNA into proteins, typically quantified by comparing the density of ribosome-protected mRNA fragment (RPF) to the mRNA abundance for that feature (see Materials and Methods)."

(2) Page 7, line 209: "The translational efficiencies (TEs) of the conserved uORFs were highly correlated between the two species across all developmental stages and tissues examined, with Spearman correlation coefficients ranging from 0.478 to 0.573 (Fig. 2A)." However, the authors did not analyze the correlation of translation efficiency of conserved CDSs between the two species, and compare this correlation to the correlation between the TEs of CDSs. These analyses will further support the authors conclusion regarding the role of conserved uORFs in translation regulation.

In the revised manuscript, we have incorporated a comparison of translational efficiency (TE) correlations for conserved CDSs between the two species. We found that CDSs exhibit significantly higher interspecific TE correlations than uORFs, with Spearman's ρ ranging from 0.588 to 0.806. This suggests that uORFs tend to show greater variability in TE than CDSs, consistent with our model in which uORFs buffer fluctuations in downstream CDS translation. The updated results were included in the revised manuscript (Lines 223-227) as follows:

"In contrast, TE of CDSs exhibited a significantly higher correlation between the two species in the corresponding samples compared to that of uORFs, with Spearman's ρ ranging from 0.588 to 0.806 ($P = 0.002$, Wilcoxon signed-rank test; Figure 2A). This observation is consistent with our simulation results, which indicate that uORFs experience greater translational fluctuations than their downstream CDSs."

(3) Page 8, line 217: "Among genes with multiple uORFs, one uORF generally emerged as dominant, displaying a higher TE than the others within the same gene (Fig. 2C)." The basis for determining dominance among uORFs is not explained and this lack of clarification undermines the interpretation of these findings.

Thank you for pointing this out. We apologize for the confusion. In our study, a "dominant" uORF is defined as the one with the highest translation efficiency (TE) among all uORFs within the same gene. This designation is based solely on TE, which we consider a key metric for uORF activity, as it directly reflects translational output and potential regulatory impact. We have revised the manuscript to clarify this definition (Lines 232-244), now stating:

"Among genes with multiple uORFs, we defined the uORF with the highest TE as the dominant uORF for that gene, as TE is one of the most relevant metrics for assessing uORF function 45,67..... These results suggest that genes with multiple uORFs tend to retain the same dominant uORF across developmental stages, indicating that the dominant uORFs may serve as the key translational regulator of the downstream CDS."

(4) According to the simulation, the translation of uORFs should exhibit greater variability than that of CDSs. However, the authors observed significantly fewer uORFs with significant TE changes compared to CDSs. This discrepancy may be due to lower sequencing depth resulting in fewer reads mapped to uORFs. Therefore, the authors may compare this variability specifically among highly expressed genes.

Thank you for this thoughtful observation. We agree that the lower proportion of uORFs showing significant TE changes compared to CDSs, as reported in Table 1, appears inconsistent with our conclusion that uORFs exhibit greater translational variability. However, this discrepancy is largely attributable to differences in sequencing depth and feature length—uORFs are generally much shorter and more weakly expressed than CDSs, resulting in fewer mapped reads and reduced statistical power (Figure S18A).

To address this issue, we first followed your suggestion and restricted our analysis to genes with both mRNA and RPF RPKM values above the 50th percentile in *D. melanogaster* and *D. simulans*. While this filtering increased the total proportion of features with significant TE changes (due to improved read coverage), the proportion of significant uORFs still remained lower than that of CDSs (Table R1). This suggests that even among highly expressed genes, the disparity in read counts between uORFs and CDSs persists (Figure S18B), and thus the issue is not fully resolved.

To better capture biological relevance, we compared the absolute values of $\log_2(\text{TE changes})$ between *D. melanogaster* and *D. simulans* for uORFs and their corresponding CDSs. Across all samples, uORFs consistently exhibit larger TE shifts than their downstream CDSs, supporting our model that uORFs act as translational buffers (Figure 3B).

We have made relevant changes to report the new analysis in this revised manuscript. Specifically, in our original submission, we stated this observation with the sentence “The smaller number of uORFs showing significant TE changes compared to CDSs between *D. melanogaster* and *D. simulans* likely reflects their shorter length and reduced statistical power, rather than indicating that uORFs are less variable in translation than CDSs.” To make this point clearer, in the revised version (Lines 275-284), we rephrased this sentence which read as follows:

“Note that due to their shorter length and generally lower TE, uORFs had considerably lower read counts than CDSs, limiting the statistical power to detect significant interspecific TE differences for uORFs. This trend consistently holds whether analyzing all expressed uORFs (Figure S18A) or only highly expressed genes (Figure S18B). Thus, the fewer uORFs showing significant TE divergence likely reflects lower read counts and statistical sensitivity rather than reduced translational variability relative to CDSs. In fact, the absolute values of $\log_2(\text{fold change})$ of TE for uORFs between *D. melanogaster* and *D. simulans* were significantly greater than those observed for corresponding CDSs across all samples ($P < 0.001$, Wilcoxon signed-rank test; Figure 3B), suggesting that the magnitude of

TE changes in CDSs is generally smaller than that in uORFs, due to the buffering effect of uORF.”

Author response table 1.

Proportion of uORFs and CDSs with significant TE changes before and after selecting HEGs

Sample types	Original				Highly expressed genes			
	# of expressed uORFs	$\beta_u \neq 1$ (%)	# of expressed CDSs	$b_c \neq 1$ (%)	# of expressed uORFs	$\beta_u \neq 1$ (%)	# of expressed CDSs	$b_c \neq 1$ (%)
0-2 h embryo	7,704	1,193 (15.49)	7,934	4,189 (52.80)	3,531	703 (19.90)	2760	1,393 (50.47)
2-6 h embryo	7,822	567 (7.25)	9,795	3,063 (31.27)	4,666	456 (9.77)	3910	1,926 (49.26)
6-12 h embryo	10,400	1,040 (10.00)	10,643	2,924 (27.47)	5,718	866 (15.15)	4271	1,544 (36.15)
12-24 h embryo	11,365	535 (4.71)	11,158	3,537 (31.70)	5,454	423 (7.76)	4304	1,852 (43.03)
Larva	10,008	464 (4.64)	11,831	3,554 (30.04)	5,247	377 (7.18)	4546	2,149 (47.27)
Pupa	12,309	635 (5.16)	12,209	4,087 (33.48)	6,028	480 (7.96)	4487	2,196 (48.94)
Male body	10,894	197 (1.81)	12,284	2,432 (19.80)	4,422	135 (3.05)	4461	1,390 (31.16)
Male head	10,904	144 (1.32)	10,447	1,151 (11.02)	5,644	123 (2.18)	3981	605 (15.20)
Female body	9,809	340 (3.47)	11,002	3,605 (32.77)	4,231	255 (6.03)	4064	2,047 (50.37)
Female head	10,935	332 (3.04)	10,545	1,270 (12.04)	5,510	269 (4.88)	4019	618 (15.38)

(5) If possible, the author may need to use antibodies against bicoid to test the effect of ATG deletion on bicoid expression, particularly under different developmental stages or growth conditions.

According to the authors' conclusions, the deletion mutant should exhibit greater variability in bicoid protein abundance. This experiment could provide strong support for the proposed mechanisms.

Thank you for this excellent suggestion. We fully agree that testing Bcd protein levels across developmental stages or stress conditions using antibodies would be a strong validation of our model, which predicts greater variability in Bcd protein abundance upon uORF deletion.

In fact, we attempted such experiments in both wild-type and mutant backgrounds. However, we encountered substantial difficulties in obtaining a reliable anti-Bcd antibody. Some Bcd antibodies referenced in the published literature were homemade and often shared among research groups as gifts [1-3] and some commercially available antibodies cited in previous studies are no longer supplied by vendors [4-6]. We managed to obtain a custom-made antibody from Professor Feng Liu, but unfortunately, it produced inconsistent and unsatisfactory results. Despite considerable effort—including during the COVID-19 pandemic—we were unable to identify a reagent suitable for robust and reproducible detection of Bcd protein.

As an alternative, we used sucrose gradient fractionation followed by qPCR to directly measure the translation efficiency of *bicoid* in vivo. We believe this approach offers a clear and quantitative readout of translational activity, and it avoids potential confounding from protein degradation, which may vary across conditions and developmental stages. Nonetheless, we recognize the value of antibody-based validation and will pursue this direction in future work if reliable antibodies become available. We have added this limitation to the revised Discussion section (Lines 563–568) as follows:

“We demonstrated that the *bcd* uORF represses CDS translation using sucrose gradient fractionation followed by qPCR—an approach that directly measures translation efficiency while minimizing confounding from RNA/protein degradation. However, detecting Bcd protein levels with antibodies across developmental stages or conditions in the mutants and wild-type controls would provide an even stronger validation of our model and should be explored in future studies.”

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The authors should provide a more detailed explanation for the modifications made to the ICIER model. Specifically, an explanation of the biological or mechanistic rationale behind the ability of the 80S ribosome to cause upstream 40S ribosomes to dissociate from mRNA would help clarify this aspect of the model.

Thank you for this suggestion. In the original submission, we described our modifications to the ICIER model in the section titled “An extended ICIER model for quantifying uORF buffering in CDS translation” (Lines 88-124 of the revised manuscript).

To further clarify the biological rationale behind this mechanism, we have now included a conceptual model figure (Figure 9) illustrating mechanistically how uORF translation can buffer downstream translation within a single mRNA molecule. Additionally, we expanded the Discussion to summarize the current understanding of how collisions between translating 80S ribosomes and scanning 40S subunits may lead to dissociation, referencing known initial ribosome quality control (iRQC) pathways. These revisions provide a clearer mechanistic framework for interpreting the buffering effects modeled in our simulations. The relevant part is reproduced from Discussion (Lines 540-557) which reads as follows:

“Ribosome slowdown or stalling on mRNA due to rare codons [56,96-98] or nascent blocking peptides [99-102] frequently triggers ribosome collisions genome-wide [103-105]. Such collisions, especially among elongating 80S ribosomes, often activate ribosome quality control (RQC) pathways that recognize collision interfaces on the 40S subunit, leading to ribosomal subunit dissociation and degradation [106-108]. In mammals, ZNF598 specifically identifies collided ribosomes to initiate ubiquitin-dependent protein and mRNA quality control pathways [109-113]. Analogously, yeast employs Hel2-mediated ubiquitination of uS10, initiating dissociation via the RQC-trigger complex (RQT) [114]. Furthermore, the human RQT (hRQT) complex recognizes ubiquitinated ribosomes and induces subunit dissociation similarly to yeast RQT [115]. However, transient ribosome collisions can evade RQC by promoting resumed elongation through mechanical force provided by trailing ribosomes, thereby mitigating stalling [116]. Beyond 80S collisions, evidence increasingly highlights a distinct collision type involving scanning 40S subunits or pre-initiation (43S) complexes. Recently, an initiation RQC pathway (iRQC) targeting the small ribosomal subunit (40S) has been described, particularly involving collisions between scanning 43S complexes or between stalled 43S and elongating 80S ribosomes (Figure 9B) [117,118]. During iRQC, E3 ubiquitin ligase RNF10 ubiquitinates uS3 and uS5 proteins, resulting in 40S degradation [118]. This mechanism aligns closely with our ICIER model, proposing collision-driven 43S dissociation

in the 5' UTRs. Future studies exploring these mechanisms in greater detail will clarify how uORFs modulate translational regulation through buffering effects.”

(2) The figure legend references Figure 5C; however, this figure appears to be missing from the document.

We apologize for the oversight. The missing panel previously referred to as Figure 5C has now been incorporated into the revised Figure 6A. The figure and its corresponding legend have been corrected accordingly in the updated manuscript.

Reviewer #2 (Recommendations for the authors):

This is an important study that enhances our understanding of the roles of uORFs in translational regulation. In addition to the suggestions provided in the public review, the following minor points should be addressed before publication in eLife:

(1) Page 7, line 207: "We identified 18,412 canonical uORFs shared between the two species (referred to as conserved uORFs hereafter)." The term "canonical uORFs" requires clarification. Does this refer to uORFs with specific sequence features, conservation, or another defining characteristic?

Thank you for pointing this out. We apologize for the lack of clarity. In our study, a canonical uORF is defined as an open reading frame (ORF) that initiates with a canonical AUG start codon located in the 5' untranslated region (UTR) and terminates with a stop codon (UAA, UAG, or UGA) within the same mRNA. Conservation of uORFs is defined solely based on the presence of AUG start codons at orthologous positions in the 5' UTR across species, regardless of differences in the stop codon.

To clarify this definition, we have revised the sentence as follows (Lines 213-219): “We focused on canonical uORFs that initiate with an ATG start codon in the 5' UTR and terminate with a stop codon (TAA, TAG, or TGA). Because the ATG start codon is the defining feature of a canonical uORF and tends to be more conserved than its downstream sequence [67], we defined uORF conservation based on the presence of the ATG start codon in the 5' UTR of *D. melanogaster* and its orthologous positions in *D. simulans*, regardless of differences in the stop codon. Using this criterion, we identified 18,412 canonical uORFs with conserved start codons between the two species.”

(2) Page 8, line 227: "Furthermore, the dominant uORFs showed a higher proportion of conserved uATGs than the other translated uORFs." There appears to be a typographical error. Should "other uATGs" instead read "other uORFs"?

Thank you for pointing this out. As we addressed in response to your previous concern, in this study, we defined uORF conservation primarily based on the presence of their start codon (uATG) both in *D. melanogaster* and the orthologous sites of *D. simulans*, as the start codon is the defining feature of a uORF and tends to be more conserved than the remaining sequence, as demonstrated in our previous study [7]. We used the term “conserved uATGs” to reflect this definition and believe it accurately conveys the intended meaning in this context.

(3) Page 8, line 240: "uORFs exhibited a significant positive correlation with the TE of their downstream CDSs in all samples analyzed ($P < 0.001$, Spearman's correlation)." A Spearman's rho of 0.11 or 0.21 may not practically represent a "significant" positive correlation. Consider rephrasing this as "a positive correlation."

Thank you for the suggestion. We have revised the sentence in the manuscript to read (Lines 257-259): “uORFs exhibited a modest, yet statistically significant, positive correlation with the

TE of their downstream CDSs across all samples analyzed ($P < 0.001$, Spearman's correlation)."

(4) Page 9, line 269: The analysis of interspecific TE changes between uORFs and their corresponding CDSs is a crucial piece of evidence supporting the authors' conclusions. Presenting this analysis as part of the figures, rather than in "Table 1," would improve clarity and accessibility.

Thank you for this suggestion. In Table 1, we originally presented the number of uORFs and CDSs that showed significant differences in TE between *D. melanogaster* and *D. simulans* during various developmental stages. One key point we aimed to emphasize was that, although TE changes in uORFs and their downstream CDSs are positively correlated, there is a notable difference in the magnitude of these changes. To better convey this, we have summarized the core findings of Table 1 in graphical form.

In Figure 3B of the revised version, we compared the absolute values of interspecific TE changes between CDS and uORF, showing that CDSs consistently exhibit smaller shifts than their upstream uORFs. This result further supports the translational buffering effect of uORFs on downstream CDS expression. We have included the updated results in the revised manuscript (Lines 281-284) as follows:

"In fact, the absolute values of $\log_2(\text{fold change})$ of TE for uORFs between *D. melanogaster* and *D. simulans* was significantly greater than that observed for corresponding CDSs across all samples ($P < 0.001$, Wilcoxon signed-rank test; Figure 3B), suggesting that the magnitude of TE changes in CDSs is generally smaller than that in uORFs, due to the buffering effect of uORF."

(5) Page 9, line 279: The phrase "dominantly translated" needs clarification. Does it refer to Figure 2C, where one uORF is dominantly translated within a gene, or does it mean that the uORF's translation is higher than that of its corresponding CDS?

We apologize for the obscurity. The phrase "dominantly translated" means one uORF with the highest TE compared to other uORFs within a gene. We have rephrased the relevant sentence in the revised version (Lines 299-304), which now reads:

"To investigate how the conservation level and translation patterns of uORFs influence their buffering capacity on CDS translation, we categorized genes expressed in each pair of samples into three classes:

Class I, genes with conserved uORFs that are dominantly translated (i.e., exhibiting the highest TE among all uORFs within the same gene) in both *Drosophila* species; Class II, genes with conserved uORFs that are translated in both species but not dominantly translated in at least one; and Class III, the remaining expressed genes."

(6) The sequencing data and analysis code should be made publicly available before publication to ensure transparency and reproducibility.

Thank you for this suggestion. As described in the Data availability section, all deepsequencing data generated in this study, including single-ended mRNA-Seq and Ribo-Seq data of 10 developmental stages and tissues of *Drosophila simulans* and paired-end mRNA-Seq data of 0-2 h, 26 h, 6-12 h, and 12-24 h *Drosophila melanogaster* embryos, were deposited in the China National Genomics Data Center Genome Sequence Archive (GSA) under accession numbers CRA003198, CRA007425, and CRA007426. The mRNA-Seq and Ribo-Seq data for the different developmental stages and tissues of *Drosophila melanogaster* were

published in our previous paper [8] and were deposited in the Sequence Read Archive (SRA) under accession number SRP067542.

All original code has been deposited on GitHub: https://github.com/lujlab/uORF_buffer; https://github.com/lujlab/Buffer_eLife2025.

Response reference

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